

A morphology-matched forward-readout protocol for SPARC rotation curves

Jozsef Olcsak

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Abstract

This paper tests whether residual-blind morphology/readout information can choose a rotation-curve readout family before the residuals are inspected. Standard rotation-curve tests usually compare baryonic Newtonian predictions with universal or weakly morphology-dependent correction laws. This paper develops a different, claim-bounded protocol: before scoring a rotation curve, assign a residual-blind morphology/readout state, select the corresponding readout family, freeze source-side scales and amplitudes where possible, and compare the matched family against wrong-family, shuffled-label, Newtonian, MOND-like, RAR, and fixed predecessor-readout controls. The protocol is motivated by three observations that are made self-contained here: external structural disturbance can correlate with residual structure; residual-shape diagnostics are useful but inverse; and a forward test must choose the readout family without using endpoint residuals. Using SPARC rotation curves as the common numerical arena, the accompanying reproducibility code finds a retrospective internal label-formula compatibility signal: a source-native readout-formula preflight beats wrong readout-formula families in 0.886 of the 44 holdout galaxies. For the shuffled-label null, the mean matched-minus-wrong statistic gives $p \simeq 0.002$ in the 1000-shuffle endpoint run, while the beat-fraction statistic is reported separately below. The matched-versus-wrong signal is preserved in a Newtonian-carrier amplitude-freeze stress test, supporting carrier robustness but not baseline superiority. However, a later cross-fitted audit of the same 45-galaxy Paper 1–3 label packet finds no stable projection-contrast increment after endpoint-residual-free observability/baryonic covariates and MOND/RAR-common residual structure are supplied. In addition, a source-frozen seven-galaxy EDGE-CALIFA matched-tracer stress test fails its predeclared external morphology-alignment gate (mean $D = -0.0587$, exact one-sided $p = 0.6016$, four of seven positive). Further public-data routes do not reverse that verdict: a prospective 14-galaxy LITTLE THINGS replay gives only mixed one-family transfer; low-order PHANGS CO-H α structure either obeys the morphology-orthogonal null or fails source-label specificity; and a higher-dimensional PHANGS confirmatory packet stops at its predeclared spatial-support gate without releasing a score. A robust one-galaxy NGC4254 $m = 2$ source coordinate survives internal source controls, but its terminal gain and primitive action curvature are non-identifiable from the present source state. Thus the internal preflight has not transferred into conditional specificity or external confirmation. Moreover, the same preflight beats a fixed TPG/v6 predecessor comparator and MOND-like comparators in only 0.477 of holdout galaxies each, so it is not evidence that the proposed framework outperforms existing baselines as a population law. A separate all-SPARC 175-galaxy morphology triage identifies 39 source-freeze candidates, 61 source/kernel precision cases, 18 morphology/subfamily review cases, and 57 control or low-quality cases. Four inspected single-galaxy readout cases also show encouraging matched-family behavior, but remain heterogeneous and caveated. An unchanged-law transfer from the strongest NGC4088 case to the source-native outer warp of UGC08490/NGC5204 preserves a weak morphology-ordering signal—the correct onset gives 29.70 km s^{-1} RMSE versus 31.64 km s^{-1} for the wrong NGC4088 onset—but fails badly

against fixed MOND (8.78 km s^{-1}) and two-parameter halo fits ($1.20\text{--}1.55 \text{ km s}^{-1}$). This is a reproducible negative result for the present one-component universal warp-law transfer, not for morphology information or Tau Core in general. A second unchanged-law replication on the source-selected warp of UGC03580/UGC3580 is stronger: the source-correct onset gives 38.04 km s^{-1} full-curve RMSE, worse than Newtonian baryons (36.01), fixed MOND (27.17), and an earlier wrong-onset control (24.56). The uncapped kernel also overshoots beyond R_{HI} . Together these transfers reject the present uncapped one-component law as a universal warp-class rotation law. An endpoint-blind follow-up replaces the scalar UGC03580 onset proxy by a piecewise geodesic path through all published supported tilted-ring orientations. It reconstructs the independent inner/outer plane angle. The resulting defect-ratio family is bounded, but an explicit counterfamily proves that the body alone does not select its power, amplitude, sign, carrier, or physical occupation. A standard fixed-inclination projection control reaches 1.09685 in squared speed on the supported rings and must be audited and, where applicable, accounted for before any Tau attribution. No rotation endpoint is used or rescored in this derivation. A later source-frozen NGC 4062 H I–H α endpoint is null in both resolutions and changes sign under the alternate published velocity zero point; it is therefore a preserved calibration-limited negative endpoint, not a Tau-specific detection or an accepted framework-level true negative. The analysis otherwise contains branch-level negative controls and morphology-underdetermined weak negatives, but no source-complete endpoint-grade true negative. The main conclusion is therefore methodological: morphology-matched forward readouts define a falsifiable endpoint protocol, not a validated gravity theory. A practical next step is a residual-blind readout morphology catalogue that freezes source morphology, projection, H I support, vertical structure, and morphology-trajectory/phase observables before endpoint scoring. The result should be read as a reproducible protocol and internal preflight signal, not as evidence for a new gravitational law.

1 Introduction

The SPARC data set provides high-quality rotation curves and baryonic mass models for 175 disk galaxies [19]. It is a standard testbed for Newtonian baryonic predictions, MOND-like dynamics, and empirical radial-acceleration relations [26, 25, 22]. These comparisons are usually organized around scalar acceleration laws or globally fitted mass-to-light choices. They are powerful precisely because they are simple.

The comparison set in this paper has two different kinds of baselines. The Newtonian, MOND-like, and RAR-like references are external literature-facing comparators. The TPG/v6 curve, by contrast, is an internal predecessor readout/closure comparator from the same research programme. It is included because it is frozen in the reproducibility package and is often a strong smooth baseline. It is not treated here as an independently published reference model, and it is not used to choose or tune the morphology-matched readout family.

The present paper asks a different question. Suppose that some residual structure is not described well by a universal scalar correction, but depends on which morphology/readout state the galaxy occupies: regular disk, extended scale-tail disk, compact finite-source system, vertical/projection-dominated system, ring/resonance structure, warp/trajectory structure, or a mixture of such components. Then the relevant test is not merely whether a flexible curve can fit a galaxy. The test is whether the morphology-matched readout family, chosen before endpoint scoring, performs better than wrong morphology families and shuffled labels, and only then whether it is competitive with conventional baselines.

The forward kernel is deliberately a terminal hypothesis. A morphology label, a fitted rotation residual, or a successful matched-family score does not construct a smooth parent source arrow or optical/body connection. Current Tau Core source accounting permits such non-scalar transport

only after an independently sourced full body connection or a varying, constant-rank, source-owned access projector has been supplied. This added boundary does not alter the scoring problem below; it prevents a terminal success from being reinterpreted as identification of the upstream transport mechanism.

This paper makes that test self-contained. The framework is used only through operational definitions. In Tau Core terminology the intrinsic source object is the *morphological body* M_τ , while the data used here are only projected and source-frozen handles on that body: a galaxy has a projected morphology handle K_{obs} , a possibly different readout-relevant proxy K_{readout} , and a family of local readout kernels that generate a forward correction $\delta v_K^2(R)$ or equivalent closure response. No external theory paper is required for the protocol. The definitions, allowed inputs, forbidden inputs, endpoint score, negative-result taxonomy, and numerical status are stated below.

1.1 Reader’s roadmap

What must be accepted to read the paper: only the operational protocol defined here. A reader does not need to accept a completed new gravity theory.

What is tested: whether a residual-blind morphology/readout label can choose a restricted readout family that performs better than wrong families and shuffled labels.

What is not claimed: population-level superiority over Newtonian, MOND/RAR-like, or TPG/v6-like baselines; a final morphology catalogue; or a validated gravitational theory.

Where the evidence appears: Section 3 defines the scoring protocol; Section 4 defines the readout families; Section 5 gives the reproducibility and leakage-prevention path; Section 6 reports population preflights, the 175-galaxy triage, four inspected rotation-curve cases, and the unchanged-law UGC08490 transfer; Section 7 explains how to read failures.

Equations (1)–(3) make this taxonomy falsifiable. A representation-level failure may motivate a new source-only representation, but it cannot be relabeled as a body-level success and it cannot be repaired on the opened endpoint. Conversely, a source-complete frozen representation that satisfies the terminal factorization premise and still fails its declared controls is a genuine negative for that completed model class.

1.2 Status of the underlying theory language

The terminology “readout”, “closure”, and “morphological body” is motivated by an unpublished broader theory programme. That programme is not assumed as a published premise of this manuscript. The present paper therefore uses only the operational layer defined here: residual-blind morphology labels, predeclared readout families, frozen source-side inputs, endpoint scores, wrong-family controls, shuffled-label controls, and conventional comparators. Any reader can evaluate the claims below without accepting the unpublished theory motivation.

1.3 Theory motivation isolated as a testable consequence

The broader motivation can be summarized without importing the full derivation. In the underlying readout picture, the morphology seen in 4D is not assumed to be the complete state that controls the effective gravitational response. It is a projected handle on a deeper readout configuration. Consequently, a galaxy’s current visual class K_{obs} may differ from the readout-relevant class K_{readout} , especially when projection, H I support, vertical structure, warp/trajectory evidence, rings, bars, or lopsidedness are present.

In the companion bridge language, this deeper source object is called the *morphological body* M_τ . The name does not mean a visible galaxy shape. It denotes a Tau-side organizing body whose

four-dimensional readouts may include visible morphology, matter distribution, gravity response, time/clock structure, observer/path appearance, and quantum/coherence structure. The present paper uses only the operational consequence of that idea: the observed 4D morphology label is a source handle, not automatically the active readout-family label.

This language should not be read as a claim that the present rotation-curve tests recover quantum mechanics or that galaxies are quantum-mechanical wavefunctions. The intended architectural point is weaker: in the broader Tau Core programme, quantum, gravitational, lensing, galactic, and cosmological phenomena are treated as typed terminal readouts of a shared body-conditioned extraction, not as automatically independent physical channels. The present paper tests only the galaxy-scale operational consequence: whether residual-blind morphology and projection handles select useful forward readout families. Full Hilbert-space, Born-rule, or quantum-field recovery is outside the scope of this manuscript.

The observer is likewise not inserted as a detached endpoint after this extraction. In the current theory convention a body-side carrier becomes an operational observer only together with a regular local rank-four Lorentzian descent, a stable observer–source resolution map, and a nonzero record/effect. The operational observer and its accessible four-dimensional world are thus co-readouts of one body-conditioned solution. This paper begins downstream of that closure: its galaxy measurements presuppose one operational observer–source packet and test only whether source-frozen morphology selects additional terminal readout-family information. A successful score would not by itself prove observer production or the parent origin of that packet.

This distinction is also how the observer/projection intuition enters the paper. The intrinsic Tau-side organization of a galaxy is not treated as a mere observer-dependent illusion. However, the 4D morphology and dynamics available to an observer are projected readouts of that organization. Thus the working hierarchy is

$$K_{\text{intrinsic}}^{\tau} \longrightarrow K_{\text{projected}}^{4D}(O_{OS}^{\text{op}}) \longrightarrow K_{\text{readout}},$$

where O_{OS}^{op} denotes the already realized operational observer–source packet, not a generatively prior observer or a separate physical channel layer. A fuller projection-enriched version also allows a source-frozen morphology trajectory/phase variable Θ_{morph} , representing the possibility that 4D past, present, and future morphology are different readout slices of one deeper Tau-side configuration. The present paper does not model that full observer-time response. It uses this hierarchy only to justify why the observed morphology label should not automatically be identified with the readout family tested against the rotation curve.

The current Tau Core theory formulation sharpens this hierarchy as a source-factored kernel rule. In that language a stabilized source class M_{τ}^{stab} is read through branch maps,

$$K_i(s_0) = \tilde{K}_i(M_{\tau}^{\text{stab}}),$$

and an observer’s clock/order variable is an internal host-sector-conditioned readout,

$$t_{\text{obs}}^{\text{gal}} = R_{\text{time}}^{\text{obs,gal}}(M_{\tau}^{\text{stab}}, OI_{\text{gal}}, o_{\text{path}}^{\text{gal}}),$$

not an external Tau-time in which the source body evolves. The galaxy-sector index is part of the claim boundary: a rotation-curve observer/path ordering is not automatically the same as a cosmological, lensing, or quantum detector clock. Operationally this paper can only approximate the protected source quotient by residual-blind source proxies. Hence the morphology labels used below are proxy coordinates for M_{τ}^{stab} , not final Tau-side source classes.

The operational consequence is the only part tested here:

$$K_{\text{obs}} \longrightarrow K_{\text{readout}} \longrightarrow \mathcal{F}_{K_{\text{readout}}} \longrightarrow \delta v_{K_{\text{readout}}}^2(R).$$

If this readout interpretation carries empirical information, then a residual-blind morphology-matched family should perform better than wrong morphology families and shuffled labels. This is a weaker and safer statement than claiming a completed gravity theory. The full derivation may motivate the choice of language, but the present paper tests only this forward-readout consequence.

1.4 Mini-glossary

Table 1: Plain-language glossary for the main protocol terms.

Symbol or term	Meaning in this paper
K_{obs}	The residual-blind morphology handle suggested by observed structure: disk, warp, ring, bar, compact support, tail, or lopsidedness.
K_{readout}	The morphology/readout state used to choose the formula family. It may differ from K_{obs} when projection, H I support, vertical structure, or trajectory/phase evidence matters.
Matched family	The readout family selected before endpoint scoring.
Wrong family	A deliberately mismatched family used as a morphology-specific control.
Shuffled labels	A null test where morphology labels are randomized.
TPG/v6	A frozen internal predecessor comparator, not an external published baseline and not a tuning input.
Endpoint	A scored test after the readout family and source inputs have been frozen.
True negative	A source-complete, residual-blind endpoint failure against the relevant baselines and controls.

2 Operational framework

2.1 Observed morphology versus readout morphology

Let K_{obs} denote a residual-blind observed morphology handle: for example thin disk, thick disk, warp, ring, bar, compact support, extended H I tail, or lopsidedness. The quantity used to choose a readout family is K_{readout} . It may equal K_{obs} , but it need not. A present 4D visual class can be only a projection of the readout-relevant state; edge-on projection, vertical structure, H I support, or morphology-trajectory evidence can change which readout family should be tested.

In this sense, the morphology component is partly observer/projection dependent, but not because the galaxy’s source structure is assumed to be arbitrary. The source-side organization is the target; the observer sees a 4D projection of it. Inclination, line-of-sight geometry, waveband, H I extent, vertical support, and trajectory/phase indicators can change the available projected proxy. The protocol therefore treats K_{obs} as a residual-blind observational handle and K_{readout} as the source-disciplined class used for forward scoring. The current kernels are projection-aware morphology proxies, not full observer-projection Tau Core readouts; observer/path/time-projection effects enter only through frozen source proxies such as inclination, vertical support, H I extent, warp/trajectory flags, and projected readout-family selection. The kinds of source fields needed for this distinction are standard observational products rather than exotic new observables: rotation-curve and mass-model catalogues [19], H I synthesis and velocity-field studies [41, 44], photometric/kinematic late-type galaxy studies [39], and lopsidedness or asymmetry catalogues [43].

For each family K , let $\mathcal{F}_K$ be a source-first readout shell and let $K_{gK}(R)$ be the radial kernel assigned to galaxy g . The generic velocity-squared form used in this paper is

$$\delta v_{gK}^2(R) = A_{gK} K_{gK}(R),$$

where the amplitude A_{gK} must be either source-derived or frozen by a predeclared rule. The rotation curve v_{obs} may be read only by the separate scoring step.

2.2 No-curve-fitting rule

The construction ladder is

$$\begin{aligned}
K_{\text{obs}} &\longrightarrow K_{\text{readout}} \\
&\longrightarrow \{O_{gK}\}_{\text{source}} \\
&\longrightarrow \mathcal{F}_K, K_{gK}(R) \\
&\longrightarrow A_{gK}^{\text{source}} \text{ or predeclared amplitude rule} \\
&\longrightarrow \delta v_{gK}^2(R).
\end{aligned}$$

Allowed inputs are residual-blind morphology and source quantities: disk scale, H I radius, outer-tail onset, compact support radius, vertical thickness, projection state, bar/ring/lopsidedness evidence, and environment or trajectory/phase context. Forbidden construction inputs are the observed endpoint residual, the best baseline score, inverse diagnostics that ask what preparation would be needed to absorb the residual, and post-hoc family selection.

The common bridge object behind this ladder is not a fitted rotation curve. It is the conditional Tau-side quotient residual

$$\delta g_{\text{morph}}(x) = R_{g,x}^{4D} [\rho_D(\mathcal{M}_\tau(x))(Q_D^\tau(x))],$$

where Q_D^τ is the ordered-disk readout direction, $\mathcal{M}_\tau$ is the local readout/closure mismatch generator, $\rho_D = \pi_B \circ \mathcal{M}_\tau|_{D_{\text{diag}}}$ projects away the Newtonian, ordinary-disk, and gauge sectors, and R_g^{4D} is the weak-field 4D readout map. Appendix A derives the operational kernels by reducing this same bridge equation to source-frozen one-dimensional radial readouts. This is why the families below are treated as bridge specializations rather than as independent empirical templates.

2.3 Readout family registry

The current first-pass registry is deliberately small and test-oriented. It is not asserted to be a final physical taxonomy.

Table 2: Claim-bounded morphology/readout families used in the present tests.

Family	Morphology/readout handle	Representative shell	Status
K_{tail}	scale-tail or diffuse LSB disk	$\delta v^2 = \xi_{\text{tail}} \Lambda_{\text{tail}} I_2(R)/R$	1D testable
K_{exp}	regular exponential disk	$\delta v^2 = A_{\text{exp}} y^2 [I_0 K_0 - I_1 K_1], y = R/(2R_s)$	1D testable
K_{compact}	compact finite support	exterior finite-source response, $\delta g_R \sim -I_\infty/R^2$	1D testable
K_{flare}	thick/flared or projection-sensitive disk	$v_{\text{carrier}}^2 [1 - \Gamma_{\text{vp}} K_{\text{vp}}(R)]$	1D proxy
K_{ring}	ring or resonance structure	localized radial kernel near R_{ring}	1D proxy
$K_{m=2}$	barred spiral	non-axisymmetric harmonic response	velocity-field preferred
$K_{m=1}$	lopsided disk	non-axisymmetric harmonic response	velocity-field preferred

The entries in Table 1 are not arbitrary curve templates. They are the minimal source-first kernels retained after the readout motivation is reduced to one-dimensional rotation-curve observables. The reduction has three steps:

morphology/readout handle $\rightarrow$ allowed radial source scale $\rightarrow$ restricted kernel shape.

For example, a scale-tail or diffuse LSB handle permits an extended outer-load kernel; an exponential-disk handle permits the standard finite exponential-disk radial carrier shape; a compact-support handle permits an exterior finite-load falloff; and a thick/flared or projection-sensitive handle permits a vertical/projection attenuation kernel. Ring, bar, and lopsided handles are listed because they are part of the morphology registry, but in the present one-dimensional SPARC analysis they remain proxy or velocity-field-preferred families rather than full endpoint kernels.

Table 3: Current sample coverage of the source-native readout families. Counts refer to the 175-row SPARC preflight label table. The examples are illustrative, not a complete list.

Family	Total	Holdout	Example assigned galaxies	Kernel origin in this paper
$K_{\text{tail}} / K_{\text{scale_tail_spiral}}$	80	20	CamB, F561-1, F563-1, IC2574, NGC4214, NGC6789	outer-load / scale-tail source handle reduced to a radial tail kernel
$K_{\text{exp}} / K_{\text{exponential_disk}}$	32	7	ESO116-G012, NGC4183, UGC02259, UGC07151, UGC07261, UGC07603	regular disk handle reduced to a finite exponential-disk carrier shape
$K_{\text{compact}} / K_{\text{compact_finite}}$	29	7	IC4202, NGC4013, NGC5985, NGC6946, UGC03580, UGC08699	compact finite-source handle reduced to an exterior finite-load response
$K_{\text{flare}} / K_{\text{thick_flared}}$	34	10	F568-1, NGC0024, NGC2683, NGC3726, NGC3949, NGC3972	vertical, edge-on, or projection-sensitive handle reduced to an attenuation proxy

The current population scoring uses the four one-dimensional families in Table 3. The registry also records ring, barred, and lopsided families, but those require either stronger source fields or two-dimensional velocity-field information before they should be promoted to the same endpoint status. This is why the paper’s numerical claims are stated for the source-native 1D preflight families and for four inspected single-galaxy cases, not for the full conceptual registry. Appendix A gives the common bridge equation and the explicit readout-kernel derivation ledger for the morphology families used here, including the dimensional status and the precise claim boundary for each family.

Kernel-audit verdict. The audit status of the present Paper 8 kernel set is deliberately modest. The scale-tail, exponential, compact, and thick/flared shells are accepted as *source-native one-dimensional preflight kernels*: they may be used for matched-versus-wrong-family and shuffled-label tests under the frozen manifest, but they are not claimed to be final physical laws or complete Tau Core readout operators. Ring, barred, and lopsided entries are retained as registry/control families because one-dimensional SPARC curves do not contain enough source information to freeze their non-axisymmetric readout operators. Projection-sensitive and thick/flared rows remain proxy-level unless their orientation, vertical, or H I support fields are source-frozen before scoring. All active terms must obey the source-support rule introduced above: one protected source coordinate can supply at most one quotient contribution. If a future morphology, projection, or time-like correction reuses the same source proxy as an active kernel, it must be merged into a shared quotient term or kept as a diagnostic control.

MORPHOLOGY-MATCHED-FORWARD-READOUT-GATE

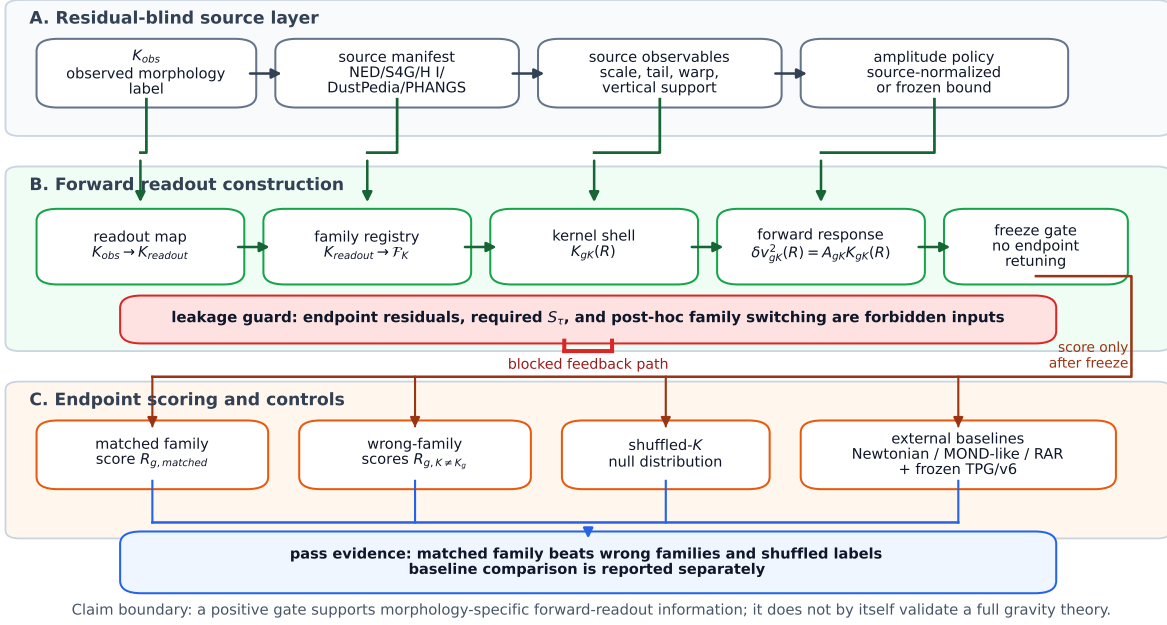


Figure 1: Morphology-matched forward-readout gate. Residual-blind source fields first determine the observed morphology handle, readout-relevant proxy, source observables, and amplitude policy. Only after the readout family, kernel shell, forward response, and freeze gate are fixed may endpoint scores be computed against matched, wrong-family, shuffled-label, and baseline controls. The red guard marks forbidden leakage from endpoint residuals or required- S_r diagnostics back into family selection.

3 Endpoint protocol

For galaxy g , define $R_{g,K}$ as the residual score after applying family K . In the present numerical work this is an RMSE in rotation speed. The matched score is

$$R_{g,matched} = R_{g,K_{readout,g}},$$

and the wrong-family mean is

$$R_{g,wrong} = \frac{1}{|\mathcal{K}| - 1} \sum_{K \neq K_{readout,g}} R_{g,K}.$$

The morphology-specific endpoint is

$$\Delta_{matched} = \text{RMS}_g(R_{g,wrong}) - \text{RMS}_g(R_{g,matched}).$$

The matched family should also rank well among all candidate families:

$$\text{rank}_g(K_{readout,g}) = \text{rank}(R_{g,K_{readout,g}} \text{ among } \{R_{g,K} : K \in \mathcal{K}\}).$$

A positive result requires comparison not only to wrong families, but also to shuffled morphology labels and conventional baselines. A morphology-specific win against wrong families is therefore weaker than a full all-baseline population win.

4 Data, controls, and scoring protocol

The numerical arena is SPARC [19]. The current scripts use the available SPARC rotation-curve points and the baseline columns already present in the reproducibility package: Newtonian baryonic, TPG/v6, and a MOND-like comparator. The analysis is organized into a train/holdout split with 131 train galaxies and 44 holdout galaxies for the population preflights. Where population amplitudes are needed, they are estimated on the train split only and then evaluated on the holdout split. This is weaker than a fully source-frozen amplitude theorem, but stronger than fitting each holdout galaxy independently.

The controls have separate roles:

- *Wrong-family controls* test morphology specificity within the readout family registry.
- *Shuffled-label controls* test whether the morphology labels carry information beyond generic flexibility.
- *Newtonian, TPG/v6, and MOND-like controls* test whether the selected readout is competitive with conventional or already-available comparators.
- *Single-galaxy endpoint controls* test whether an object-specific source-frozen protocol is internally nonleaky before any broader claim is made.

4.1 Status of the TPG/v6 comparator

The TPG/v6 column requires a separate caveat. It is not a standard external literature baseline in the same sense as Newtonian, MOND-like, or RAR-like comparisons. It is a fixed predecessor readout from the author’s earlier closure-oriented programme. In this manuscript it is used operationally: a galaxy either beats the frozen TPG/v6 curve or it does not. The endpoint code does not rederive TPG/v6, does not retune it per galaxy, and does not use it to select K_{readout} .

TPG/v6 use in this paper: frozen before the endpoint runs; not used for label selection; not retuned per galaxy; included only as an internal predecessor-control; not claimed as an external literature baseline.

This distinction matters for interpretation. If a morphology-matched readout beats wrong families but not TPG/v6, the result still supports morphology specificity but not baseline superiority. If it beats TPG/v6 in a source-frozen endpoint, that is evidence against a strong predecessor-control curve, not evidence against an independently established external theory.

4.2 Score definitions

The main score is RMSE in rotation speed, quoted in kms^{-1} . Lower scores are better. For difference scores such as matched-minus-wrong, negative values mean that the matched readout improves on the comparator. The permutation nulls use 1000 deterministic shuffles in the current executable analysis. The reported p -values are therefore preparation-level finite-shuffle estimates, not asymptotic discovery claims.

For a galaxy g , model m , and measured radii R_i , the plotted and tabulated endpoint score is

$$\text{RMSE}_{g,m} = \left[\frac{1}{N_g} \sum_{i=1}^{N_g} (v_m(R_i) - v_{\text{obs}}(R_i))^2 \right]^{1/2}.$$

The endpoint residuals shown below are

$$\Delta v_m(R_i) = v_m(R_i) - v_{\text{obs}}(R_i).$$

Thus a visually smooth curve is not sufficient; it must reduce the signed residual structure without having been selected from those residuals.

5 Reproducibility and leakage prevention

The numerical claim in this manuscript is meaningful only if the morphology labels, family choices, and source scales are fixed before endpoint residuals are scored. The reproducibility package therefore separates source artifacts, amplitude artifacts, scoring artifacts, and paper-generation artifacts.

5.1 Exact data sources and frozen artifacts

The released package uses relative repository paths. One upstream SPARC/TPG point table was read from the author’s local run directory while preparing this draft; in a release archive it should be copied or symlinked into the package under a relative path before reproduction. The exact inputs used by the source-native readout-formula preflight are:

- *Rotation curves and baseline columns:* the SPARC-based point table `data/external/tau_rotation_curve_frozen_proxy_runner_v0_points.csv` in the released package. In the author’s local run this was read from the TPG results directory. This table supplies R , v_{obs} , σ_v , Newtonian/proxy columns, TPG/v6, and MOND-like comparator columns used by the executable preflight.
- *SPARC metadata:* `data/derived/external_sparc_master_table.csv`, derived from the SPARC master table [19]. This supplies galaxy type, distance, inclination, luminosity, disk scale, surface brightness, H I mass/radius where available, and quality flags.
- *Residual-blind morphology parameter manifest:* `data/derived/morphology_parameter_manifest.csv`. This is the source manifest consumed by the formula endpoint script.
- *Label-freeze artifact:* `data/derived/source_native_readout_formula_labels.csv`. It records the train/holdout split, formula family, manifest confidence, source proxies, amplitude policy, parameter source, caveats, and forbidden fields for all 175 SPARC galaxies.
- *Amplitude-freeze artifact:* `data/derived/source_native_readout_formula_amplitudes.csv`. It records one family-level amplitude per readout family, estimated on train galaxies only.

The population split used in the preflight is the frozen split column consumed from the point/label artifacts: 131 train galaxies and 44 holdout galaxies. The endpoint scripts do not redraw this split for the headline result. The alternative-split robustness audit reported below is a separate stress test, not the primary endpoint.

5.2 Amplitude freeze rule and forbidden endpoint fields

For the source-native formula preflight, the amplitude rule is train-only and family-level:

$$A_K = \arg \min_A \sum_{g \in \text{train}, K_g=K} \sum_i \left[v_{\text{obs}}^2(R_{gi}) - v_{\text{TPG/v6}}^2(R_{gi}) - A K_{gK}(R_{gi}) \right]^2.$$

The fitted A_K is then applied unchanged to holdout galaxies. This is not a source-derived amplitude theorem; it is a preparation-level amplitude freeze. The TPG/v6 curve is used only as the frozen carrier for this amplitude-residual preflight. It is not used to select labels or readout families, and holdout scoring still reports matched-versus-wrong, TPG/v6, and MOND-like comparisons separately. The paper therefore treats the population result as evidence for morphology-specific formula choice, not as final empirical validation.

Forbidden fields for label and family construction are:

v_{obs} , $v_m - v_{\text{obs}}$, RMSE, best baseline score, required S_τ , post-hoc family choice.

The label-freeze artifact records this explicitly in its `forbidden_inputs` column. Endpoint residuals may be used only after K_{readout} , source proxies, kernels, and amplitude policy have been written to disk.

5.3 Scripts, seeds, and outputs

The shortest path to reproduce the headline source-native result is:

```
python scripts/reproduce.py
python scripts/run_source_native_readout_formula_endpoint.py
python scripts/build_source_native_readout_robustness.py
python scripts/run_source_native_carrier_robustness.py
cat data/derived/source_native_readout_formula_endpoint_summary.csv
```

The corresponding robustness table is written to `data/derived/source_native_readout_formula_robustness_summary.csv`; the carrier stress test is written to `data/derived/source_native_carrier_robustness_summary.csv`.

The executable path is:

1. `scripts/build_morphology_parameter_manifest.py`: builds the residual-blind morphology/source manifest.
2. `scripts/run_source_native_readout_formula_endpoint.py`: builds readout kernels, fits train-only family amplitudes, scores train and holdout, and runs the 1000-shuffle null.
3. `scripts/build_source_native_readout_robustness.py`: runs the 10k shuffle, bootstrap, error-weighted, alternative-split, and family-balanced robustness checks.
4. `scripts/run_source_native_carrier_robustness.py`: repeats the train-only family-amplitude freeze with TPG/v6 and Newtonian baryonic carriers while keeping labels, kernels, source proxies, and splits fixed.
5. `scripts/build_paper_summary_figures.py` and `scripts/generate_paper8_artifacts.py`: regenerate paper figures and protocol diagrams from saved CSV artifacts.
6. `scripts/reproduce.py`: repository-level reproduction driver.
7. `scripts/build_arxiv_source.py`: builds the arXiv source package.

The random seeds are fixed: 31415 for the original 1000 shuffled-label null, 271828 for the 10k shuffled-label robustness null, 271829 for the bootstrap confidence interval, and 271830 for the alternative-split robustness audit.

The output artifacts used by the paper are:

- `data/derived/source_native_readout_formula_scores_by_galaxy.csv` and `data/derived/source_native_readout_formula_endpoint_summary.csv`;
- `data/derived/source_native_readout_formula_shuffled_null.csv` and `data/derived/source_native_readout_formula_shuffled_null_summary.csv`;
- `data/derived/source_native_readout_formula_robustness_summary.csv` and `data/derived/source_native_readout_formula_error_weighted_scores.csv`;
- `data/derived/source_native_carrier_robustness_summary.csv`, `data/derived/source_native_carrier_robustness_scores_by_galaxy.csv`, `data/derived/source_native_carrier_robustness_by_family.csv`, `data/derived/source_native_carrier_robustness_freeze_invariance_audit.csv`, and `data/derived/source_native_carrier_robustness_shuffled_null_summary.csv`;
- `data/derived/all_galaxy_morphology_precision_triage.csv`;
- `data/derived/four_case_endpoint_status_cases.csv`.

Leakage can be checked by verifying that the label artifact contains no endpoint residual or score columns, that its `parameter_source` fields are source/proxy descriptors, that the `forbidden_inputs` column lists endpoint residuals and required- S_τ diagnostics as forbidden, and that the amplitude artifact contains only family-level train-only amplitudes. The holdout scores in `source_native_readout_formula_scores_by_galaxy.csv` are produced only after those artifacts exist.

5.4 All-galaxy morphology triage

In addition to the 44-galaxy holdout preflights and the four inspected endpoints, the reproducibility package contains an all-SPARC morphology precision triage over the full 175-galaxy SPARC table. This step is not an endpoint validation. It is a worklist compiler: for every galaxy it assigns a current readout family, records which source observables are missing, scores the currently compiled proxy against available rotation data, and classifies the case by what should happen next.

The important methodological point is that this triage is deliberately one-way. The readout family and missing-source diagnosis are not promoted from endpoint residuals. Endpoint velocities are used only after the current compiled protocol exists, in order to decide whether the case is already useful, needs source/kernel precision, needs morphology or subfamily review, or should be kept as a low-quality/control object. The resulting classes are summarized in Table 4.

Table 4: All-SPARC morphology precision triage. These 175 rows are a source-acquisition and protocol-prioritization product, not endpoint validation.

Triage class	Galaxies	Proxy baseline flag fraction	Interpretation
Current morphology good enough for source-freeze review	39	1.000	Promising candidates for predeclared replay or source-frozen endpoint work
Morphology signal present, needs source/kernel precision	61	0.000	Matched family often beats wrong families, but not the best baseline yet
Morphology or subfamily review needed	18	0.000	Likely readout-family, subfamily, or kernel mismatch; not a true negative
Control or low quality, do not prioritize	57	0.281	Useful mainly as controls or after rotation/source-quality improvement

This table is the paper’s broadest survey layer. It says that the current readout proxies carry morphology information on many galaxies, but also that the limiting factor is often source completeness and readout-subfamily precision rather than the existence of a fitted curve. In particular, the 39 “good enough” cases are not claimed as accepted endpoints; they are the highest-priority queue for residual-blind source freezing. The 18 morphology/subfamily-review cases are also not

failures of the framework; they are the clearest places where the present readout label may be too coarse.

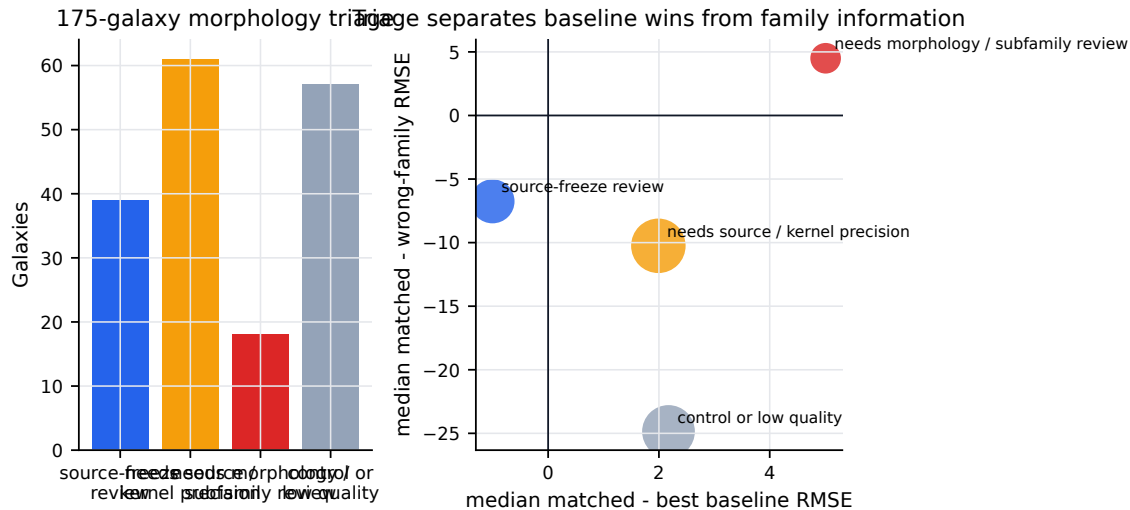


Figure 2: All-SPARC morphology precision triage. Left: the 175 galaxies are partitioned into source-freeze candidates, source/kernel precision cases, morphology/subfamily review cases, and control/low-quality cases. Right: median score differences show why the classes should not be collapsed into a single win/loss statistic. Some classes carry strong wrong-family information without yet beating the best baseline.

5.5 How morphology labels were assigned

The morphology labels used in the present paper have different evidence grades. This is why the manuscript distinguishes population preflights, all-galaxy triage, and inspected single-galaxy endpoints. The broad population layers use available residual-blind metadata and proxy source fields; the four inspected cases use object-specific source ledgers. Table 5 summarizes the assignment logic.

Table 5: Morphology/readout source audit. “Proxy” means sufficient for preflight or triage but not for endpoint validation.

Layer or galaxy		Readout label status	Source basis	Claim boundary
44-galaxy preflights	holdout	proxy family labels	SPARC rotation/mass-model meta-data, type, inclination, radial coverage, surface-brightness and gas/bulge proxy fields; train-only amplitude policies	Preparation-level screen only; useful for matched-vs-wrong and shuffled-label tests, not endpoint validation
175-galaxy SPARC triage	all-	compiled current readout family	Accepted-manifest table plus missing-observable ledger: source status, confidence, caveats, missing fields, fallback policy, and next source gate	Worklist and prioritization layer; endpoint velocities are used only after the compiled protocol exists
NGC4013		mixed exponential-disk / warp-vertical overlay lane	H I and vertical-structure literature context [48, 5]; later fresh-source review supports the lane, but the score remains retrospective	Protocol-ready and replay-needed; not an accepted population endpoint
NGC5907		projection-dominated mixed readout	Warp, edge-on, and interaction/projection context [35, 37, 46]	Accepted preliminary single-galaxy control; population validation still open
NGC7331		exponential/vertical/outer warp mixed lane	H I warp/trajectory and vertical-scale context [2, 29]; pure branch retained as negative control	Caveated accepted mixed case; broad outer window and pure-branch negative remain explicit caveats
NGC4088		warp/trajectory readout	H I/interaction and warp/asymmetry source packet from the object-specific ledger; B1 provenance and B2/B3 normalization remain caveated	Strongest single inspected case; caveated endpoint, not a general law claim

Thus the morphology determination is not uniform across all rows. The broad screens deliberately use weaker but residual-blind proxies to find where the protocol is informative. The inspected galaxies use stronger source packets, but they still carry case-specific caveats. This is why the paper does not collapse the evidence into a single “Tau wins” or “Tau fails” statement.

6 Numerical evidence

Claim levels used below.

- **Strong internal-preflight signal:** source-native readout formulas beat wrong morphology families on the holdout split more often than shuffled labels would predict.
- **Moderate case evidence:** four inspected galaxies show local matched-family improvement, but each remains caveated.
- **Not shown:** population-level superiority over Newtonian, MOND/RAR-like, or TPG/v6-like baselines.

6.1 Population preflights

The current executable analysis contains several preparation-level screens. They are useful because they separate morphology specificity from baseline superiority.

Table 6: Main population-level preflight outcomes. These are internal preflight results, not empirical validation.

Test	Holdout/sample	Beats wrong family	Beats TPG/v6	Beats MOND-like
Metadata readout proxy	44 holdout	0.568	0.636	0.614
Naive radial shell proxy	44 holdout	0.500	0.477	0.477
Source-native readout formulas	44 holdout	0.886	0.477	0.477
Readout-mixture proxy	44 holdout	0.455	0.500	0.455
L1 source-light screen	66 screened	53/66	17/66 all-baseline	–

The strongest internal-preflight signal is the source-native readout-formula preflight: the matched formula beats the wrong-formula mean in 0.886 of the 44 holdout galaxies, with a 1000-permutation shuffled-label null $p \simeq 0.002$ for the mean matched-minus-wrong statistic and $p \simeq 0.006$ for the beat-fraction statistic. This records internal label-formula compatibility in the frozen packet and motivates a specificity hypothesis; it does not establish conditional or externally replicated morphology specificity. It does not support a stronger claim that the present readout formulas beat TPG/v6, MOND-like, or RAR-like baselines as a population law.

Table 7 records robustness checks for the same source-native formula result. These checks are not independent discoveries; they are stress tests of the headline matched-versus-wrong statistic.

Table 7: Robustness checks for the 0.886 holdout matched-versus-wrong fraction. All rows use the frozen source-native formula labels and are generated by `scripts/build_source_native_readout_robustness.py` or `scripts/run_source_native_carrier_robustness.py`.

Test	Result	p -value	Detail
Original holdout	0.886	–	44 holdout galaxies; mean matched-minus-wrong = $-17.684 \text{ km s}^{-1}$.
10k shuffled-label null	0.886	0.0093	Seed 271828; beat-fraction statistic; null beats-wrong median = 0.773.
Bootstrap CI	0.886	–	Seed 271829; 10000 galaxy bootstrap interval [0.795, 0.977].
Error-weighted RMSE	0.886	–	Uses SPARC velocity-error weights; mean weighted matched-minus-wrong = $-17.758 \text{ km s}^{-1}$.
Alternative split median	0.841	–	Seed 271830; 100 family-stratified splits; central 90% range [0.773, 0.909].
Family-balanced score	0.832	–	Mean of holdout per-family beat fractions; compact 0.571, exponential 0.857, scale-tail 1.000, thick/flared 0.900.
Newtonian-carrier amplitude freeze	0.955	0.0010	Train-only amplitudes fit around v_n^2 rather than TPG/v6; mean matched-minus-wrong = $-51.131 \text{ km s}^{-1}$, median $-53.425 \text{ km s}^{-1}$; family-balanced beat fraction = 0.929; beat-fraction null $p = 0.0010$.

The Newtonian-carrier row is important because it asks whether morphology specificity survives when the train-only amplitude residual is defined around the baryonic Newtonian carrier v_n rather than the TPG/v6 carrier. It does: the matched-versus-wrong holdout fraction remains high, the median margin has the same sign as the mean, and the family-balanced score remains high. The freeze-invariance audit in `data/derived/source_native_carrier_robustness_freeze_invariance_audit.csv` records that labels, kernels, source proxies, and train/holdout splits are unchanged. This strengthens the carrier-robustness of the internal matched-versus-wrong statistic, but it still does not establish population-level superiority over TPG/v6, MOND-like,

RAR-like, or Newtonian baselines.

This outcome is not treated as a surprise to be hidden. It is part of the protocol. If a galaxy is morphologically regular, approximately scalar in its effective residual structure, or dominated by a smooth closure-like response, then Newtonian, MOND/RAR-like, or TPG/v6-like curves can be strong. In the language of the present paper, those baselines may be good low-dimensional readout limits. The morphology-matched programme becomes distinctive only where the source morphology selects a different readout subfamily and that choice improves against wrong families and shuffled labels before any endpoint repair.

Holdout source-native readout formula landscape

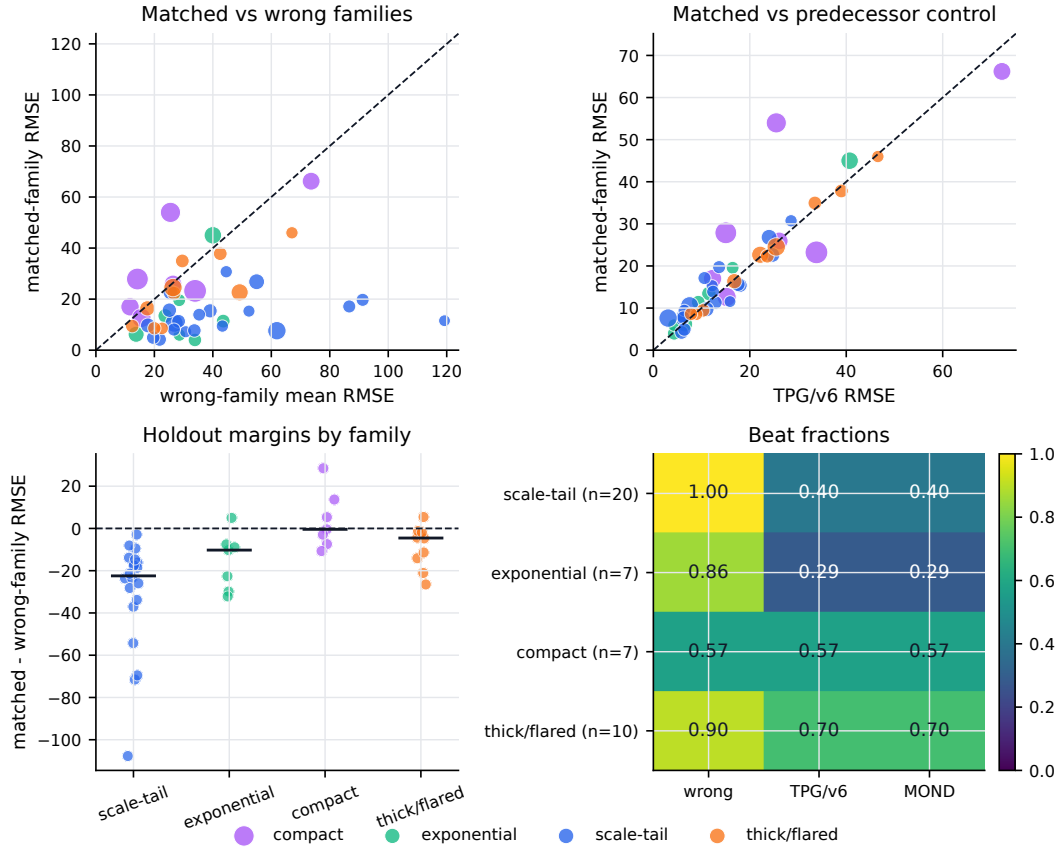


Figure 3: Holdout source-native readout formula landscape. The upper panels compare matched-family RMSE against wrong-family and TPG/v6 controls for each holdout galaxy. Marker color denotes readout family and marker size tracks the number of rotation-curve points. The lower panels show family-level margins and beat fractions. The figure makes the main result visual: the matched families often beat wrong morphology families, while baseline superiority remains mixed.

Shuffled-label null: observed readout labels carry information beyond random labels

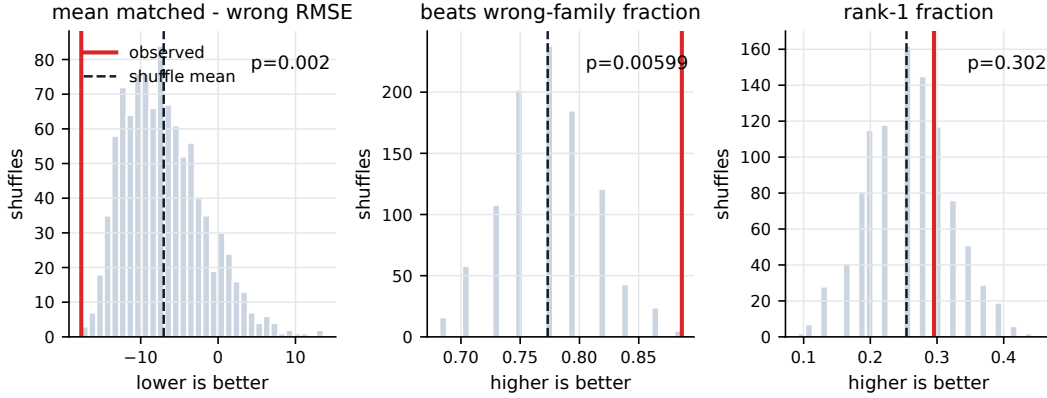


Figure 4: Shuffled-label null distributions for the source-native readout formulas. Grey histograms show 1000 morphology-label shuffles; red lines show the observed holdout statistic. The observed matched-minus-wrong mean and beats-wrong fraction lie in the favorable tail of the shuffled null, while the rank-1 statistic is weaker.

The negative readout-mixture proxy is also informative. It shows that a mixture layer cannot be supplied by coarse present-day proxy heuristics. Mixture weights need accepted source observables: projection, H I support, vertical structure, interaction/history context, and a residual-blind normalization rule.

Family-level coverage and performance: specificity is not the same as baseline dominance

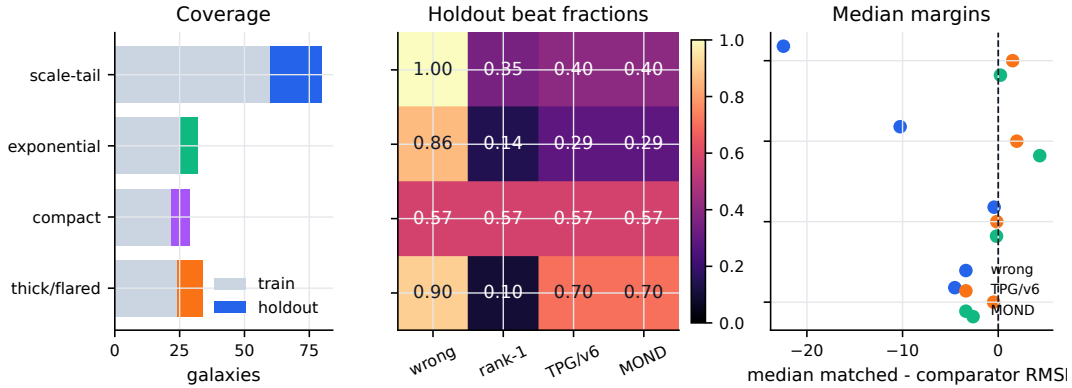


Figure 5: Family-level coverage and performance. Left: train/holdout coverage of the four one-dimensional source-native families. Middle: holdout beat fractions against wrong-family, rank-1, TPG/v6, and MOND-like comparators. Right: median matched-minus-comparator RMSE margins. The panel shows that coverage, morphology specificity, and proxy baseline flags are distinct quantities. Baseline-comparator beat fractions are proxy-level flags, not endpoint validation.

6.2 Conditional-information and external matched-tracer audits

Two later audits test whether the internal SPARC preflight can be promoted. First, a repeated stratified cross-fitted audit on the frozen 45-galaxy A/C packet supplies endpoint-residual-free observability and baryonic covariates, then asks whether the projection contrast

$$X_{\text{proj}} - \frac{1}{2}(X_{\text{MOND}} + X_{\text{RAR}})$$

adds held-out predictive information. At the primary regularization the baseline AUC is 0.87395, while the contrast changes AUC by -0.01891 , log-loss improvement by -0.03994 , and Brier improvement by -0.01189 . Both proper scores worsen for both declared baselines and all three frozen regularizations. This is a negative conditional predictive-increment audit, not a proof that mathematical conditional mutual information is exactly zero.

Second, the public EDGE–CALIFA matched-resolution packet provides an external operational homolog using CO(1–0), H α , stellar, dispersion, line-ratio, and star-formation maps [21, 47]. The 17 galaxies in the published CO–H α rotation analysis were excluded before cohort selection. A first source-only protocol admitted no object and was preserved as a source-coverage failure. A separately versioned, source-developed protocol admitted eight galaxies before either terminal velocity field was opened: NGC2639 for development and seven untouched objects for confirmation. One score-blind development support failure changed only the requirement from all six to at least five of six occupied azimuthal sectors; the source matrices, galaxies, five radial zones, rank-four covariance gate, controls, velocity conversions, and decision thresholds remained fixed.

The velocity convention is material. EDGE CO velocities use the radio convention in the LSRK frame, whereas the CALIFA H α velocities use the optical convention. The frozen endpoint first applies the public LSRK-to-heliocentric correction and converts both tracers separately to the relativistic convention, following Appendix A of Levy et al. [21]; only then is the contrast formed. A per-zone intercept absorbs an arbitrary constant zero point.

For a declared source span S , the endpoint uses

$$Q(S) = y^T P_{\perp}(S)^T \left[P_{\perp}(S) \Sigma P_{\perp}(S)^T \right]^+ P_{\perp}(S) y,$$

and the per-galaxy matched-versus-wrong contrast

$$D_g = \frac{1}{3} (Q_{\text{radial}} + Q_{\text{phase}} + Q_{\text{cross}}) - Q_{\text{correct}}.$$

This construction has an exact identifiability boundary. Because each source matrix has full column rank, every invertible basis change R obeys

$$P_{\perp}(SR) = P_{\perp}(S), \quad Q(SR) = Q(S).$$

The score can therefore test a source subspace, but cannot select an oriented coefficient vector or the sign of a terminal prediction inside it. In particular S and $-S$ are score-equivalent while an otherwise unspecified fixed coordinate vector would give opposite candidate terminal vectors. A physical prediction must instead supply the signed load and ordered forward descent before endpoint access, schematically

$$H_B \delta M = j_s, \quad z_A = J_{A \leftarrow M} \delta M + J_{A \leftarrow OS} \xi_{OS}, \quad c_{OS} = Q_{OS}(z_A), \quad h_{\tau} = R_T(c_{OS}) - R_T(Q_{OS}(0)), \quad h_{\perp} = P$$

Here the stabilized body response is solved and frozen before the post-body observer–source channel. The irreducible observer–source load ξ_{OS} and source-derived stable resolution map Q_{OS} are not

part of a simultaneous body solve. On a stable linearized branch these maps may be compressed into the usual Jacobian notation, but the body-driven product $J_T J_O H_B^{-1} j_s$ is only the $\xi_{OS} = 0$ restriction. A Tau-specific first-order direction also requires $h_\perp \neq 0$ relative to a complete frozen standard comparator. The existing conditional CHDF/ordered-disk completion does not supply this packet: the same pure source shells admit a direct-product control with zero disk-CHDF mixed derivative, so physical occupation and the signed terminal law remain open. All correct and wrong-family constructions have source rank 16 and leave a rank-four projected terminal covariance. The primary population test is the exact one-sided shared-sign permutation of $\bar{D}$ over all 2^7 sign vectors; radial-reversal, phase-rotation, and cyclic cross-galaxy controls use the corresponding studentized max- T distribution.

Table 8: Frozen EDGE-CALIFA seven-galaxy confirmatory endpoint. A positive D_g favors the matched morphology over the mean equal-rank wrong-family control.

Quantity	Result
Mean D_g	-0.058679
Median D_g	+0.110202
Positive galaxies	4/7
Exact one-sided primary p	0.6015625
Radial/phase/cross max- T adjusted p	0.9766/0.9453/0.3359
Frozen decision	prevalidation fail

The mean, positive-fraction, primary-significance, and all three control gates fail; only the positive median condition holds. No post-opening replacement, mask change, alternate sector rule, or new score is permitted in this packet. An independent implementation rebuilds all 7×20 terminal coefficients from the checksum-matched public HDF5 file, with maximum absolute coefficient difference $3.55 \times 10^{-15} \text{ km s}^{-1}$, and reproduces all 32 declared hash, rank, projection, score, and inference checks. It also records that the local hash chain does not independently prove freeze chronology, the original endpoint JSON does not bind the coefficient CSV, and the public raw packet is referenced rather than vendored. The result rejects this development-informed morphology-alignment proxy as an external promotion route. It neither falsifies the broader Tau Core architecture nor identifies standard astrophysics as complete: the declared standard span is not exhaustive and no unique Tau coefficient law was supplied. Conversely, that limitation would also have prevented a positive result from being called Tau-specific. A claim-raising successor must use an independent survey or release, a source-derived signed h_τ , a complete declared standard comparator, and a new freeze before endpoint access.

6.3 External rotation-search ladder beyond the EDGE endpoint

Several already-frozen public-data routes constrain what a successor may claim. They are kept separate because they test different objects and do not form one combined significance. The LIT-TLE THINGS lane uses published dwarf mass models [28]. The PHANGS lanes combine public CO kinematics, MUSE ionized-gas products, and residual-blind stellar-structure maps [18, 9, 30].

Table 9: Current external rotation-search ladder. A numerical feature is not promoted when its source-integrity, specificity, identifiability, or predeclared support gate fails.

Route	Frozen numerical result	Claim-safe decision
LITTLE THINGS	$N = 14, 313$ points; matched-minus-TPG mean RMSE -0.101 km s^{-1} , matched-minus-MOND $+0.649 \text{ km s}^{-1}$	Mixed prospective transfer of one frozen scale-tail family; below the $N \geq 15$ replication gate and not a multi-family attribution test.
PHANGS low-order orthogonal	Three eligible galaxies; $\chi^2 = 23.17/30$, $p = 0.8082$	The declared morphology-orthogonal low-order component is consistent with zero; the earlier wrong-family hint does not replicate.
PHANGS two-body population	Frozen $m = 2$: $\chi^2 = 77.606/20$, $p = 9.97 \times 10^{-9}$; $m = 1$ control also non-null, $p = 8.26 \times 10^{-11}$	Numerical structure is preserved, but the source-label audit contradicts bar absence for NGC4321 and leaves IC5332 unvalidated; no morphology-conditioned promotion.
PHANGS radial body projection	Four confirmatory galaxies	The mandatory 12-sector occupancy gate fails in the outer common-support zones; no individual or aggregate Q is released, and the 19-galaxy product family is exhausted for replacement.
NGC4254 source coordinate	Direct beam-safe $m = 2$ candidate has source range $[0.0466, 0.1017]$, measurement 95% interval $[0.0612, 0.1229]$, and sign probability 0.996; the $m = 1$ control fails	Internally robust source-side candidate only. A free terminal gain exactly saturates a scalar endpoint, and identical source states admit primitive witness gains 0.3333–0.6667; endpoint scoring remains closed.

The PHANGS radial source matrices have an additional source-only property: a unique Euclidean rank-four approximation captures 0.865–0.939 of their normalized Frobenius energy in the five development galaxies. This is a useful compression diagnostic, not evidence for physical four-dimensional descent, because the source-coordinate metric, CHDF roles, body connection, terminal map, and occupation law are not derived by the singular-value decomposition.

Taken together, these routes locate two real but non-promotable candidates: the PHANGS low-order differential field and the NGC4254 source coordinate. Neither supplies a signed, unit-calibrated terminal vector outside a complete standard tangent space. The next valid endpoint must therefore begin from the forward packet above rather than from another freely oriented source span or a post-open repair of these samples.

6.4 Finite morphology representation versus the parent body

The negative ladder tests finite morphology representations, not direct access to a source-complete parent body. Let $\Phi_k : \mathcal{M}_\tau \rightarrow \mathcal{Z}_k$ be the representation frozen for endpoint k , and let T_k denote the complete body–channel–observer–resolution–terminal composition. A descended law $T_k = f_k \circ \Phi_k$ exists only if

$$\Phi_k(M_1) = \Phi_k(M_2) \implies T_k(M_1) = T_k(M_2). \quad (1)$$

If the terminal differs within one Φ_k fibre, the representation is non-identifying even if a richer body exists. In the linear source case,

$$P_\perp(S) = I - S(S^\top W S)^{-1} S^\top W, \quad u \in \ker(S^\top W) \implies P_\perp(S)u = u. \quad (2)$$

Thus an omitted physical component cannot be recovered by coefficients inside the declared source span. The seven EDGE source matrices are 20×16 , so each leaves four linear directions outside that representation. This does not show that Nature occupies an omitted direction; it shows why failure of a finite proxy is not logically identical to falsification of every possible M_τ .

The corresponding anti-rescue result is equally exact. Once terminal vector y has been inspected, augmenting the source matrix to $S' = [S, y]$, with the weighted Moore–Penrose projector used if S' is rank deficient, gives

$$P_\perp(S')y = 0. \quad (3)$$

An endpoint-informed morphology refinement can therefore fit the same packet by construction and has no confirmatory force. The only admissible iteration is to construct a richer Φ_{k+1} , its Q_{OS} , sign, units, and terminal descent from source-only information, freeze them, and evaluate a new untouched survey or release.

6.5 Inspected single-galaxy cases

The reproducibility code also contains four inspected cases. They are heterogeneous and should not be combined into a population validation claim.

Table 10: Four inspected readout cases. RMSE values are in km s^{-1} .

Galaxy	Matched	Best baseline	Best wrong	Status	Interpretation
NGC4013	10.61	10.88	11.37	retroactive/caveated	protocol-ready, replay needed
NGC5907	16.37	16.79	16.85	accepted preliminary	mixed projection readout
NGC7331	22.26	23.47	22.67	caveated accepted	mixed branch positive; pure branch negative
NGC4088	11.62	25.40	37.87	caveated accepted	strongest single case; low-level caveats

Across these four inspected rows, the source- or morphology-matched readout beats the best local baseline and the inspected wrong-family controls in all four cases. This is a retrospective case-level compatibility pattern. It is not population validation, and these cases were not selected as a statistically independent validation sample: NGC4013 remains retrospective, NGC7331 carries a broad outer-window caveat, and NGC4088 still has source-provenance and low-level normalization caveats. The object-level morphology context is residual-blind and source-ledger based: NGC4013 is supported by H I and vertical-structure studies [48, 5]; NGC5907 by warp, edge-on, and interaction/projection context [35, 37, 46]; NGC7331 by H I warp/trajectory and vertical-scale context [2, 29]. These references motivate the readout lanes and caveats; they are not by themselves endpoint evidence.

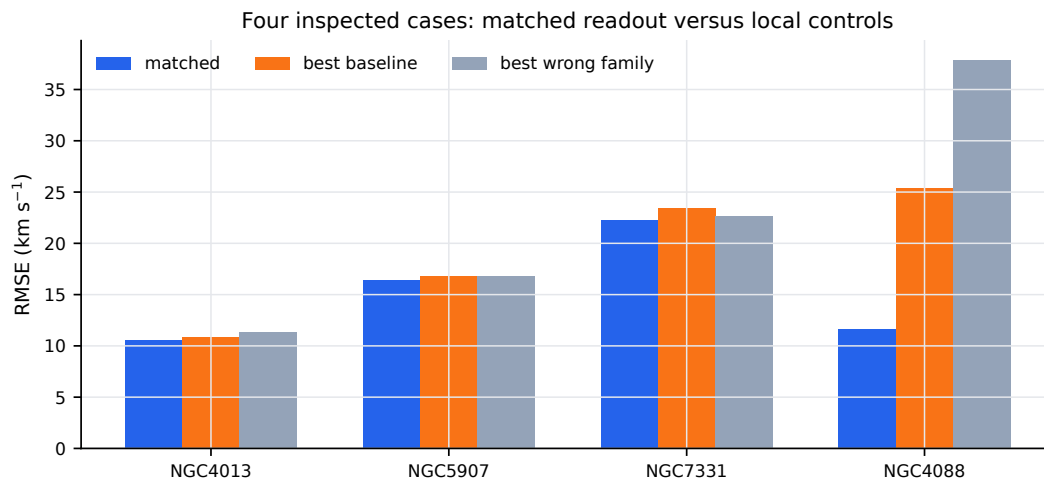


Figure 6: Four inspected cases. The matched readout is better than the best local baseline and best inspected wrong-family control in all four rows. The claim remains case-level and caveated because the evidence packets are heterogeneous.

Inspected rotation-curve cases: observed data and matched readouts

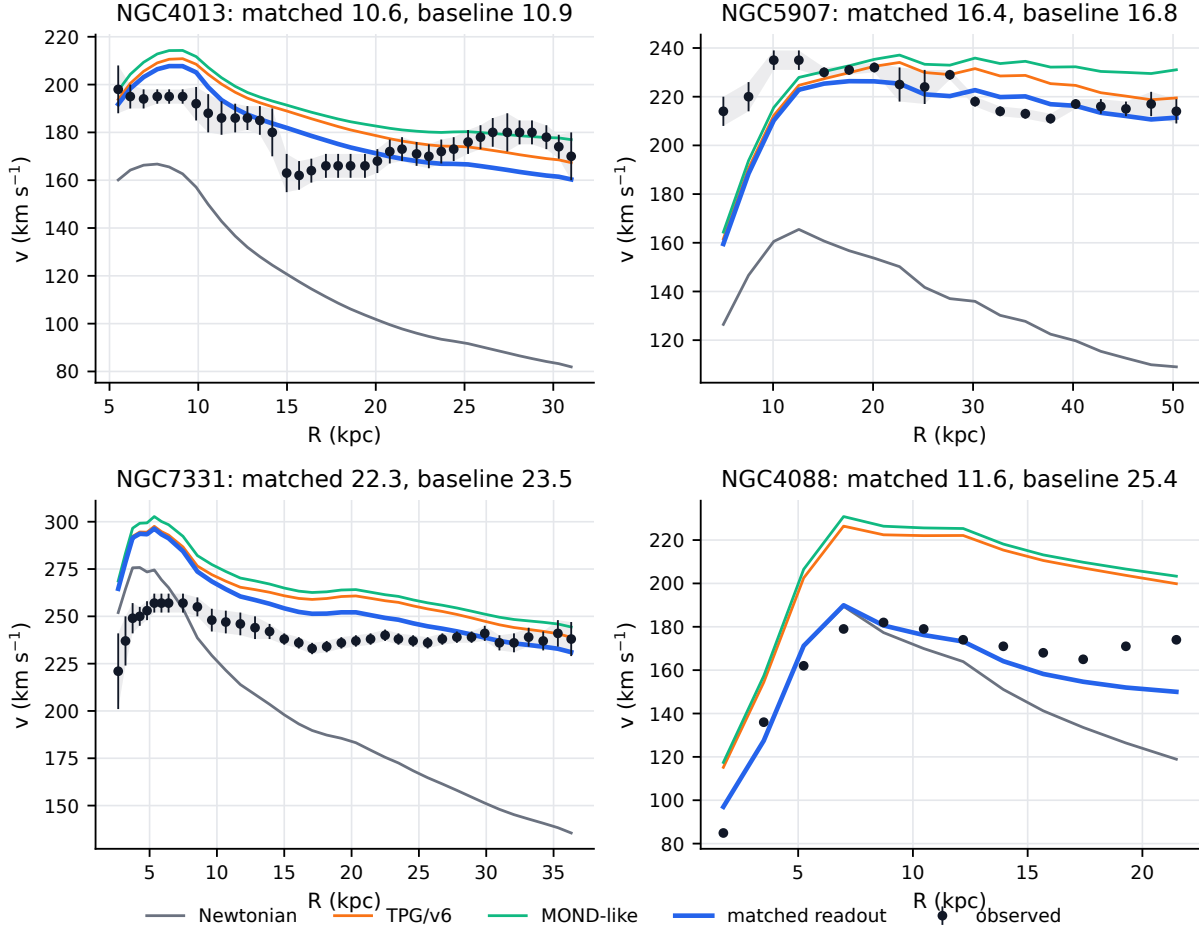


Figure 7: Rotation-curve view for the four inspected endpoint cases. Points are observed SPARC rotation speeds with reported error bands. The matched readout curve is compared with the Newtonian baryonic curve, the fixed TPG/v6 predecessor comparator, and the MOND-like comparator available in the reproducibility package. Panel titles report matched and best-baseline RMSE values.

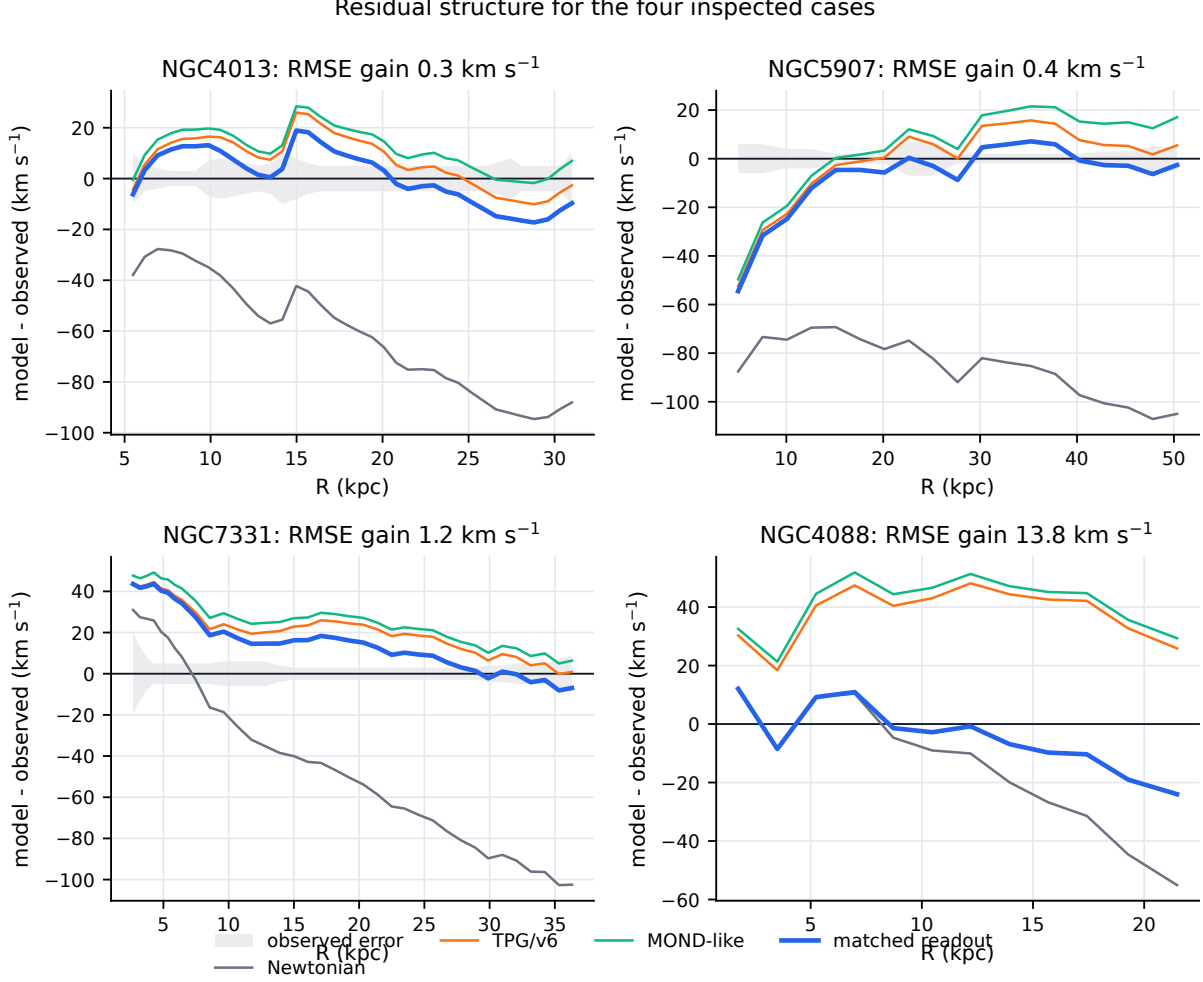


Figure 8: Signed residuals for the same four inspected cases, $\Delta v = v_{\text{model}} - v_{\text{obs}}$, with the observational error band shown around zero where available. Residual panels are important because an endpoint can look visually reasonable while leaving systematic radial structure. Panel titles report the matched improvement over the best local baseline in km s^{-1} .

These rotation and residual panels are the most direct visual summary of the present single-galaxy evidence. NGC4088 shows the clearest endpoint-style gain: the matched warp/trajectory readout reduces a large baseline mismatch over much of the measured radial range. NGC5907 and NGC7331 are more modest: the matched mixed readout improves the score, but the curves remain close enough to strong smooth controls that the result should be read as a source-frozen case-improvement, not as a new law. NGC4013 is scientifically useful for a different reason: it shows how a mixed vertical/warp readout can become competitive, while also illustrating why retrospective cases need replay before promotion.

The TPG/v6 curves in these panels are therefore not a disposable comparison. They are a serious predecessor-control: when they are already close to the observations, a new morphology-matched readout must explain why it adds source-specific information rather than merely reproducing a smooth closure shape. This is why the paper separates three claims: morphology specificity against wrong families, competition against TPG/v6 or MOND-like controls, and full endpoint-grade validation.

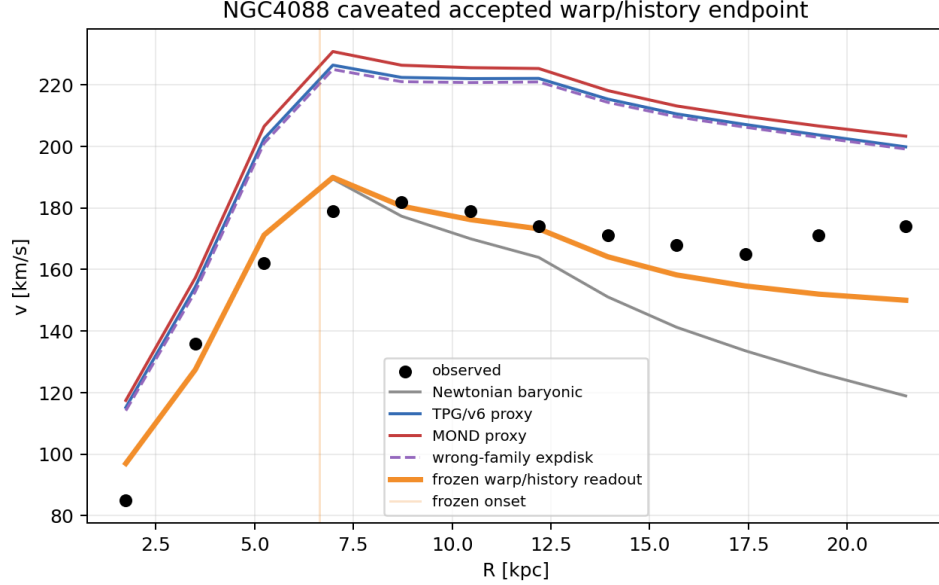


Figure 9: NGC4088 caveated warp/trajectory endpoint. This is the strongest single-galaxy result in the current code, but remains a preliminary control endpoint, not population validation.

An object-specific freeze note, `reports/ngc4088_object_specific_freeze_note.md`, records the residual-blind source fields, current blockers, and claim boundary for this case. The note is included because NGC4088 is visually compelling but should not be allowed to carry a population-level claim by itself.

6.6 Unchanged-law transfer to UGC08490/NGC5204

The strongest direct test of whether the NGC4088 result generalizes is to keep its formula class unchanged and replace only the source-owned galaxy inputs. For UGC08490/NGC5204, a three-dimensional TiRiFiC tilted-ring analysis supplies a source-native two-plane H I warp: the first nonzero tip occurs at 156 arcsec and the inner and outer planes differ by 41.1 ± 5.7 degrees [17]. Converting the angular onset on the SPARC distance scale and dividing by the published H I radius gives the frozen value

$$x_w = 0.4508763.$$

Before opening the pointwise SPARC curve, the transfer retained $q_{\text{warp}} = 1$, $\sigma_{\text{warp}} = +1$, $p = 1$, and $\lambda_w = x_w V_{\text{flat}}^2$, producing $\lambda_w = 2785.50 \text{ km}^2 \text{ s}^{-2}$. The manifest was then hashed and the endpoint was scored by a separate script. Since UGC08490 had been opened in earlier, different tracer analyses, this is a retrospective frozen transfer, not a prospective blind endpoint. Moreover, the published SPARC R_{HI} and V_{flat} are observed summary inputs, so the test is not a zero-information dimensional-amplitude prediction.

Table 11: UGC08490 unchanged NGC4088-class transfer. No target parameter is fitted to the pointwise endpoint, except in the explicitly fit-aided halo comparators. RMSE values are in km s^{-1} .

Model or control	RMSE	Role
NFW halo + baryons	1.20	two-parameter endpoint fit
Pseudo-isothermal halo + baryons	1.55	two-parameter endpoint fit
Fixed MOND	8.78	fixed standard comparator
Fixed TPG/v6	12.75	fixed predecessor comparator
Frozen source-correct warp transfer	29.70	matched transfer
Wrong NGC4088 onset	31.64	morphology control
Wrong quadratic turn-on	31.90	morphology control
Newtonian baryons	44.87	fixed carrier limit

The correct source onset therefore performs better than the wrong onset and wrong turn-on controls, while reducing the Newtonian RMSE by 15.17 km s^{-1} . That ordering is suggestive but not decisive: a circular block bootstrap gives only 0.9436 probability of a lower mean squared error than the wrong-onset control, with a 95% interval that crosses zero. The endpoint-informed best-onset diagnostic is $x_w = 0.4937$, close to the source value, but improves RMSE by only 0.074 km s^{-1} ; it is recorded and not used to repair the frozen law.

The standard-comparator failure is unambiguous. Ten of the thirty points lie inside the warp onset, where the transferred law is exactly Newtonian although the large rotation discrepancy is already present. Even outside the onset, the frozen transfer has 26.10 km s^{-1} RMSE, compared with 7.13 km s^{-1} for fixed MOND. Thus the outer warp is a real and informative morphology component, but it is not the complete carrier of this galaxy’s rotation discrepancy. The result rejects the current one-component NGC4088 warp law as a universal complete rotation law. It does not reject a multi-component source-frozen readout, because the source description itself also contains an inner irregular/ringed disk that the transferred kernel does not represent. No same-endpoint enrichment is permitted.

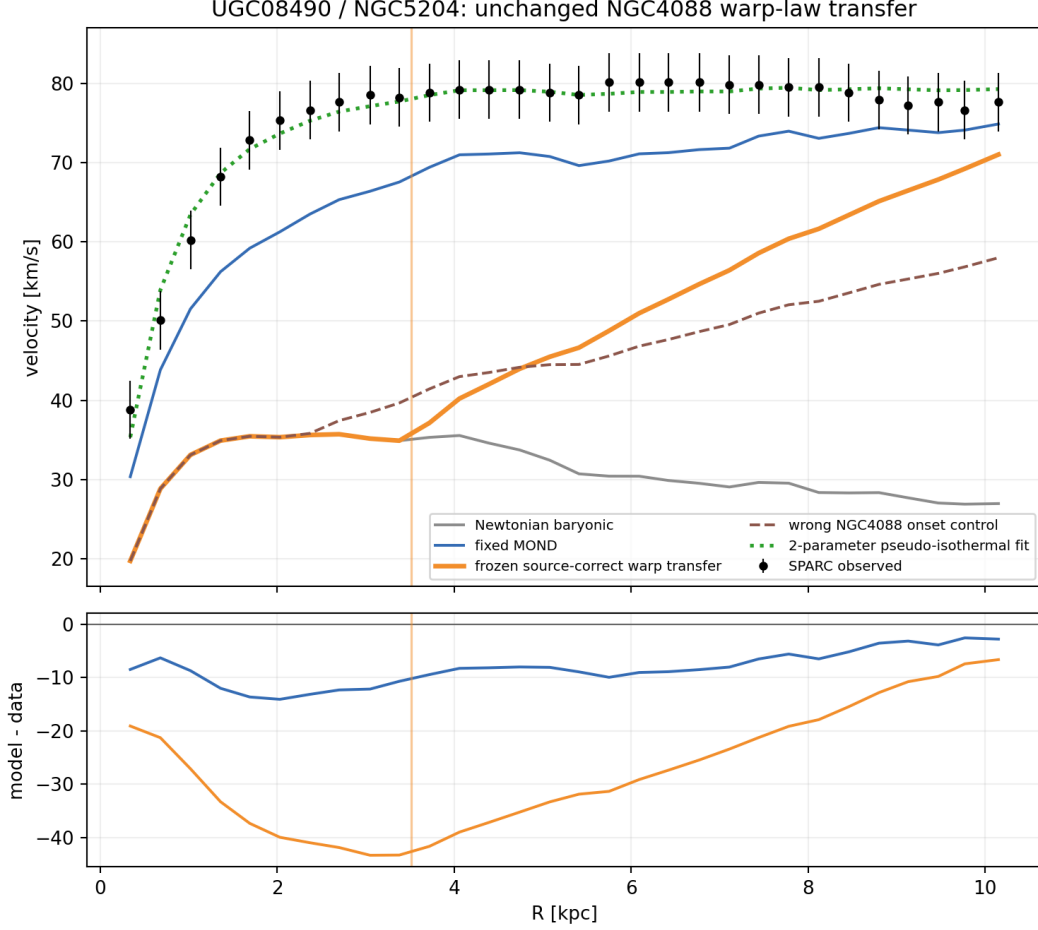


Figure 10: Retrospective unchanged-law transfer to UGC08490/NGC5204. The source-correct warp onset beats the wrong-onset morphology control but remains far below the observed curve and is decisively worse than fixed MOND and fit-aided halo comparators. The lower panel shows signed residuals.

6.7 Second unchanged-law replication: UGC03580/UGC3580

The same Jozsa tilted-ring study supplies an independent member of the source-selected warp class. For UGC03580/UGC3580 the inner reference plane is $i = 67.2$ degrees, $PA = 96.3$ degrees; the first nonzero source tip is 5.5 ± 3.9 degrees at 180 arcsec, and the mean inner/outer mutual inclination is 13.9 ± 2.5 degrees [17]. The source explicitly calls this warp less significant than those of NGC2541 and NGC5204 and treats a direct rotation-rise/warp connection cautiously. Those caveats are retained rather than converted into a stronger morphology label.

On the SPARC distance and H I-radius scale the frozen onset is

$$x_w = 0.9045639, \quad \lambda_w = x_w V_{\text{flat}}^2 = 14406.48 \text{ km}^2 \text{ s}^{-2}.$$

The class-law sign, binary activation and linear turn-on were copied unchanged, and the freeze script did not open the local pointwise curve. UGC03580 had, however, appeared in earlier repository endpoint analyses, so this remains a retrospective replication. A source-native angular-denominator

sensitivity $x_w = 180/240 = 0.75$ and the published outer-speed normalization were also declared before this scoring script.

Table 12: UGC03580 unchanged NGC4088-class replication. Full-curve RMSE values are in km s^{-1} . The two halo rows are explicitly fit-aided.

Model or control	RMSE	Role
Pseudo-isothermal halo + baryons	14.99	two-parameter endpoint fit
NFW halo + baryons	17.39	two-parameter endpoint fit
Wrong UGC08490 onset	24.56	morphology control
Wrong NGC4088 onset	25.29	morphology control
Fixed TPG/v6	25.49	fixed predecessor comparator
Fixed MOND	27.17	fixed standard comparator
Source-H I denominator sensitivity	27.68	predeclared sensitivity
Newtonian baryons	36.01	fixed carrier limit
Frozen source-correct transfer	38.04	matched transfer

The source-correct onset does not preserve the morphology ordering seen in UGC08490. Its full-curve RMSE is worse than Newtonian baryons, fixed MOND, and both earlier-onset morphology controls. Against the UGC08490-onset control, the circular block-bootstrap interval for the primary-minus-control mean squared error is strictly positive (58.3 to $2088.2 \text{ km}^2 \text{ s}^{-2}$); only 0.00015 of draws favour the primary. Inside R_{HI} , the matched transfer modestly improves Newtonian RMSE from 31.66 to 29.68 km s^{-1} , but remains worse than fixed MOND at 28.07 km s^{-1} .

Three SPARC points lie beyond R_{HI} . There the unchanged kernel

$$C_{\text{warp}}(x) = \max\left(0, \frac{x - x_w}{1 - x_w}\right)$$

continues above unity, reaching 4.72 , and strongly overshoots. This is a structural failure of the uncapped class law, not a licence to cap it after seeing the endpoint. A bounded or plateauing two-plane kernel is a new law that must be derived from source geometry and tested on another target. Thus the current replication rejects the uncapped one-component NGC4088 law as a universal warp-class rotation law while leaving the broader morphology/readout programme open.

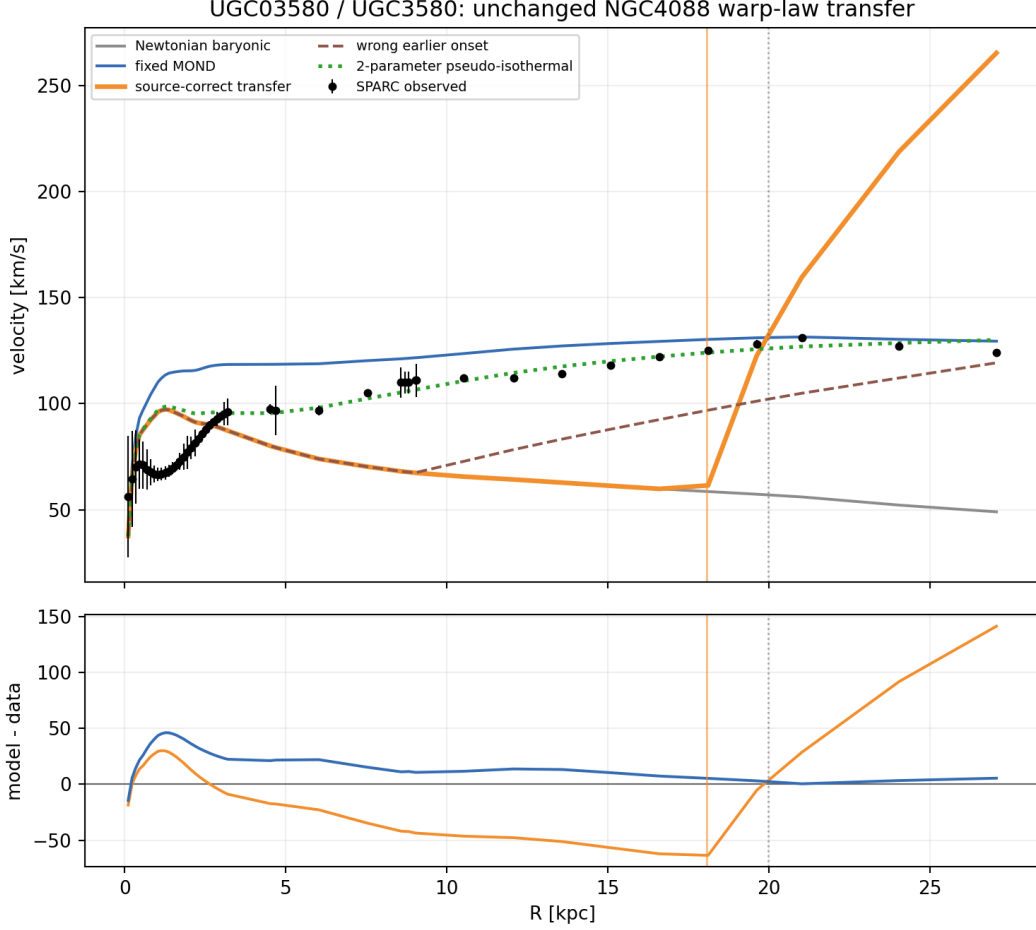


Figure 11: Retrospective UGC03580/UGC3580 replication. The source-correct onset activates only in the outermost region and the uncapped continuation overshoots beyond R_{HI} . Earlier wrong-onset controls fit the full curve better, so the UGC08490 morphology-ordering signal does not replicate.

6.8 Source-only radial two-plane morphology refinement

The failed transfer identifies two distinct objects that the scalar proxy had conflated: the source morphology and the terminal law. The endpoint cannot be used to repair either one. We therefore return only to the source table and transcribe its radius, H I surface density, inclination, position angle, spin normal, tip and line-of-nodes fields. The published rotation-velocity columns and the SPARC endpoint are excluded from this construction. Seventeen rings lie inside the published terminal support at 375 arcsec; the 420-arcsec row is retained only as a declared support sensitivity.

At radius R , the local disk plane is represented by the source orientation normal

$$\mathbf{n}(R) = (\sin i(R) \sin \text{PA}(R), -\sin i(R) \cos \text{PA}(R), \cos i(R)) \in S^2.$$

The inner reference plane is fixed by the published $i_0 = 67.2$ degrees and $\text{PA}_0 = 96.3$ degrees. As a source-only summary, the outer plane $\mathbf{n}_{\text{out}}$ is *defined*, rather than uniquely derived, as the unweighted normalized chordal mean of the five ring normals on the published 240–375-arcsec outer range. Adjacent source rings are connected by shortest great-circle interpolation. If $\theta_j =$

$\arccos(\mathbf{n}_j \cdot \mathbf{n}_{j+1})$ and $\eta \in [0, 1]$, the segment is

$$\mathbf{n}_j(\eta) = \frac{\sin[(1 - \eta)\theta_j]}{\sin \theta_j} \mathbf{n}_j + \frac{\sin(\eta\theta_j)}{\sin \theta_j} \mathbf{n}_{j+1}.$$

This gives a radial orientation field without Euclidean interpolation off the unit sphere. A bounded source morphology invariant is then

$$\delta_{\text{tilt}}(R) = 1 - \mathbf{n}_0 \cdot \mathbf{n}(R), \quad 0 \leq \delta_{\text{tilt}} \leq 2,$$

and the local connection magnitude between sampled rings is

$$\kappa_j = \frac{\theta_j}{R_{j+1} - R_j}.$$

Neither quantity is identified here with a velocity correction.

The reconstructed inner/outer plane separation is 14.297 degrees, versus the independent Table 4 value 13.9 ± 2.5 degrees, a difference of only 0.159 quoted standard deviations. The maximum supported discrepancy between the orientation-derived and tabulated normals is 0.085 degrees; the maximum reconstructed-tip discrepancy is 0.093 degrees. The inner defect is exactly zero through the published flat inner region, and sampled great-circle interpolation preserves unit norm to 1.2×10^{-16} . Source-only alternatives move the outer mean plane by 1.487 degrees for H I-density weighting, 1.806 degrees when the beyond-support ring is included, 2.217 degrees under every-other-ring thinning, and at most 1.365 degrees under leave-one-outer-ring-out deletion. All four shifts remain below the published 2.5-degree Table 4 uncertainty.

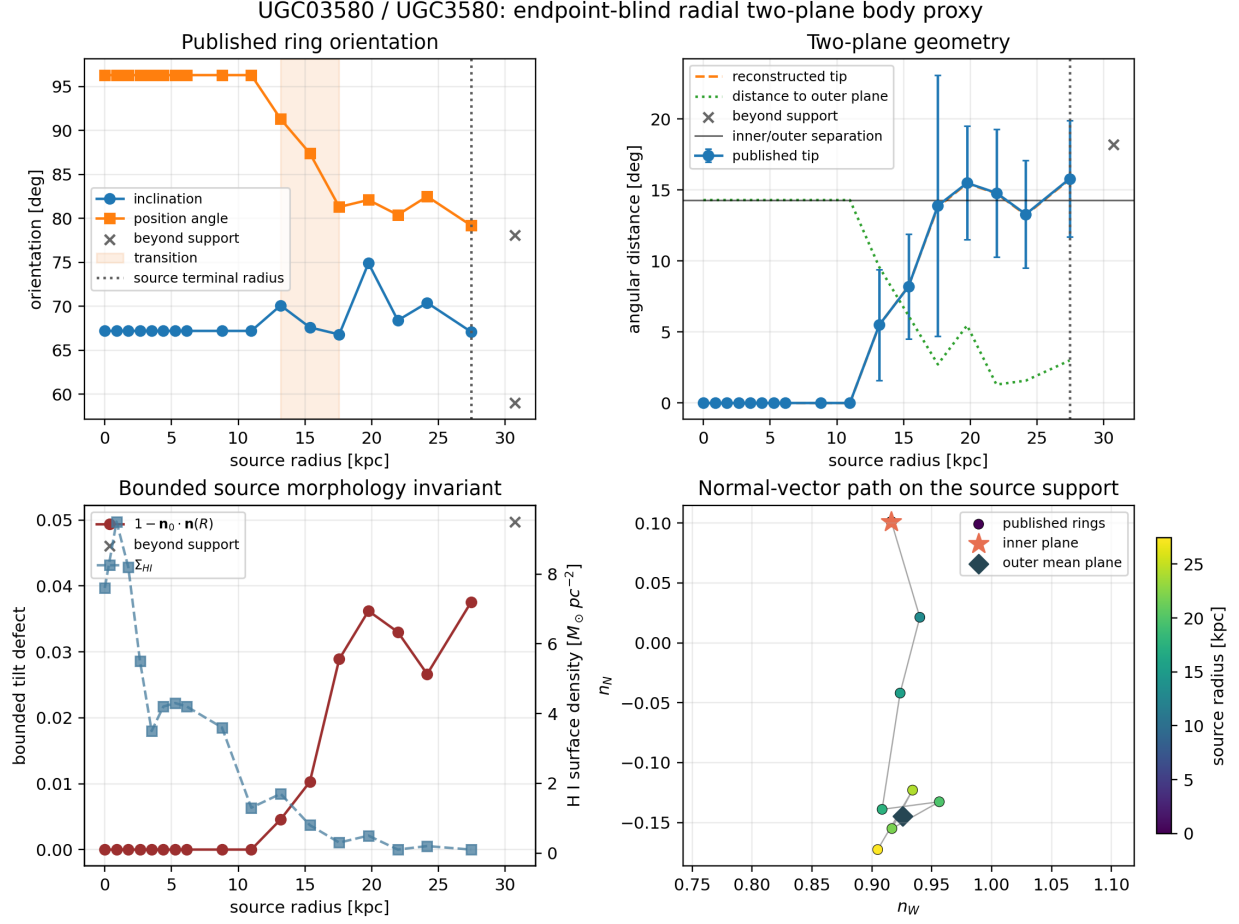


Figure 12: Endpoint-blind UGC03580/UGC3580 radial morphology refinement. The upper panels show the published ring orientation and its separation from the source-defined inner and outer planes. The lower panels show the bounded tilt defect, H I support and the normal-vector path. Crosses lie beyond the source terminal support and do not enter the primary outer-plane estimate.

This construction closes a source-representation defect: UGC03580 is no longer reduced to one onset number. It does not repair the failed endpoint, select a bounded readout law, or show that warp geometry explains the rotation curve. In Tau Core terminology it is a localized galaxy-scale descriptor of a downstream subsystem, not a separately seeded universe-level morphological body M_7 . A bounded two-plane terminal shell remains a new source-side hypothesis and must be frozen before a genuinely new endpoint.

6.9 Bounded two-plane coordinate and terminal non-identifiability

The radial body is sufficient to define a bounded morphology coordinate, but not a unique physical readout. Let $\mathbf{n}_{\text{in}}$ and $\mathbf{n}_{\text{out}}$ denote the source-defined reference planes and set

$$d_{\text{in}}(R) = 1 - \mathbf{n}_{\text{in}} \cdot \mathbf{n}(R), \quad d_{\text{out}}(R) = 1 - \mathbf{n}_{\text{out}} \cdot \mathbf{n}(R).$$

For any $p > 0$, define

$$K_p(R) = \frac{d_{\text{in}}(R)^p}{d_{\text{in}}(R)^p + d_{\text{out}}(R)^p}.$$

Both defects are nonnegative. For distinct reference planes their powers cannot vanish simultaneously, hence

$$0 \leq K_p \leq 1, \quad K_p(\mathbf{n}_{\text{in}}) = 0, \quad K_p(\mathbf{n}_{\text{out}}) = 1.$$

Exchanging the plane labels gives exactly $1 - K_p$. If the two reference planes coincide, we define $K_p = 0$, because no active two-plane morphology then exists. These statements are algebraic consequences of the frozen body. The choice $p = 1$ is only the minimal affine-defect convention; the source does not select it physically.

The remaining ambiguity can be proved constructively. For every $p > 0$, $0 \leq a \leq 1$, $s \in \{-1, +1\}$, and nonnegative carrier $v_c^2(R)$, the family

$$v_{p,a,s}^2(R) = v_c^2(R) [1 + s a K_p(R)]$$

is dimensionally valid and nonnegative and recovers the carrier when $K_p = 0$ or $a = 0$. Different p , a , or s generally produce different terminal curves while sharing the same source body. Thus the two-plane morphology alone cannot determine the terminal exponent, amplitude, sign, carrier, or physical occupation. This is an explicit non-identifiability counterfamily, not a failed numerical fit.

For UGC03580, the plane separation gives the source-only diagnostic scale

$$a_{\text{sep}} = \frac{1 - \mathbf{n}_{\text{in}} \cdot \mathbf{n}_{\text{out}}}{2} = \sin^2\left(\frac{\Theta}{2}\right) = 0.015484869.$$

On the supported rings, K_1 ranges from 0 to 0.992276495, while the largest transition-zone difference between K_1 and K_2 is 0.149874019. The envelope $v_c^2[1 \pm a_{\text{sep}}K_1]$ is therefore retained only as a geometry-scale control; no source theorem identifies a_{sep} with a physical coupling.

There is also an exact standard-geometry control. If a circular ring with inclination $i(R)$ is reduced using the fixed inner inclination i_0 , its inferred squared speed is multiplied by

$$G_{\text{proj}}(R) = \left[\frac{\sin i(R)}{\sin i_0} \right]^2.$$

The supported UGC03580 range is 0.994090714–1.096849451. This factor is not Tau-specific and is applicable only if the terminal reduction actually used the fixed inner inclination. It must therefore be audited as a standard projection null before any morphology-readout claim.

The endpoint-blind construction and independent audit pass all declared checks without reading a rotation endpoint. The resulting status is `FORMULA_SHELL_DERIVED_ENDPOINT_BLOCKED`: a bounded morphology coordinate is derived, but terminal scoring remains inadmissible until a parent/source law independently freezes the body registration, complete coupling, gravity/observer calibration, and projection applicability.

This blocker can be stated more exactly using the source-action response theorem. After the body has been solved and frozen, let $Z_*(K)$ be the regular post-body state, $H_Z > 0$ its reduced Hessian, $B_{ZK} = D_Z D_K \mathcal{A}_{\text{post}}$ the source/body-response mixed block, and L_G the typed gravity-terminal covector. The implicit-function theorem then gives

$$D_K Z_* = -H_Z^{-1} B_{ZK}, \quad \Gamma = -L_G H_Z^{-1} B_{ZK}, \quad \delta v_G^2(R) = [\Gamma K](R).$$

Coercivity of H_Z establishes a regular stable response but does not fix the sign or normalization of B_{ZK} , L_G , or their contraction. In particular, replacing B_{ZK} by $-B_{ZK}$ or by λB_{ZK} , $\lambda > 0$, preserves the body and the positive Hessian while reversing or rescaling the terminal response. The executable audit contains this same-body countermodel.

Moreover, the scalar shell above is a special factorized case, not the generic consequence of the action response. A radius-independent a exists only if

$$\frac{[\Gamma K_p](R)}{v_c^2(R)K_p(R)} = s a$$

is constant on the nonzero support of K_p . Otherwise the correct local law contains a radial response operator or gain function. The exact exponent p likewise requires the complete source-owned defect law to be proportional to d^p on the support; the first nonzero analytic source jet selects only a local leading order. Finally, v_c^2 must be derived as the calibrated zero-morphology physical terminal branch, not chosen as an empirical residual baseline.

The existing common-action construction sharpens this list. Let $\hat{D} = (d_{\text{in}}, d_{\text{out}}, \chi)$ be the complete signed descriptor (the retention result is proved below), and let a source-owned, full-support registration $I_{\hat{D}} = \iota_{I \leftarrow \hat{D}}(\hat{D})$ place it in the Parent invariant used by one frozen JCSEL/BRAC plus Einstein–Newton completion. The chain rule then fixes

$$B_{Z\hat{D}} = B_{ZI}J_{I \leftarrow \hat{D}}, \quad \Gamma_{\hat{D}} = -L_G H_Z^{-1} B_{ZI}J_{I \leftarrow \hat{D}}, \quad v_c^2 = O_G[Z_*(I_{\hat{D}} = 0)].$$

Thus $B_{Z\hat{D}}$, $L_G = D_Z O_G$, and the carrier are not three independent gains inside that declared completion. If $Z = (g, \varphi)$, exact elimination of the occupied internal field gives

$$H_G^{\text{eff}} = H_{gg} - H_{g\varphi}H_{\varphi\varphi}^{-1}H_{\varphi g}, \quad B_{G\hat{D}}^{\text{eff}} = B_{g\hat{D}} - H_{g\varphi}H_{\varphi\varphi}^{-1}B_{\varphi\hat{D}},$$

and reproduces the complete response. Dropping the internal-field Schur term would therefore change the terminal rather than simplify the same law.

The remaining obstruction is upstream. Two distinct Parent registrations can have the same visible K projection but differ along a terminal-visible hidden direction. The endpoint-free finite witness gives terminal gains -0.163083062345 and -0.136881148198 for two such lifts, with exactly zero visible-projection discrepancy; exact Schur reduction reproduces both. Consequently the visible two-plane coordinate cannot reconstruct its own Parent lift. The expanded audit passes all 39/39 checks. The current source packet still does not derive the signed-descriptor registration, exact full-support exponent/profile, any further full-Parent kernel safety, calibrated occupied galaxy packet, or Nature-level selection of the common completion. Endpoint scoring remains blocked.

The hidden fibre can be characterized exactly for the source geometry rather than left as an arbitrary Parent direction. Let

$$c = \frac{\mathbf{n}_{\text{in}} \times \mathbf{n}_{\text{out}}}{\|\mathbf{n}_{\text{in}} \times \mathbf{n}_{\text{out}}\|}, \quad \chi(R) = c \cdot \mathbf{n}(R).$$

The raw pair $D = (d_{\text{in}}, d_{\text{out}})$ fixes the projection of $\mathbf{n}$ into the span of the two reference normals but generically leaves the reflected pair

$$\mathbf{n}_{\pm} = \mathbf{n}_{\parallel} \pm \sqrt{1 - \|\mathbf{n}_{\parallel}\|^2} c.$$

They have equal D and opposite χ . Conversely $(d_{\text{in}}, d_{\text{out}}, \chi)$ reconstructs the oriented normal uniquely, and

$$\chi^2 = 1 - p^T G_{\text{anchor}}^{-1} p, \quad p = (1 - d_{\text{in}}, 1 - d_{\text{out}})^T.$$

This is a discrete fibre: away from $\chi = 0$, $D : S^2 \rightarrow \mathbb{R}^2$ already has local rank two. Therefore a local Jacobian-null test cannot prove terminal descent; every claimed terminal must be exactly

equal on the mirror pair, or χ must be retained in the shared descriptor. On the supported UGC03580 rings the source-only control spans $-0.047813169 \leq \chi \leq 0.095201374$, includes both signs, and satisfies the displayed identity with maximum independent residual below 2.4×10^{-15} . The source resolves the quotient decision for the complete standard projection stack. Table 6 of Józsa (2007) publishes the Cartesian components of the oriented spin normal toward west, north and the observer, and the UGC3580 figure captions identify the approaching side as southeast [17]. Let ℓ be the line-of-sight unit vector and $R_c \mathbf{n} = \mathbf{n} - 2\chi c$. For the standard fixed-inner-inclination projection

$$G_\ell(\mathbf{n}) = \frac{1 - (\ell \cdot \mathbf{n})^2}{1 - (\ell \cdot \mathbf{n}_{\text{in}})^2},$$

direct expansion gives

$$G_\ell(R_c \mathbf{n}) - G_\ell(\mathbf{n}) = \frac{4\chi(\ell \cdot c)[(\ell \cdot \mathbf{n}) - \chi(\ell \cdot c)]}{1 - (\ell \cdot \mathbf{n}_{\text{in}})^2}.$$

This terminal is not mirror-even on the 17 supported rings. The maximum absolute difference is 0.142726674696, the RMS difference is 0.046534679366, and the analytic identity residual is below 2×10^{-16} . At $R = 270$ arcsec the source and reflected projections are 1.096849451016 and 0.954122776319, respectively. Consequently χ must be retained in the complete descriptor for this lane. This is a standard projection and representation result; it is not evidence for fundamental Parent handedness, a Tau-specific coupling to χ , or a new rotation effect.

The mixed-Hessian migration makes this standard control executable without selecting a Tau correction. For the Bregman graph registration of the complete signed descriptor, $\Gamma = J_{OS} = D_{\hat{D}} \mathcal{N}$, while the ordinary projection terminal has covector $P_R = (0, 0, -2n_{\text{los}}/\sin^2 i_{\text{global}})$. Hence

$$D_{\hat{D}} G_{\text{proj}} = P_R J_{OS}.$$

A source-only freeze using the published oriented ring normals and the SPARC global distance and inclination fields verifies this chain rule to 1.21×10^{-10} without reading pointwise rotation velocities. A separate diagnostic scorer then applies $v_{\text{pred}} = \sqrt{G_{\text{proj}}} v_{\text{carrier}}$ to the already opened 47-point UGC03580 terminal. The central-catalog RMSE changes are -0.617 km s^{-1} for Newtonian baryons, $+3.102 \text{ km s}^{-1}$ for TPG/v6, and $+3.531 \text{ km s}^{-1}$ for fixed MOND. Their signs remain unchanged over the declared distance/inclination uncertainty grid and linear/nearest interpolation control. Thus this ordinary projection sensitivity does not explain the residual remaining after TPG/v6 or MOND in this galaxy. This is a post-open standard-systematics diagnostic, not a Tau endpoint. If the published terminal already contains the same tilted-ring correction, its reapplication would be double counting.

The same construction has also been frozen prospectively at the source side for NGC2541 (UGC4284), without reading a rotation-velocity endpoint. The published oriented-ring geometry gives 25 supported rings over 30–590 arcsec. For $\hat{D} = (d_{\text{in}}, d_{\text{out}}, \chi)$ the frozen decoder is

$$\mathcal{N}(\hat{D}) = A^{-1}(1 - d_{\text{in}}, 1 - d_{\text{out}}, \chi)^{\text{T}}, \quad J_{OS} = A^{-1} \text{diag}(-1, -1, +1).$$

Its basis condition number is 6.57269, the maximum reconstruction error is 8.40×10^{-16} , and the standard fixed-inclination chain-rule check closes to 1.24×10^{-10} . The resulting reference projection ratio ranges from 0.81865 to 1.03107, but remains a sensitivity diagnostic, not a selected Tau correction. No endpoint values or fit parameters enter this freeze. Its status is therefore source-geometry ready but terminal-calibration and untouched-endpoint blocked; no score is reported.

We subsequently executed the frozen calibration protocol on independently reduced HALOGAS DR1 high-resolution moment maps [13, 14]. The protocol and runner were hashed before endpoint access; the first execution stopped before producing any statistic because the FITS WCS and fixed centre used different celestial frames, and the lock records the sole coordinate-frame compatibility fix. On 23 common eligible rings, the variable source geometry changes the mean approaching–receding side discrepancy from 13.2069 to 12.5729 km s^{−1}, or −0.6341 km s^{−1}, with no fitted parameter. A required nuisance audit demotes this favorable direction: replacing the optical/NED centre by the same source’s fitted kinematic centre changes the delta to approximately +0.3799 km s^{−1}; a four-arcsec eastward shift gives +0.0888 km s^{−1}, so even sub-beam centre changes can reverse its sign. Moreover, the fixed and variable reductions selected different sky pixels, the 12-arcsec inner ring spacing is smaller than the 14–17.5-arcsec beam, and several accepted sides contain less than one synthesized beam. The earlier ring-iid bootstrap interval and sign-flip $p = 0.2144$ are therefore naive descriptive diagnostics, not calibrated population evidence. We classify this as a post-open moment-1 side-symmetry diagnostic with unstable calibration, not a standard-calibration detection. It neither selects a Tau-specific q_R nor tests a dark-matter replacement.

This audit fixes the design of the next population test. NGC 2541 is a development case and is excluded from confirmation. NGC 3198 is likewise excluded because its diagnostic and replay scores were already opened, while NGC 5055 is excluded because it belonged to the Paper 8 SPARC training population. The galaxy is the inference unit, every candidate geometry is evaluated on one common sky footprint, and beam-block covariance, centre alternatives, source orientation uncertainties, and numerical convergence are compulsory. Two galaxy-level gains are frozen: improvement of the morphology geometry over fixed geometry, and improvement over equal-complexity source-side wrong-family geometries. Exact shared-sign permutations, a max- T multiplicity control, and leave-one-galaxy-out sign stability are required. This source-side protocol freeze is not itself a population result or a physical q_R selection.

An endpoint-blind audit of the checksum-verified 24-object HALOGAS DR1 manifest leaves eight moderate-inclination, previously unopened candidates after these exclusions and the current edge-on cut are applied [13, 14]. Thus HALOGAS alone is four galaxies short of the frozen minimum of 12. NGC 925 was audited first. The primary THINGS analysis publishes its centre, mean geometry, and adopted radial inclination and position-angle curves [8]. Both curves are extracted from the immutable 561-by-756 source JPEG into 93 three-arcsecond rings over 15–291 arcsec. This corrects an earlier partial freeze that treated the printed 66° mean inclination as a constant radial law. The digitized inclination and PA means are 64.861° and 286.608°, within 1.139° and 0.008° of the printed summaries. The largest 90-percent raster-sensitivity widths are 7.771° and 4.195° respectively; these are digitization diagnostics, not physical covariance.

An independent THINGS fit supplies 15 orientation pairs with marginal errors from 100-step residual-rescrambling Monte Carlo fits [36]. That source also states that the geometry inside 250'' is non-identifiable because nearly solid-body rotation does not separate inclination from rotation speed. At the only source-claimed robust ring shared with the de Blok extraction (270''), the inclination gap is 0.878° (1.23 Schmidt marginal standard deviations), while the PA gap is 3.773° (2.91 such standard deviations). These ratios are diagnostics, not cross-source significances, because de Blok physical errors, cross-ring covariance and joint inclination–PA covariance are unavailable.

The six source-robust outer rings have at least 11.89 ideal projected half-annulus beam areas even for the broadest published HALOGAS pilot beam, so an outer-only route is not geometrically impossible before masking. It fails the current clean-terminal source gate: the same outer disk is described as a minor-merger gas stream with twisted isovelocity contours, warp/spiral ambiguity and anomalous H I [13]. NGC 925 is therefore retained only for a separately frozen disturbed-system robustness or stronger-terminal lane, and NGC 4062 becomes the next clean source-acquisition

target. No HALOGAS endpoint pixel or score was opened in reaching this decision. This remains a sampling and source-readiness result, not a physical q_R selection or Tau signal.

NGC 4062 was subsequently frozen and opened without changing that source-first order. GHASP gives $i = 68 \pm 2^\circ$, a receding $PA = 99 \pm 2^\circ$, and $v_{\text{sys}} = 758 \pm 1 \text{ km s}^{-1}$ [10]. The independent HALOGAS 3DBarolo reduction gives median $i = 67.1^\circ$ and $PA = 100.1^\circ$ and does not include NGC 4062 among the six discs it flags as substantially warped [24]. Two non-overlapping LR-beam-sized common rings were therefore frozen at $42''$ and $84''$, with HR as primary, LR as mandatory replication, and centre, $i \pm 5^\circ$, $PA \pm 2^\circ$, and both published systemic velocities as compulsory nuisance variants. The opened cross-tracer side-odd contrast is consistent with zero: the HR GLS mean is $4.27 \pm 10.23 \text{ km s}^{-1}$ ($p = 0.470$), and the LR mean is $3.22 \pm 10.50 \text{ km s}^{-1}$ ($p = 0.556$). Their primary signs agree, but changing from the HALOGAS catalogue velocity of 769 km s^{-1} to the GHASP value reverses the inferred sign in both resolutions. Hence the confirmatory endpoint fails rather than supporting a nonzero readout effect. After importing this result and the published interaction, warp, complexity, and beam-smearing flags for the other seven moderate-inclination unopened objects, the current HALOGAS pool contains no further clean candidate under this protocol. Expansion must therefore use an independently source-frozen survey family or a newly preregistered, stronger terminal; post-open repair of NGC 4062 is forbidden.

The resulting calibration obstruction can nevertheless be stated exactly for future work. Let $d \in \mathbb{R}^n$ be a radial contrast, let C be its positive-definite covariance, and let the columns of G be source-frozen nuisance tangents, including the velocity-zero-point direction. Then the covariance-weighted residual maker

$$P_\perp = I - G(G^T C^{-1} G)^+ G^T C^{-1}$$

obeys $P_\perp G = 0$, $P_\perp^2 = P_\perp$, and $P_\perp(d + G\eta) = P_\perp d$. Thus it removes the declared additive nuisance without fitting the endpoint signal. The construction also exposes an identifiability condition: a target template s survives only when $\text{rank}[G, s] = \text{rank } G + 1$. If both G and s are the common two-ring vector, the entire putative signal is removed. A post-open audit of NGC 4062 leaves only one of two radial degrees of freedom and retains 9.11×10^{-4} (HR) and 2.52×10^{-5} (LR) of the covariance-weighted information in a constant radial template. Its projected diagnostic remains consistent with zero, but this is not a new confirmatory score: the secant nuisance tangent was itself estimated from the two opened systemic-velocity reductions. A new survey must therefore freeze a source-owned or independently calibrated G , covariance, target template, rank tolerance, and minimum retained-information rule before opening the contrast.

The already opened, independently sourced NGC 3726 GHASP–Ursa Major development case provides a six-radius method control. Under its published fixed-systemic-velocity side convention, $dO_t/dv_{\text{sys},t} = -2$, so the two tracer zero points span only one constant radial nuisance direction. The same covariance-weighted projector therefore leaves five shape degrees of freedom. All operator identities pass, and the projected statistic equals the previously reported residual about the best constant, $\chi^2 = 9.1784$ for five degrees of freedom ($p = 0.1022$); the shape-only zero null is not rejected. This closes the constant-zero-point algebra in that fixed-side model but not the physical endpoint: a constant radial signal is annihilated, beam/centre/PA and radial covariance remain incomplete, and no nonconstant morphology template has been source-frozen on an unopened sample.

A further residual-blind sensitivity control found a previously missed machine-readable Ursa Major side table for the interacting system NGC 3893. Before reading its velocities, four common radii ($20'', 40'', 60'', 80''$), the literature disturbance onset above $75''$, the nonconstant template $(0, 0, 0, 1)$, and the same zero-point projection were frozen. The opened matched amplitude is $7.78 \pm 11.40 \text{ km s}^{-1}$ ($p = 0.495$); the full projected shape statistic is $\chi^2 = 0.964$ for three degrees

of freedom ($p = 0.810$). Thus this terminal does not detect even the predeclared conventional disturbed-system control. Because the source literature analysed related kinematics, this was never an independent Tau endpoint; its negative result instead weakens the demonstrated morphology sensitivity of the current side-odd instrument. No post-open template, radius or covariance repair is made.

We also froze an independent two-dimensional H I sensitivity control on an exact three-disturbed/three-quiet VIVA cohort [4]. The operator uses the standard harmonic projection fact that an in-plane morphology mode m contributes to line-of-sight velocity sidebands $k = m - 1$ and $k = m + 1$, consistent with nearby-galaxy Fourier velocity-field practice [33]. For the frozen $m = 1$ construction the lower sideband overlaps the nuisance sector, so the test retains the two quadratures of $k = 2$, with four preregistered annuli, held-out azimuthal sectors and three wrong-template controls. After cube opening, every galaxy failed the frozen requirement of at least 24 beam-independent samples and eight occupied sectors in every annulus (total retained sample counts 29, 4, 81, 28, 27, 45). The pipeline therefore emitted no velocity score and the opened cohort is not repaired post hoc. This is a resolution/capacity failure of this implementation, not evidence against Tau, a parent-loss measurement, a rotation correction or a dark-matter alternative.

We therefore preregistered a higher-resolution successor on the public LITTLE THINGS H I products [16, 15]. The source intake used the full 40-object table intersection rather than disturbed/quiet labels. Twenty-five robust-weighted moment-0 maps survived the catalogue geometry/support screen, 22 passed the image-level source-support gate, and velocity-blind rank/injection calibration retained 20. Two objects were excluded because one held-out azimuth block supplied no target novelty. Before velocity access we also removed an invalid negative control: for a two-quadrature target a global phase rotation spans exactly the same subspace. The valid replacement applies a nonconstant annular phase scramble.

The frozen 20-galaxy moment-1 endpoint is negative. Matched-minus-best-wrong specificity has mean -0.107551 , median -0.056207 , positive fraction 0.25, and exact one-sided sign-flip $p = 0.983505$. No individual wrong-template family yields a significant matched advantage, and all leave-one-galaxy-out primary means remain negative. Thus the concrete same-tracer $m = 1 \rightarrow k = 2$ terminal does not demonstrate morphology-specific kinematic sensitivity on this sample. The result weakens this terminal, not the full parent/channel architecture: conventional lopsided gas dynamics, tidal response, feedback, projection, and shared moment-map construction remain sufficient alternatives. It is not a Tau detection, parent-loss measurement, rotation correction, gravity result, or dark-matter replacement.

This endpoint is an auxiliary source-terminal sensitivity control, not the primary test of observer-centred parent-distance loss. The latter requires a separately derived observer–emission-point corridor $\mathfrak{C}_{O \leftarrow x}$, its parent transfer $T_{P|Ox}$, and a frozen observer access, noise, quantization, and calibration chain. Neither projected galactocentric radius nor the H I $m = 1$ mode is a substitute for that object. The present negative score therefore neither tests nor weakens the open corridor-loss hypothesis. The four-image, two-transition SDP.81 program below is structurally closer because it holds one source behind several lens paths. The conditional Hessian-to-depth compiler is now derived, but no physical occupied parent Hessian or path lift is present. Its current lens/source diagnostics are therefore not yet a Tau parent-loss score.

An endpoint-free standard comparator has now been constructed from the frozen Inoue smooth-lens model. For all four q1 paths it records relative Fermat potential, the local lens Jacobian and its singular values, signed magnification, and parity without reading a spectral or velocity endpoint. All seven algebraic/provenance checks pass; the relative Fermat span is $0.743341 \text{ arcsec}^2$, the absolute magnification range is 3.730–17.887, and the centered path descriptor has rank three. These are standard lens/corridor nuisance and comparison variables, not Tau parent distance. Endpoint

authorization remains false until a source-owned physical $T_{P|Ox}$, parent-depth calibration, full covariance, and stable observer terminal are independently frozen. The largest q1 lens-Jacobi singular value is 1.038708, one path therefore exceeds unit operator norm, and two paths reverse parity. Together with the different angular-coordinate domain, this forbids identifying the standard lens Jacobian itself with the positive-metric parent-loss contraction; it remains part of the conventional comparator.

The parent-loss generator is now frozen before any SDP.81 endpoint access. Under the explicitly conditional post-body row-refinement principle PCRR-P1, continuity, row locality and the inherited local half-unit uniquely select

$$\Gamma_e = \frac{P_e K_{\text{src}} P_e^*}{2A_* \ell_*}.$$

The accompanying theory audit passes all fourteen checks: square and saturating spectral filters fail same-mode row-splitting invariance even though they pass ordinary direct-sum naturality, while a quadratic source Hessian alone still permits alternative positive stage metrics. Forward decision PCRR-D1 therefore treats the linear law as the primary conditional family and predeclares those alternatives as wrong-family controls. Endpoint failure may demote this completion but may not select a replacement filter. Physical parent depth, registration/orientation, source overlap, terminal visibility and stable-cell crossing remain required before scoring is authorized.

In the common commuting-mode branch this frozen choice gives a concrete nuisance-reduced multipath invariant. If the corrected positive amplitudes are $z_{in} = c_i s_n \exp(-\eta_i \lambda_n/2)$, double-centering $\log z$ over paths and modes cancels arbitrary common source-mode amplitudes, path-scalar gains, and the additive depth origin. The remainder must be

$$-\frac{1}{2}(\eta_i - \bar{\eta})(\lambda_n - \bar{\lambda}),$$

and therefore has rank one. The theory audit passes all twelve checks. Rank one alone is not a Tau signature: nonlinear spectral filters share it, while path-and-mode-dependent differential magnification, beam, radiative-transfer, or calibration terms survive. A future SDP.81 score must test alignment with the full frozen outer product after those conventional terms, not merely low rank.

The absolute scalar parent-distance unit is therefore not an eligibility requirement for the first dimensionless score. If a source-side positive parent interface Hessian supplies an invertible contraction C_γ , the integrated morphology-depth operator

$$\Omega_\gamma = -\log[(C_\gamma C_\gamma^*)^{1/2}]$$

and the polar orientation are unique and sufficient. A separate factorization $\Omega = \eta F$ is not identifiable. Within PCRR-D1, a scalar η is admissible only if Ω lies exactly on the registered K_{src}/A_* ray; the twelve-check theory audit recovers its coefficient and rejects an off-ray control. Consequently the remaining SDP.81 blocker is the physical source-owned parent interface Hessian/transfer and its occupation, not an SI calibration of parent length.

The ordered segment product is itself conditional. It follows from one source-owned simultaneous innovation action with primitive residuals $x_e - V_e x_{e-1}$; its triangular stationary solve yields $V_N \cdots V_1$ without introducing parent time. Generic positivity does not imply this law. An additional positive internal susceptibility changes the endpoint transfer and must be retained by exact Schur elimination. The supporting audit passes all twelve checks. Thus the SDP.81 source freeze must supply either this innovation factorization or the complete global corridor Hessian; multiplying local contractions is otherwise unauthorized.

The missing transfer formula is nevertheless conditionally closed by the existing FULLCONE body law. After all coupled corridor-interior morphology is eliminated from one positive causal-body Hessian, split its positive boundary Schur operator into source and observer blocks and define

$$C_\gamma = -S_{OO}^{-1/2} S_{OS} S_{SS}^{-1/2}.$$

A second Schur complement proves strict contraction, and the corresponding theory audit passes all twelve checks. Paper 8 therefore no longer lacks a mathematical $H_{\text{corridor}} \mapsto C_\gamma$ map. It lacks physical source selection and occupation of that causal-body Hessian, together with a source-owned co-registration between the four optical SDP.81 paths and four complete parent-corridor restrictions. Those objects must be frozen before spectral pixels are read.

CSCR-T32h/PATHLIFT-C1 makes this co-registration condition exact. An occupied EBOA parent path already fixes its restriction and observer trace, but one observed 4D ray fixes a unique FULLCONE transfer only if that transfer is constant, up to canonical endpoint-frame isometries, over the entire fibre of parent lifts above the ray. Positive same-ray countermodels yield different transfers and integrated morphology depths. The source-only SDP.81 audit passes all eight method checks and supplies four unique optical path keys without reading spectral pixels, but materializes none of seven physical lift requirements. Neither a common occupied parent-Hessian key with a fibre-basiness certificate nor four occupied parent-lift keys under one such packet is present. The result is therefore `PREFLIGHT_NOT_ENDPOINT`; standard Fermat potentials, parities and lens Jacobians cannot select the missing parent lifts.

CSCR-T32i turns the fibre-basiness branch into an executable local certificate. On every connected regular parent-lift fibre, the endpoint-frame-covariant vertical derivative of C_γ must vanish. It is computed source-forward from the occupied Hessian jet by differentiating the interior Schur complement and solving the Sylvester equations for the inverse positive square roots. Disconnected fibres additionally require matching component anchors. The twelve-check theory audit verifies these identities, but the SDP.81 packet contains no occupied smooth parent-Hessian family, $D\pi_{4D}$, vertical kernel or fibre-topology certificate. The required source schema is sharper, while endpoint status remains unchanged.

CSCR-T32j gives a finite reduction only if the occupied parent packet supplies a connected Lie-group action transitive on every regular lift fibre. Its fundamental generators must span the complete vertical kernel at every orbit point, and the endpoint-frame-covariant transfer defect must vanish on every generator. A one-point check, a partial generator list, or an identity-component calculation is insufficient; discrete components and discrete isotropy require separate anchor or holonomy checks. The SDP.81 packet contains neither this group action nor occupied generator jets, so the refinement leaves `PREFLIGHT_NOT_ENDPOINT` unchanged.

CSCR-T32k–T32l close the next source-side fork. A physically occupied, fibre-transitive action that preserves the complete typed parent Hessian up to one common positive scale is sufficient: Schur functoriality makes C_γ basic and the scale cancels. Hessian positivity does not construct that action. A reconstruction must own a separately typed lift-space metric, solve the vertical symmetry algebra of the complete packet, prove complete flows, and obtain an everywhere-zero generator-span projector defect. The standard two-dimensional lens-potential Hessian is not the parent Hessian, $d\beta/d\theta$ is not $D\pi_{4D}$, four-image multiplicity is not parent fibre topology, and an optical path key is not an occupied parent-lift key. None can be promoted by relabelling, so endpoint status remains closed.

The minimum-scope physical acquisition is consequently the direct-lift route: one common physical parent-Hessian/selector-law packet, independently reviewed source ownership, typed $E_P \rightarrow E_{4D}$, and one nonzero positive actual post-body observer–source incidence measure. The four

parent-lift keys must then be derived by pushing that measure through the common selector, with $\pi_{4D}(\tilde{\gamma}_i) = \gamma_i$ certificates, frozen before any spectral endpoint read. A manifest hash proves record integrity rather than physical occupation.

CSCR-T32m–T32n prove that the keyed route is mathematically nonempty without promoting it to physical occupation. A fibre-proper, strongly convex post-body selector gives a unique smooth conditional section, and the selected positive $S/O/I$ Hessian gives the strict FULLCONE transfer. A simultaneous mixed body/lift scalar action is inadmissible because it backreacts on the frozen body. Gauge-invariant uniqueness is only orbit uniqueness unless a separate source-owned slice is supplied. The 23-check theory audit constructs four conditional keys but leaves Nature occupation false.

CSCR-T32o–T32q remove one redundant occupation layer. Given a physically realized selector law and actual positive observer–source incidence, the occupied-lift measure and four keys follow by pushforward rather than as four independent facts. Actual 4D paths alone do not select the parent section. The least exact local occurrence law is the section graph square, with zero upstream Schur remainder, so it cannot refit the frozen body. The 20-check theory audit verifies the conditional identities; SDP.81 supplies neither physical selector-law ownership nor parent-lift occupation.

The observed 4D incidence statement is now stronger. An endpoint-blind audit of fixed 0.12 arcsec apertures in the ALMA Band-7 continuum image detects all four registered q1 locations above both integrated and peak 5σ thresholds. The integrated SNR range is 21.061–29.729 and the peak range is 10.434–13.553. This supplies a positive, unweighted 4D counting measure of mass four, not a parent-lift measure. It does not select the physical parent section, so the physical count remains zero of seven and the endpoint remains closed.

CSCR-T32s–T32u make the remaining selector request finite. One physically owned complete post-body action would derive the selector, its Schur Hessian and section jet by exact elimination. In the linear positive-action class the unique coefficient-free candidate is

$$J_{\min} = H_P^{-1} L^* (L H_P^{-1} L^*)^{-1}, \quad K_{\min} = H_P|_{\ker L}, \quad L = D\pi.$$

The two theory audits pass 15/15 and 14/14 checks. The current source domain, however, has no parent-path fibre map or physically owned affine parent anchor. A vertical affine shift preserves the entire observed 4D packet while selecting a different parent lift. PLMIN-P1 is therefore selected only conditionally inside its declared class; the physical gate remains zero of seven.

CSCR-T32v–T32w require the lift selector and the $S/O/I$ transfer to descend from one source-owned post-body action rather than from separately entered Hessians. Nested Schur elimination derives the effective lift metric and the boundary transfer without double counting. It also distinguishes a source-selected lift held fixed under terminal variation from a lift that relaxes jointly with that variation. The two boundary Hessians differ by a positive-semidefinite Schur term for nonzero effective coupling, and the normalized transfers generally differ, so this semantics must be source-frozen before endpoint access. The 16-check theory audit proves the identities, but no current SDP.81 artifact owns the common action, correctly typed affine path descent, anchor or Nature occupation. The physical gate therefore remains zero of seven.

The corresponding scoring path is now executable and fail-closed. A source-only compiler reduces each authorized pathwise (B, u, I) Hessian to the selected fixed-lift or relaxed-lift transfer, freezes a five-dimensional centered-log spectral terminal and pathwise covariance, and declares a leave-one-path-out common-source statistic. Three paths reconstruct one latent source-mode vector and predict the fourth. Lossless identity transfer and all wrong path-to-transfer assignments are frozen controls. A separate endpoint runner is the only component permitted to open the CO(8–7) cube. With no physical manifest, both stages return `FORMULA_SHELL_DERIVED_ENDPOINT_BLOCKED`;

the endpoint-read flag is false and no score is emitted. Seven targeted tests verify authorized synthetic compilation, source recovery, wrong-assignment discrimination, common-gain invariance, nested-Schur positivity, foldwise source identifiability and perturbation stability.

Any eventual score on this SDP.81 cube remains retrospective because its spectral values were inspected before the common-action formula freeze. It may serve as an informative diagnostic but not as a novel prediction. A confirmatory test requires a new untouched same-source multipath endpoint.

We nevertheless execute one explicitly nonphysical sensitivity control. The conventional relative Fermat coordinate is retained in its 4D comparator type, normalized to $r_i \in [0, 1]$, and mapped through the unit-normalized path-graph Laplacian K of the six ordered spectral channels, $C_i = \exp[-(1 + r_i)K]$. This formula has no fitted coefficient and is frozen before the spectral values are reread, but it is not identified with parent distance or a Nature-occupied Tau action. Its held-out identity-metric quadratic sensitivity is 4.46913, slightly worse than the lossless value 4.35135; it does not beat the wrong-path-assignment median, and 17 of the 23 nonidentity assignments are no worse. We therefore retain `FIXED_4D_PROXY_NOT_PREFERRED` as a negative retrospective diagnostic. Since the identity covariance is not a measured noise covariance, these values are not calibrated chi-square statistics. The result rejects only this fixed 4D proxy family, not the broader parent-loss hypothesis.

A second diagnostic propagates channel covariance measured from fixed blank aperture grids into the centered-log modes. Across nine annulus/grid geometries the matched proxy beats the lossless model in six and the wrong-assignment median in all nine, but it beats every wrong assignment in none; between 39.1% and 47.8% of wrong assignments remain no worse. Moreover, the frozen unit strength is the descriptive grid optimum in only two geometries, whereas strength 0.5 is optimal in seven. Post-endpoint retuning is forbidden. Thus covariance weighting changes the result from a simple negative norm comparison to weak path-order compatibility, but not to model selection or physical evidence.

We next freeze a source-constrained diagnostic rather than fitting that post-endpoint scale. The independently pre-existing conditional CSCR-T32c/PCRR-D1 refinement rule fixes the linear generator factor to one half, giving

$$C_i^{(\text{PCRR}|4D)} = \exp[-(1 + r_i)K_{\text{ch}}/2].$$

Across the same nine covariance geometries this candidate beats lossless in nine and the wrong-assignment median in nine, but it beats every wrong assignment in none; 34.8%–39.1% of wrong assignments remain no worse. The exact rank p-values among all $4! = 24$ assignments are 0.375–0.417, and no geometry reaches 0.05. These are conditional ranks under the exchangeable path-label null, not population-level significance values. We therefore record PCRR half-shape compatibility without identification. This is a retrospective intersection between a source-pre-existing conditional factor and an already exposed endpoint, not a prediction: r_i is still a conventional Fermat proxy, not a derived parent distance.

A companion diagnostic inserts the WR-T19 five-mode stiffness spectrum

$$H_C = \text{diag}(1.25, 1.5, 2, 2.5, 3),$$

normalized by its frozen maximum. The increasing-stiffness alignment leaves only 4.3%–30.4% of wrong assignments no worse and gives exact rank p-values 0.083–0.333, but never wins against every wrong assignment or reaches 0.05. The eigenmode-to-Helmert-mode alignment is not derived and its reversal is retained as a control. It is therefore a terminal-calibration hypothesis, not a physical source identification. Both diagnostics leave the physical gate at zero of seven.

A stricter follow-up removes the arbitrary ordering of the Helmert columns. The six ordered channels define the path graph P_6 . Its centered normalized Laplacian has five simple eigenvalues, hence its eigenvectors provide a canonical mode basis up to signs. Every self-adjoint terminal operator that commutes with this Laplacian is diagonal in that basis. This does not determine which WR-T19 stiffness belongs to which graph frequency: every permutation of the five stiffnesses still commutes. The predeclared monotone assignment gives exact path-assignment rank $p = 0.375$ – 0.417 , while its reversed control gives 0.083 – 0.333 and an isotropic control gives 0.333 – 0.375 . No candidate beats every wrong assignment or reaches 0.05 . We therefore derive the canonical basis but do not support the monotone assignment. The better reverse control is not promoted: this is a retrospective comparison using the same 4D Fermat proxy, not a physical parent-to-terminal calibration. Exhausting all $5! = 120$ stiffness-to-graph-mode permutations places the monotone assignment only at rank 39–81 of 120 ($p = 0.325$ – 0.675) across the nine covariance geometries; the reverse ranks 55–97. Three different permutations are best in at least one geometry. The endpoint therefore selects neither monotonicity nor a stable alternative assignment.

The remaining calibration can be written without guessing another basis. A source-owned post-body joint action gives $W = -H_Z^{-1}B_{ZC}$. For each lensed path i , let Φ_i be the independently sourced positive-flux composition of CO line formation, lens propagation, instrument response, and aperture extraction. At a positive baseline F_i^0 , the logarithms use fixed positive same-unit flux references, which drop out on differentiation. The local centered-log calibration is

$$C_{ST,i} = -U_0^\top \text{diag}[(F_i^0)^{-1}] D_Z \Phi_i H_Z^{-1} B_{ZC}.$$

Whitening this operator by the centered-log covariance and source metric and taking its SVD defines the calibrated terminal and parent modes. The P_6 basis follows only when the resulting whitened terminal Gram commutes with the graph Laplacian. Positive graph-aligned and rotated same-reduct completions show that the present parent metrics, graph, and WR-T19 spectrum do not select B_{ZC} or the CO radiative port. The formula is therefore derived locally, but the four physical path-specific Jacobians are absent and the endpoint remains sealed.

The ordinary downstream part of those Jacobians can be factored more sharply. Let $S_{nk}(Z)$ denote source-plane line emission in spatial basis cell n and channel k , and let L_{in} be the fixed smooth-lens plus aperture response of path i . Then

$$F_{ik}(Z) = \sum_n L_{in} S_{nk}(Z), \quad [D_Z \Phi_i]_{ka} = \sum_n L_{in} [D_Z S]_{nka}.$$

An endpoint-blind quadrature on the four registered q1 paths and the frozen 49-cell source basis gives a rank-four path matrix, leaving 45 spatial null directions per channel. If $S_{nk}(Z) = p_n f_k(Z)$, each path differs only by a scalar gain, which centered log channels cancel exactly; the maximum numerical Jacobian spread is 3.69×10^{-16} . A nonseparable spatial-spectral countermodel under the identical smooth lens gives spread 0.2869 without parent loss. Hence standard lens geometry is explicit, but a source-owned CO(10–9) radiative mixed jet $D_Z S$, its line-specific instrumental response, and covariance remain open. Spatial-spectral differential magnification is a mandatory standard-physics control for any later parent-loss score.

The unresolved radiative jet admits a second exact local factorization, $D_Z S = (D_\xi S)P$ with $P = D_Z \xi$. Standard molecular-line calculations take gas temperature, density, column density, linewidth, and background radiation as physical inputs [42, 23]; they determine $D_\xi S$ only after the gas state and transfer closure are fixed, and do not determine the cross-type map P . For a uniform slab,

$$S_k = \eta \Delta J \left(1 - e^{-\tau_0 \phi_k(v,w)} \right).$$

Centered logs annihilate both multiplicative amplitude columns, leaving rank at most three, or at most two in the optically thin limit. Thus one homogeneous component cannot calibrate five modes. A two-component thin Gaussian mixture, after common-gain quotienting, has relative amplitude, two centroid, and two width coordinates. The endpoint-blind witness has rank five and minimum singular value 0.231306; coincident components collapse to rank two. This proves only that a minimal radiative completion is nonempty. The source-owned full-rank parent/body-to-gas bridge P , CO(10–9) gas baseline, line-specific instrument response, and covariance remain open. The retrospective CO(8–7) two-component preference cannot select them.

For the common-volume direction there is nevertheless one exact standard transfer derivative. At fixed source function,

$$I_j = S_j(1 - e^{-\tau_j}), \quad \tau_j(\zeta) = e^\zeta \tau_j \quad \implies \quad \left. \frac{\partial \log I_j}{\partial \zeta} \right|_0 = \frac{\tau_j}{e^{\tau_j} - 1}.$$

This positive response is strictly decreasing with positive optical depth. Equal channel opacities, and the leading optically thin uniform gain, vanish under centered logs; unequal opacities can survive as a line-shape mode [23]. A varying source function contributes a separate $\partial_\zeta \log S_j$ term.

The two already exposed transitions do not identify this derivative. For any positive opacity pair and measured positive intensities, setting $S_j = I_j/(1 - e^{-\tau_j})$ reproduces the same observations while changing the volume response. Two positive counterfamilies reproduce the published intrinsic $r_{8-7/5-4} = 0.30 \pm 0.04$ but yield different centered responses. Moreover, the reconstructed CO(5–4) and CO(8–7) distributions are known to have excitation-dependent morphology [32]. The expanded endpoint-blind audit passes all thirteen checks and proves non-identifiability, not a source freeze. Independent channel-wise excitation/source-function or opacity information, with filling and differential magnification, remains necessary before CO(10–9) can be scored.

We have now executed the source-acquisition part of that route using the official 200-pc SDP.81 multi-line products [31], while excluding CO(10–9) from both inputs and fit. A solar-metallicity **wk2006** PDR fit to FIR, [C II], CO(3–2), CO(5–4), and CO(8–7) has 288 pixels detected at S/N at least three in every tracer. Eighty pixels have reduced $\chi^2 < 5$. The blind median CO(10–9)/CO(5–4) prediction is 0.231, with median relative 68% half-width 0.363; its median change between 0.05-dex and 0.10-dex grids is 6.0%, narrowly failing the frozen 5% stability check. This is therefore a partial integrated-amplitude prior, not a channel-wise opacity certificate.

The distinction is exact. Write the source-cell emission as $S_{nk} = A_n p_{nk}$ with $p_n > 0$ and $\sum_k p_{nk} = 1$. A nonzero sufficiently small perturbation $p_m \mapsto p_m + \epsilon h$, with $\mathbf{1}^\top h = 0$, preserves all integrated amplitudes. For every path with $L_{im} A_m > 0$, however,

$$\delta F_i = \epsilon L_{im} A_m h, \quad \delta y_i = \epsilon L_{im} A_m U_0^\top \text{diag}(F_i)^{-1} h,$$

which is generically nonzero. At derivative level, $D_Z S_{nk} = (D_Z A_n) p_{nk} + A_n D_Z p_{nk}$ leaves the independent zero-sum profile derivative unresolved. Thus the integrated PDR result cannot be split into six channels by convention. A source-owned channel profile, its response, filling and differential magnification remain required; an isotopologue or other direct opacity tracer would be cleaner.

The available open products do not supply that missing lift. Their official source reconstructions are two-dimensional velocity-integrated maps rather than source-plane spectral cubes. The current image-plane inverse reconstruction promotes none of six channel-common source fits, while its three-path-to-fourth-path test improves only one of 24 folds and has median predictive $R^2 = -21.187$. The cross-transition projected rank also remains unpromoted. We therefore reject simple inheritance of the CO(8–7) normalized profile as a CO(10–9) profile. This rejects one candidate completion, not a richer standard line-formation law or Tau Core.

The cross-type bridge can be reduced one layer further while preserving the body-first direction. Let Z be an already solved and frozen post-body response and let the gas state be the stationary section of a conditional functional, $D_\xi \mathcal{E}_{\text{gas}}(\xi_*(Z); Z) = 0$. If $H_\xi = D_\xi^2 \mathcal{E}_{\text{gas}} > 0$ and $B_{\xi Z} = D_\xi D_Z \mathcal{E}_{\text{gas}}$, the implicit-function theorem gives

$$P = D_Z \xi_* = -H_\xi^{-1} B_{\xi Z}, \quad \text{rank } P = \text{rank } B_{\xi Z}.$$

This is an ordered conditional response, not a simultaneous body/gas variation. In a jointly varied symmetric quadratic action, eliminating ξ changes the upstream Hessian to $H_Z - B_{\xi Z}^T H_\xi^{-1} B_{\xi Z}$ and therefore generically reselects the frozen response. The endpoint-blind audit verifies the implicit formula, rank and conditioning bounds, finite differences, a singular control, and nonzero backreaction. A quadratic same-Hessian counterfamily realizes arbitrary distinct bridges, so stability proves neither the physical $B_{\xi Z}$ nor Nature occupation. The remaining source object is the typed, occupied conditional gas law at the frozen line baseline; CO(10-9) stays sealed. The terminal-registration validator now reproduces both $P = -H_\xi^{-1} B_{\xi Z}$ and every pathwise $D_Z F_i = (D_\xi F_i)P$; a directly asserted raw flux Jacobian fails closed.

The available symmetries do not select the remaining mixed derivative. A genuinely common source-owned group action would imply $P\rho_Z(g) = \rho_\xi(g)P$. The ordered P_6 reflection splits the centered five-space into parity multiplicities $2 + 3$, leaving a $2^2 + 3^2 = 13$ dimensional intertwiner space with distinct full-rank members. A hypothetical standard S_6 action has scalar commutant, but it is not a symmetry of the ordered graph, is not physically represented on the gas tangent, and still leaves the susceptibility scale free. Component exchange is not a local stabilizer at a generic distinct-component baseline; its fixed locus has centered-log radiative rank two. The endpoint-blind audit passes all fourteen checks. Thus a common source representation/intertwiner and one nonzero susceptibility anchor, or an independently derived nonsymmetric interaction, remains necessary.

The proposed four-dimensional intermediate can be stated exactly, but only as a conditional mediation theorem. If a gauge-quotiented environmental descriptor $e(Z)$ carries all dependence of the gas functional on the frozen post-body coordinate, then

$$P = D_e \xi_* D_Z e = -H_\xi^{-1} B_{\xi e} D_Z e.$$

For fixed $G = D_Z e$, an R with $P = RG$ exists if and only if $\ker G \subseteq \ker P$, equivalently $P = PG^+G$ after inner products are fixed. A scalar descriptor therefore transmits at most one mode and a four-component vector at most four; the five-mode terminal requires mediator and response ranks of at least five. Conversely, full-column rank of G makes bare factorization automatic and non-identifying. More generally a direct mixed derivative contributes $P_{\text{dir}} = -H_\xi^{-1} B_{\xi Z}^{\text{dir}}$, so pure mediation additionally requires a no-direct-coupling certificate. The endpoint-blind audit passes all thirteen checks. No present artifact owns the physical descriptor, $D_Z e$, $B_{\xi e}$, or that certificate. The terminal gate now accepts only a source-owned direct cross derivative or a reproduced gauge-quotiented mediator packet; downstream lens/path geometry cannot be substituted for it.

The existing G4D branch supplies one concrete candidate mediator. Introduce a source-owned registration $I_{ZA} : T_Z \rightarrow X_A$ from the five post-body modes to the observer-source input of the ordered G4D row. Then

$$D_Z c_* = -L_C^{-1} D_C^* W_R D_A I_{ZA}, \quad D_Z h = J_{\text{met}} D_Z c_*.$$

At the GR leaf, J_{met} is an isomorphism from the ten metric-active lapse, shift and symmetric-strain generators to the ten relative-metric components. On the unrestricted sixteen-component coframe tangent its kernel is exactly the six-dimensional infinitesimal Lorentz algebra. Hence an abstract

metric-only gas response with an access incidence of rank at least five compiles without frame contamination as

$$P_{\text{cof}} = -H_{\xi}^{-1} B_{\xi h} D_Z h.$$

The updated endpoint-blind audit passes all fifty-six checks and constructs an abstract rank-five witness. It also constructs same-G4D countermodels: I_{ZA} is not selected by the G4D row, and $B_{\xi h}$ is not selected by the coframe response. The standard covariant matter mixed Hessian fixes the relevant variational type, but not its reduction to the five phenomenological CO line coordinates. The physical G4D occupation, registration and gas-metric response remain open.

The registration need not remain a free matrix. With the post-body Z already frozen, one source-owned common relation codomain and typed rows $R_Z : T_Z \rightarrow Y_{ZA}$, $R_A : X_A \rightarrow Y_{ZA}$ and $d_A : X_A \rightarrow Y_A$ give

$$I_{ZA} = -K_A^{-1} R_A^* W_{ZA} R_Z, \quad K_A = R_A^* W_{ZA} R_A + d_A^* W_A d_A > 0.$$

The common action scale cancels and rank-five transmission is equivalent to a rank-five cross-Gram row. The ordering is essential: a simultaneous symmetric Z/a solve produces the Schur correction $B_{AZ}^* H_A^{-1} B_{AZ}$ in the already frozen Z Hessian. The permitted lower-triangular response has zero access-to- Z upper block and preserves the body solution. The endpoint-blind audit passes all fifteen checks. It also shows that an unsourced relation row can be chosen to reproduce an arbitrary target registration, so the formula is not Nature selection. The common source codomain and occupied row blocks remain open, and CO(10-9) stays sealed.

The current concrete AORS specialization does not meet that rank premise. Its access star is

$$D_A^* = (-\mathbf{1}_4, I_4) : \mathbb{R}^5 \rightarrow \mathbb{R}^4, \quad \ker D_A^* = \text{span}\{\mathbf{1}_5\}, \quad \text{rank } D_A^* = 4.$$

It follows that $\text{rank } D_Z h \leq 4$ for every registration I_{ZA} ; the rank-ten coframe cannot restore a fifth mode already removed by the access incidence. The earlier rank-five construction is therefore an abstract G4D compiler witness, not a realization by the current AORS packet. The deficit is exactly one. For the rank-four cross operator T_0 , a scalar shared row uv^* raises the rank to five exactly when u and v lie outside the current column and row images; a redundant row cannot. Its distinguished candidate direction is now fixed. The normalized AORS common-null covector $q_A = \mathbf{1}_5^*/\sqrt{5}$ pairs with the coframe determinant-line covector

$$q_V(\delta c) = D \log |\det D_O|_0[\delta c] = \delta \lambda + \text{tr } \delta S = \frac{1}{2} \text{tr}(\eta^{-1} \delta h).$$

The latter annihilates all six infinitesimal Lorentz gauge generators. Thus the natural mixed jet is $\gamma_{\text{cm}} q_V^\sharp \otimes q_A$; after registration it closes the deficit exactly when $\gamma_{\text{cm}} \neq 0$ and both legs are transverse to the existing images. Positivity of the scalar source block gives only $|\gamma_{\text{cm}}| \leq \sqrt{k_V k_A}$ and admits zero and both signs. A perfect-square source saturates the bound but still requires source-owned diagonal susceptibilities and an orientation. Including the correlated coframe self-Hessian gives the exact ordered increment

$$\Delta R = \frac{L_0^{-1} u}{1 + k_V s} [k_V u^* L_0^{-1} T_0 - \gamma_{\text{cm}} v^*], \quad s = u^* L_0^{-1} u,$$

which has rank at most one through metricization and any linear gas terminal. In the single common relation-row class the coefficient is no longer independent:

$$\gamma_{\text{cm}} = \Lambda_A \langle D_C e_V, D_A e_A \rangle_{W_R}.$$

Activation is therefore exactly nonzero, nonorthogonal occupation of the coframe-volume and common-access relation images; their weighted cosine is independent of the common action scale and their orientation fixes the sign. The standard covariant matter variation further fixes the physical type of the volume susceptibility. With $T_{\mu\nu} = -(2/\sqrt{-g})\delta S_m/\delta g^{\mu\nu}$ and $\delta g_{\mu\nu} = 2\varphi g_{\mu\nu}$, $\delta S_m = \int \sqrt{-g} \varphi T^\mu{}_\mu$ and $q_V = 4\varphi$ [3]. Cold massive gas is generically traceful, whereas classical radiation is a trace-free control. This does not force five-mode visibility because centered logs remove uniform gain; the opacity shape criterion above is still required.

The updated endpoint-blind audit passes all fifty-six checks, including orthogonal, orientation-reversed, exact-cancellation and terminal-null controls. Physical occupation, transversality, channel-wise opacity/excitation and terminal registration are not supplied by the current source packet, so the result does not open CO(10-9).

The same conditional family has a depth-free necessary consequence. After double centering over paths and channels, CSCR-T32d requires the log-spectral interaction matrix to have rank at most one for commuting scalar-depth transport. This removes common source shape, pathwise scalar gain, and the depth origin without identifying Fermat coordinate with parent distance. The observed singular values are 1.480641, 0.411652, and 0.293055, so the leading component contains 89.568% of the double-centered energy. Across nine fixed covariance geometries, rank-zero noise gives an upper-tail $p = 0.140$ – 0.163 , while the fitted-rank-one goodness-of-fit gives $p = 0.459$ – 0.532 . Neither model is rejected at 0.05. The result is therefore rank-one compatible but not detected or identifying. Nonlinear separable filters share the same rank, and path-by-mode calibration can alter it.

We also freeze the next prospective partial endpoint without opening it. The local Band-7 archive contains an unextracted CO(10-9) cube with frozen archive hash and exact member name. This is a held-out transition through the same SDP.81 lens and source, not an independent-source replication. The pre-pixel gate passes all nineteen static archive, registration, incidence, standard-path, radiative-rank, ordered parent–gas response, symmetry-no-go, four-dimensional-mediator-boundary, G4D-coframe-compiler, blind integrated PDR-amplitude and unsafe-profile-inheritance-rejection checks but zero of seven parent-physical and one of six transition-readiness requirements. The seven deterministically regenerated prerequisite hashes were explicitly re-frozen before this count. The rejected profile shortcut now has an endpoint-blind replacement route: official ALMA archive metadata separate 12 Band-4 CO(5–4) and nine Band-6 CO(8–7) executions (211.709 GB total) from all 11 Band-7 CO(10–9) executions. The permitted execution list and visibility-plane reconstruction contract are frozen, but acquisition and source-cube reconstruction have not occurred. A resource-bounded first route instead uses the two already-local restored CO(5–4)/CO(8–7) cubes, totaling only 395.928 MB, under a frozen full-arc, regularized, covariance-aware inversion with held-path/channel/grid/beam controls. That route is development-only because restored images do not retain the full visibility likelihood. Neither route therefore adds a transition-readiness item. The resource-bounded inversion has now been run: all four paths have positive median held-path improvement, but the combined median is only 2.07%, below the frozen 5% threshold. A fair reflected-source control reaches 1.61%, leaving only a 0.46 percentage-point matched advantage. The restored-image family is therefore not promoted and will not be tuned further. Grid, beam, astrometric and channel-block stability controls are not run after this primary failure and cannot be claimed as passed.

The public reduced-product hierarchy is also now explicit. The official calibrated Band-4/Band-6 measurement-set packages total 54.009 GB, while the 1.599-GB reference-image tier contains the already-tested 395.928-MB pair of allowed line cubes. The public source-plane CO(5–4) and CO(8–7) products are velocity-integrated maps rather than spectral cubes and do not supply a full channel covariance or posterior. No compact public source-plane spectral likelihood or target-and-line-only

calibrated visibility cutout was found. The next finite action is therefore an author request for one of those compact products; the 54.009-GB calibrated archive packages remain the smallest public visibility fallback. This inventory adds no transition-readiness item and does not authorize access to the held-out CO(10–9) endpoint. Neither its FITS header nor data array was read. It remains sealed until a physical common-action packet and a source-owned parent-mode-to-CO(10–9) terminal registration are frozen.

The source gate now reads physical authorization only from a separate, artifact-verified physical manifest. Strings injected into the standard corridor descriptor fail closed, and the direct route additionally requires four explicit $\pi_{4D}(\tilde{\gamma}_i) = \gamma_i$ certificates. A hash-valid self-asserted manifest also fails unless it supplies the typed post-body occurrence law, selector graph certificate, rival-lift control and independent scientific review declaration. That manifest is absent, all seven physical requirements remain false, and the endpoint remains sealed.

There is also a sharp conditional selection of the scalar coordinate. If the registration is homogeneous of degree zero, affine on the normalized positive two-defect simplex, and takes values zero and one at the inner and outer anchors, respectively, then uniquely

$$I_D = K_1 = \frac{d_{\text{in}}}{d_{\text{in}} + d_{\text{out}}}.$$

The K_p counterfamily shows that the limits, homogeneity and anchor-swap law do not imply the affine premise. Thus this theorem closes the local geometry and normalized-affine class only; physical source ownership of that class, its co-registration with the occupied JCSEL/BRAC invariant, calibration and Nature occupation remain open.

The affine uniqueness statement does not make K_1 a complete descriptor. It is constant on every raw-descriptor mirror pair, whereas the standard projection above is not, so no single-valued F can satisfy $G_\ell = F \circ K_1$ on that support. The frozen continuous source path gives an independent within-path witness. Bracketed solves on the disjoint 210–240 and 270–300 arcsec shortest-great-circle segments give

$$K_1 = 0.925 \quad \text{at} \quad R = 228.939929970 \text{ and } 275.245362641 \text{ arcsec.}$$

The corresponding signed coordinates are -0.033824288698 and $+0.074850153783$, while the fixed-inclination projection values are 0.998056542027 and 1.084700249348 . The projection difference is 0.086643707321 . Fixed neighboring levels $K_1 = 0.900$ and 0.950 reproduce the double intersection, with projection differences 0.093917224631 and 0.078771081452 , respectively. The result is stable under the declared bisection-tolerance control and uses no rotation endpoint.

Thus K_1 is not injective even on the frozen one-galaxy source path and cannot replace the complete shared descriptor. This does not prove that the physical Parent coupling is multicomponent. The range of $\hat{D}$ is a constrained two-dimensional surface, and a source-derived rank-one response through another signed scalar remains logically possible. Locally such a collapse requires

$$\Gamma_{\hat{D}} = \gamma \otimes D_{\hat{D}}u, \quad \text{rank } \Gamma_{\hat{D}} \leq 1,$$

plus the exact global terminal-fibre condition. Positive post-body stability does not imply this factorization. Until it is independently derived, the source-to-action handoff must retain $\hat{D}$ or an exactly fibre-equivalent descriptor. The historical expanded endpoint-free audit recorded 62/62 checks. A 15 September 2026 reproduction instead passes 61/62: the stored chirality-square versus raw-defect identity fails its declared numerical comparison, while the independent raw identity passes. The tolerance and frozen source inputs have not been relaxed. Consequently full numerical reproducibility of this shell is pending; it is not an accepted endpoint.

The retained descriptor also admits an endpoint-unfitted complete-registration existence candidate without using the rotation endpoint. Let A_{ab} have rows $(\mathbf{n}_{\text{in}}^T, \mathbf{n}_{\text{out}}^T, c^T)$ and define

$$\mathcal{N}(\hat{D}) = A_{ab}^{-1}(1 - d_{\text{in}}, 1 - d_{\text{out}}, \chi)^T.$$

For a typed oriented-normal register N and any positive source-frozen graph metric K_{reg} , the post-body action

$$\mathcal{A}_{\text{reg}}[N; \hat{D}] = \frac{1}{2} \langle N - \mathcal{N}(\hat{D}), K_{\text{reg}}[N - \mathcal{N}(\hat{D})] \rangle$$

has the unique zero-energy minimum $N_* = \mathcal{N}(\hat{D})$. On the physical descriptor surface the decoder differential restricted to the descriptor tangent has rank two; the graph metric changes the fluctuation cost but not the registered minimum or its Jacobian. This rank is complete only for the signed two-plane oriented-normal sector, not for the whole Parent or every physical interaction.

Because the exact decoder appears explicitly in the graph square, this is a conditional existence and consistency construction. It does not derive the decoder or physical occupation from deeper Parent source primitives by itself.

The inverse appearance can nevertheless be removed conditionally. Define

$$q(\hat{D}) = (1 - d_{\text{in}}, 1 - d_{\text{out}}, \chi)^T, \quad \rho_{\text{TPORI}} = A_{ab}N - q(\hat{D}).$$

If the post-body source owns this forward same-object incidence, then for $W > 0$

$$\mathcal{A}_{\text{TPORI}} = \frac{1}{2} \langle \rho_{\text{TPORI}}, W \rho_{\text{TPORI}} \rangle$$

has the unique zero $N_* = A_{ab}^{-1}q(\hat{D}) = \mathcal{N}(\hat{D})$ and is exactly the graph action above with $K_{\text{reg}} = A_{ab}^T W A_{ab}$. Conversely every positive K_{reg} determines one positive forward weight. Thus the decoder is derived from the forward incidence once that incidence is supplied.

Current source laws do not yet select it. A smooth endpoint-blind, proper-rotation-covariant twist of the oriented normal can fix the three source-frame axes and retain rank two while defining a distinct positive graph. Hence positivity, covariance, anchor fixing, and complete local rank do not force the same-object coidentity $\hat{D}_{ab}(N_*(\hat{D})) = \hat{D}$. The separate endpoint-free source-incidence audit passes 19/19 checks. This is a conditional source-law reduction and nonselection theorem, not physical occupation or an additional Paper 8 empirical/source-shell check.

The physical meaning of the graph square is sharper still. If N merely re-expresses the same descriptor, choose local descriptor coordinates x and register coordinates u , with decoder differential $u = Cx$. Writing $J_N = A_{ab}E_N$ and $Q = J_N C$, the linearized defect is

$$\rho = J_N u - Qx.$$

Its positive-semidefinite joint Hessian has the decoder-graph tangent as its kernel, and exact register elimination gives

$$\boxed{H_{xx} - H_{xN} H_{NN}^{-1} H_{Nx} = 0.}$$

This is the local form of $\min_N \mathcal{A}_{\text{TPORI}}[N; \hat{D}] = 0$ for every admissible $\hat{D}$. Hence a deterministic register adds no source rank, effective descriptor stiffness, force, dark-sector term, or independent readout. An independently physical register would require a separately source-owned off-graph load, typed mixed interaction, register-sensitive terminal, or calibrated fluctuation spectrum. The

separate endpoint-free Schur-neutrality audit passes 14/14 checks; current source ownership and Nature occupation remain open.

The ordered downstream solve isolates the exact remaining linear target. After $u_* = Cx$ is frozen, a response Hessian $H_Z > 0$, mixed rows B_{Zu}, B_{Zx} , and de-duplicated indirect terminal L give

$$B_{\text{eff}} = B_{Zu}C + B_{Zx}, \quad \Gamma_x = -LH_Z^{-1}B_{\text{eff}}.$$

Nonzero register coupling alone is insufficient because the two rows may cancel on the decoder graph; nonzero B_{eff} alone is insufficient because the terminal may annihilate the solved response. Replacing the register path by the direct row $B'_{Zx} = B_{\text{eff}}$ reproduces every on-graph terminal exactly. Register ontology therefore requires off-graph fluctuations or an independently typed register-sensitive terminal. The separate endpoint-free no-bypass audit passes 15/15 checks. Source ownership, standard-comparator separation, and Nature occupation remain open.

The corresponding source-selection question can also be decided relative to the current upstream reduct. Keep the same frozen graph $u_* = Cx$, and let $R : U \rightarrow Z$ be a typed post-registration incidence. For $H_Z > 0$, the positive ordered family

$$\mathcal{A}_Z^{(\lambda)}[z; u_*] = \frac{1}{2} \|z - \lambda Ru_*\|_{H_Z}^2$$

has $B_{Zx} = 0$, but its stationary response is

$$z_* = \lambda RCx, \quad B_{\text{eff}} = -\lambda H_Z RC, \quad \Gamma_x = \lambda LRC.$$

Thus zero direct descriptor-response coupling does not imply zero indirect mediation. With the same descriptor, decoder and registration law, $\lambda = 0$, rank-one R , and full-rank R realize response ranks zero, one and two, respectively; changing the sign of λ reverses the response without violating positivity, and a nonzero R may still be annihilated by L . The current source therefore selects none of activation, rank, orientation, or terminal survival. The separate endpoint-free source-selection audit passes 18/18 checks. A physical promotion requires a typed endpoint-blind incidence and source-owned activation/orientation plus a de-duplicated terminal satisfying $\lambda LRC \neq 0$, followed by standard-comparator separation and Nature occupation. This is a conditional non-emptiness and current-source non-entailment result, not evidence for a new galactic force or dark-sector component.

A scalar JCSEL/BRAC descendant $I_K = f(N)$ instead obeys

$$\text{rank } J_{I_K \leftarrow \hat{D}} \leq 1.$$

Thus there is no contradiction between a rank-two complete shared register and a rank-one scalar interaction. The actual mixed block can still transmit zero, one, or two directions. Moreover, the pullback to the single radial UGC03580 path has rank at most one, so this galaxy can reject a proposed scalar fibre but cannot establish Nature-level rank-two occupation. The separate endpoint-free graph-action audit passes 22/22 checks; it is not counted as an additional empirical/source-shell check. Physical occupation of this graph action and the map from N to the scalar BRAC invariant remain open.

7 Negative-result taxonomy

A poor score is not automatically a falsification of the morphology-readout framework. It must first be classified by source status.

How to read a failure. A failed branch says that one formula family or one component is insufficient. A weak negative says that the current morphology/source manifest is not strong enough to decide. A true negative requires a source-complete morphology manifest, a residual-blind frozen readout family and amplitude rule, endpoint scoring without retuning, and failure against relevant baselines plus wrong-family and shuffled-label controls.

1. *Branch-level negative control*: a particular readout branch fails, but another independently frozen branch may remain viable. The pure Φ_{HHC} -style interval branch for NGC7331 is an example; the mixed readout remains a separate caveated endpoint lane.
2. *Morphology-underdetermined weak negative*: the source evidence does not uniquely determine the readout family, mixture weights, projection state, or history proxy.
3. *Misclassification diagnostic*: a weak score suggests that the initial readout proxy may be wrong. This may motivate source acquisition or holdout design, but cannot promote a new family by itself.
4. *True negative candidate*: the morphology manifest is source-complete, the readout family and amplitude policy are frozen residual-blind, the endpoint is scored without retuning, and the result still fails against the relevant baselines and wrong-family controls.

At present the analysis contains branch-level negative controls and morphology-underdetermined weak negatives, but no source-complete endpoint-grade true negative against the framework. The UGC08490 transfer is stronger than a mere missing-label failure because its outer warp and formula were frozen, but it is still a component-level negative: the target was previously opened, the normalization imports observed summary scalars, and the inner source body is not represented by the one-component warp kernel. The UGC03580 replication is a stronger law-level negative because the source-correct onset loses to earlier wrong-onset controls and the uncapped continuation fails outside R_{HI} . It is still retrospective and the source labels this warp itself as less significant, so it rejects the present universal formula rather than the full morphology/readout framework.

This is not intended to make the framework unfalsifiable. It defines the minimum evidentiary standard under which a failed galaxy becomes an accepted true negative rather than a data-quality or label-quality diagnosis. In the next residual-blind catalogue campaign, a failed case will be promoted to an accepted true negative if all conditions in Table 13 are met before endpoint scoring.

Table 13: Predeclared future criteria for accepting a true negative against the morphology-readout framework.

Gate	Requirement	Failure interpretation
Source completeness	Independent morphology, H I/projection, vertical, support, and history fields are sufficient to select a readout-relevant family and subfamily.	If absent, the case is morphology-underdetermined.
Residual-blind freeze	The family, source scales, mixture weights if any, and amplitude policy are written to a freeze artifact before v_{obs} -residual scoring.	If absent, the case is diagnostic only.
Endpoint execution	The frozen readout is scored without retuning on the predeclared galaxy or holdout set.	If absent, no validation or falsification claim.
Control failure	The frozen readout fails against the relevant baseline set and does not beat wrong-family or shuffled-label controls.	If present with prior gates passed, accepted true negative.
Audit trail	The source ledger, scripts, random seeds, and output artifacts are preserved.	If absent, result is not reproducible enough to classify.

The current concrete negative and weak-negative examples are as follows.

- *Branch-level negative controls.* The pure Φ_{HHC} -style branch for NGC7331 is negative, but the mixed readout for the same galaxy improves the endpoint score. UGC08699 is best read as a compact-support branch control rather than as a general failure of the readout-morphology framework. Several weak exponential-disk population cases also behave as branch or proxy controls.
- *Morphology-underdetermined weak negatives.* NGC0247, NGC0300, NGC3917, NGC6015, NGC6503, UGC08286, and partly NGC3198 are not promoted to framework-failure cases in the present manuscript. For these objects the manifest may not yet determine the readout-relevant subfamily, projection or history component, mixture weight, or source-side amplitude strongly enough.
- *Calibration-limited negative endpoint.* The source-frozen NGC 4062 H I–H α side-odd contrast is consistent with zero in both HR and LR and fails its velocity-zero-point sign-stability gate. It is a real negative result for that frozen terminal, but not an accepted true negative against the full morphology/readout framework.
- *No accepted true negative yet.* A true negative would require source-complete morphology, a residual-blind frozen readout family and amplitude policy, endpoint scoring without retuning, failure against the relevant baselines, and wrong-family/shuffled controls that do not explain the failure. No case currently satisfies all of those requirements.

In plain operational terms, there are keys that do not open a given door, but there is not yet a source-complete case where the protocol shows that the correct key was used and the door still did not open.

8 Why a readout morphology catalogue is needed

The main bottleneck is not simply more curve fitting. The readout state of a galaxy requires information that is scattered across heterogeneous data products: NED classifications [27], S4G

decompositions [38], SPARC mass models [19], DustPedia or other multiwavelength context [6], H I maps and radial profiles, PHANGS-like resolved gas or velocity-field products where available [20], and object-specific literature. Existing surveys contain many ingredients, but they were not designed to freeze morphology/readout states before rotation-curve endpoint scoring. For example, existing H I and edge-on studies already support pieces of such a catalogue for NGC3198 [1, 12, 36] and for several exploratory disturbed or dwarf-galaxy lanes [7, 45, 34, 40].

This suggests a concrete data product: a residual-blind readout morphology catalogue. Each row should record at least

$$\{K_{\text{obs}}, K_{\text{readout}}, \text{confidence tier, provenance, source windows, blocked fields}\},$$

plus disk scales, decomposition fields, H I support and tail/onset observables, vertical/projection flags, bar/ring/lopsidedness indicators, environment or morphology-trajectory evidence, and the allowed endpoint status. The catalogue should state whether a galaxy is endpoint-ready, replay-only, candidate-only, or morphology-underdetermined.

The morphology-trajectory entry is not an extra adjustable force. It records the possibility that the present projected morphology is not the whole readout state. A galaxy can currently look like a regular disk while carrying H I warp, interaction, outer-tail, vertical-support, settling, or relaxation evidence associated with a broader Tau-side morphology trajectory. In the protocol this information is allowed only as residual-blind source evidence; it cannot be inferred from the endpoint rotation residual after the fact.

The same caveat applies to observer/projection effects. They are not an after-the-fact permission to reinterpret any failed curve. They are catalogue fields to be frozen before scoring: inclination and line-of-sight geometry, resolved H I support, vertical/projection indicators, wavelength-dependent structure, and trajectory/phase flags. Only after those fields are frozen may a galaxy be assigned to a projected/readout-relevant subfamily.

Such a catalogue is observationally realistic. A pilot could begin with roughly 20–30 source-rich galaxies having SPARC rotation curves, good H I maps or velocity fields, infrared/optical decompositions, and usable multiwavelength or literature context. The readout state would be frozen first; only then would matched-family, wrong-family, shuffled-label, and baseline endpoint scores be run. This would replace ad hoc per-galaxy data hunting with a reproducible observing and reprocessing program.

Targeted observations could make the catalogue much stronger than the present patchwork. Existing surveys were not optimized to freeze readout morphology for this protocol, so the missing fields are unsurprising. A focused campaign would prioritize H I radial profiles and velocity fields, outer warp/tail onset radii, vertical scale or flare information for edge-on systems, bar/ring/lopsidedness indicators, and multiwavelength decomposition. Those are standard astronomical observables; the new element is the residual-blind combination of them into endpoint-ready readout states.

9 Discussion

The present results have two different meanings. First, morphology-indexed readout families appear to carry information: matched formulas can beat wrong families much more often than shuffled labels would suggest. Second, the present formulas are not yet a superior galaxy-rotation law. Their baseline performance is mixed, and the strongest single-galaxy results are caveated.

This is exactly why the endpoint protocol is useful. It prevents a weak result from being silently retuned into a fit, and it prevents a positive single-galaxy result from being overpromoted into population validation. The framework becomes scientifically interesting only when a source-complete readout catalogue allows predeclared morphology-family tests on a broader sample.

9.1 Why this is not yet a curve-fitting result

The present protocol was designed because unrestricted per-galaxy fitting would be scientifically uninformative. A sufficiently flexible radial function can usually be made to follow a rotation curve. The question asked here is different: can residual-blind source morphology/readout evidence choose a restricted readout family before the endpoint residuals are inspected? For that reason the paper separates formula construction, source-field acquisition, amplitude freezing, endpoint scoring, wrong-family controls, and shuffled-label controls.

Some of the current readout formulas are still idealized. They are morphology-indexed 4D readout proxies for thin disks, thick/flared disks, compact support, rings, warps, and mixed projection/trajectory lanes. A real galaxy can occupy a readout state that is richer than its present projected shape: interaction history, future-directed relaxation or settling phase, H I extent, vertical structure, bar/ring phase, lopsidedness, and inclination can all affect the effective readout. Therefore a weak result is not automatically evidence that the framework has failed. It may instead mean that the current formula is too coarse, the readout-relevant subfamily is wrong, or the source fields needed to freeze the right subfamily are missing. Only source-complete endpoint-grade failures are eligible to become true negatives.

9.2 Outlook: projection-enriched readout kernels

The present endpoint kernels should be read as morphology-indexed 1D readout proxies, not as complete observer/path Tau Core readouts. A natural next step is to replace a single morphology kernel by a residual-blind component sum with source-frozen observer/projection visibility weights,

$$\delta v_{\text{proj}}^2(R) = \sum_j A_j(\text{source}) w_j(R, \mathcal{O}_{\text{obs/path}}, \Theta_{\text{morph}}) K_j(R; K_{\text{present}}, \Theta_{\text{morph}}).$$

Here $\mathcal{O}_{\text{obs/path}}$ denotes source-observer geometry such as inclination, position angle, line-of-sight projection, finite resolution, waveband, and velocity-field coverage, while Θ_{morph} denotes a source-frozen morphology trajectory/phase profile. The existing curves should therefore be read as present-only, observer/projection, or partial trajectory/phase approximations rather than full path-aware kernels. A still fuller path-aware layer would add a correction $\delta v_{\text{path}}^2(R; \mathcal{P})$, where $\mathcal{P}$ is the source-observer null-geodesic bundle and its causal metric environment. That path-environment term is not modeled or scored in this paper.

The component notation above must not be read as permission to add every available projection descriptor independently. A future enriched kernel must declare a source-support ledger

$$\text{SourceSupp}(\Delta K_a) \subseteq \{P_1, \dots, P_n\}$$

for every correction component. If a warp/history source datum, observer/path descriptor, or time/readout proxy supports two proposed terms, the terms must be merged into one shared quotient contribution, or an independent endpoint-blind source-side split must be supplied before scoring. Thus projection enrichment is a source-factorization problem, not a free additive rescue layer.

Five inspected systems – UGC12506, NGC4013, NGC5907, NGC7331, and NGC4088 – suggest that such projection-enriched kernels are worth testing. They are not part of the headline population endpoint here. They define the next predeclared catalogue test: source-frozen observer/projection labels should select the enriched kernel family before endpoint scoring, and the matched enriched families should be compared against simpler morphology-only proxies, wrong enriched families, shuffled observer/projection labels, and conventional baselines. The detailed derivation, candidate

audit, and validation gate are therefore separated into the companion projection-enriched readout paper.

9.3 How conventional baselines fit into the readout picture

The paper also does not assume that Newtonian, MOND/RAR-like, or TPG/v6-like curves should fail everywhere. A regular, stable, low-disturbance disk can be close to a Newtonian or smooth scalar readout. A galaxy whose residuals are well described by a single acceleration scale can be naturally strong under a MOND/RAR-like comparator. A galaxy dominated by a smooth closure response can be strong under the frozen TPG/v6 predecessor comparator. These are not contradictions of the morphology-readout programme; they are limiting cases the programme must explain.

The stronger claim would be this: after source-complete morphology labels are frozen, the correct readout subfamily should explain when those simple limits are adequate and when a richer morphology-specific kernel is needed. The present paper does not prove that claim. It sets up the endpoint machinery needed to test it.

10 Full-4D Distortion Impact on the Frozen Scores

The observer-specific full-4D Tau descent broadens the future forward model but does not alter any score reported in this paper. The current matched, wrong-family, shuffled-label, baseline, and holdout statistics were computed from a frozen morphology/readout family. Recognition of an additional model class does not change those statistics.

A future full-4D candidate must begin with a source-derived radial relative coframe $D_O(R)$, not with a rotation residual. That single object must induce the lapse, shift, spatial metric and source-observer photon transfer; the exact $D_O = I$ branch must remain a mandatory null. Time-only, spatial-only, shift-only and full-coframe ablations may then be compared under the same family-budget and wrong-family controls used here. Until that source-side contract is satisfied, the full-4D route is a blocked future family and cannot be used as a retroactive rescue term for any galaxy.

The later source-frozen compiler fixes the corresponding scoring object. If the packet induces Hessian blocks K_{HH}, C, K_{VV} , the candidate is not the positive Schur term but

$$E_K = (K_{HH} - K_{\text{std}}) - CK_{VV}^{-1}C^\dagger.$$

The graph-normal control has $C \neq 0$ but $E_K = 0$. Therefore a future Paper-8 endpoint must freeze the complete standard excess, all terminal maps, their deduplicated observable support and full covariance before scoring. No reported matched-family or holdout statistic is changed by this rule.

The latest co-registration audit sharpens the shift-only ablation of that future endpoint. Conditional on one common source-owned coframe packet, the oriented scalar row is

$$q_\tau = -\omega_{uv}P_\beta\mathcal{L}_O^{-1}B_{OS}^*W_\rho(D_B Cx + D_A a),$$

where x and a are body-side and irreducible observer-source coordinates, P_β selects the shift, and ω_{uv} is the source-frozen oriented transverse covector. If, and only if, this dimensionless row occupies the standard photon-rapidity terminal, its exact velocity increment is

$$h_\tau = c \frac{(1 - \beta_{\text{dyn}}^2) \tanh q_\tau}{1 + \beta_{\text{dyn}} \tanh q_\tau}.$$

This supplies no independent post-source gain and is only one rank-one ablation of the ten-component metric response. A single rotation terminal also cannot attribute a common signal uniquely to body morphology rather than observer access. Such attribution requires source-frozen stacked terminals whose resolved joint map separates the body/access pair after the declared stable quantizer; a duplicate or factorized terminal adds no identifying information. The present source inventory selects neither a nonzero joint incidence, a stable galactic quantizer, nor physical rapidity-terminal occupation. Consequently the formula is a conditional compiler result, not a new rotation curve, and all frozen scores above remain unchanged.

10.1 Endpoint-informed inverse required-rapidity diagnostic

Although the endpoint cannot derive its own source law, it can be used explicitly as an inverse diagnostic. For each declared baseline we therefore reconstructed the terminal-equivalent rapidity

$$q_{\text{req}}(R) = \text{artanh}\left(\frac{v_{\text{obs}}(R)}{c}\right) - \text{artanh}\left(\frac{v_{\text{base}}(R)}{c}\right), \quad u_{\text{req}} = cq_{\text{req}}. \quad (4)$$

This is an exact identity for the rapidity-composition shell, not an identification of q_τ . The frozen 131 training galaxies were used to select ridge regularization by five-fold galaxy-grouped cross-validation; the existing 44-galaxy holdout was not used for tuning. Predictor blocks were limited to radial sampling, residual-blind morphology/baryonic proxies, geometric distance, inclination and catalogue quality. Observed velocities, residuals and reconstructed required-channel values were forbidden predictors, and every training galaxy received equal total weight.

Table 14 reports the combined-model holdout result. The loss is the mean of the per-galaxy RMSE values in u_{req} ; the control predicts zero additional rapidity.

Table 14: Endpoint-informed inverse required-rapidity diagnostic. Intervals are galaxy-bootstrap 95% intervals for the paired model-minus-zero RMSE difference. The randomization p -value is one-sided for improvement. These are diagnostic results, not physical-kernel scores.

Baseline	RMSE ratio	Δ RMSE	95% interval	p
Newtonian baryonic	0.405	-25.420	[-31.295,-19.466]	0.00002
TPG/v6	0.963	-0.660	[-1.774,0.335]	0.119
fixed MOND	0.936	-1.166	[-2.723,0.349]	0.075

The Newton-relative target is strongly predictable, but ordinary baryonic scaling and the established mass-discrepancy relation are a stronger explanation than a Tau-specific channel. After TPG/v6 or MOND, the combined model gives only a small, statistically unresolved improvement and weak pointwise correlations ($r = 0.157$ and 0.169 , respectively).

Most importantly for the proposed distance-loss interpretation, adding geometric distance and distance uncertainty to the morphology block changes mean per-galaxy RMSE by only -0.107 km s^{-1} for TPG/v6 ($p = 0.323$) and $+0.039 \text{ km s}^{-1}$ for MOND ($p = 0.952$); both bootstrap intervals contain zero. Distance also adds no improvement after morphology plus orientation. Inclination/quality fields show a weak near-threshold pattern, but it is not robust and has the direct conventional alternative of beam/deprojection and catalogue-quality systematics.

The informative outcome is therefore primarily negative. The present broad source and distance descriptors do not recover a stable universal post-TPG/MOND rapidity profile on unseen galaxies. The exercise narrows the future source search, but it does not derive $D_O(R)$, a stable Q_{OS} , body-versus-observer attribution, Nature occupation or a dark-matter replacement law.

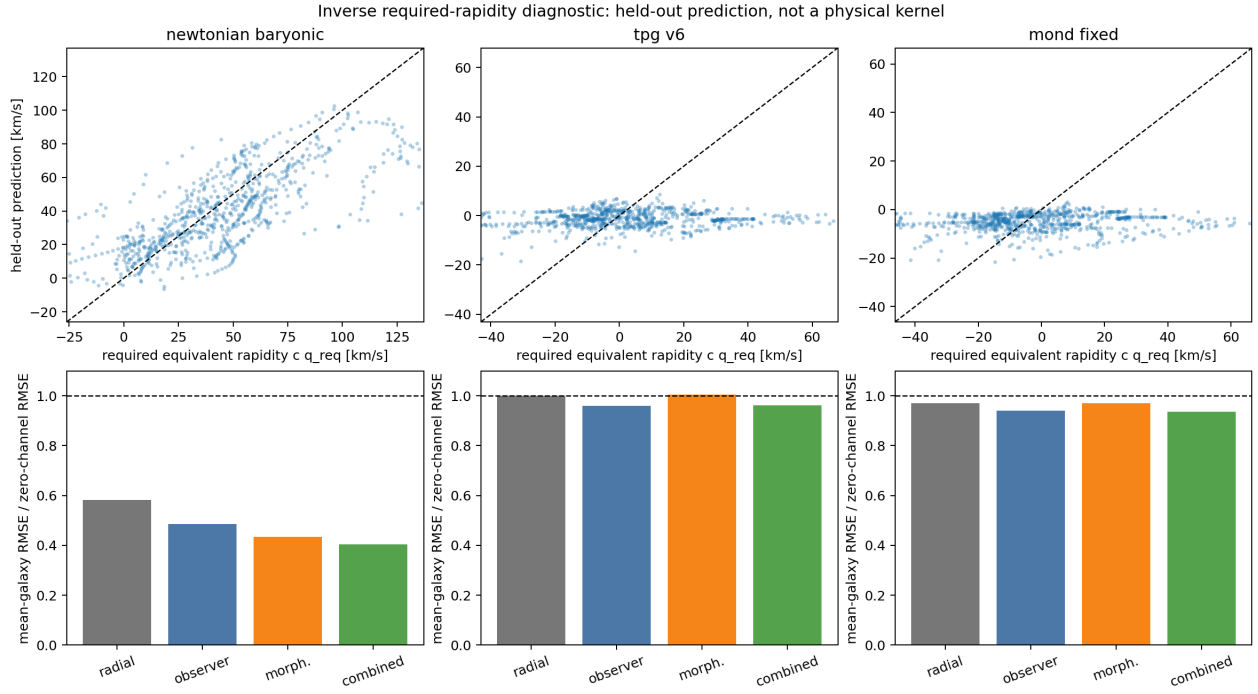


Figure 13: Held-out inverse required-rapidity diagnostic. Top: reconstructed versus combined-model prediction. Bottom: mean-galaxy RMSE relative to the zero-additional-rapidity control. The strong Newtonian result and weak post-TPG/MOND results have different interpretations; none identifies a physical Tau source.

10.2 Measurement-inverse required parent-loss target

We also reverse the empirical calculation in the simpler one-parameter shell

$$v_{\text{obs}}^2(R) = \frac{v_{\text{base}}^2(R)}{1 - \ell_{\text{req}}(R)}. \quad (5)$$

For positive measured and baseline speeds its exact inverse is

$$g_{\text{req}} = \frac{v_{\text{obs}}}{v_{\text{base}}}, \quad \kappa_{\text{req}} = \log g_{\text{req}}, \quad \ell_{\text{req}} = 1 - \frac{v_{\text{base}}^2}{v_{\text{obs}}^2}. \quad (6)$$

The scalar is unique inside this declared shell, as follows by direct substitution and injectivity on the positive-speed domain. Its physical factorization is not unique. For example,

$$\kappa_{\text{req}} = \kappa_D - \kappa_T + \kappa_M + \kappa_{\text{cal}} \quad (7)$$

maps four pointwise components to one observable and therefore has a three-dimensional null family. One rotation terminal cannot attribute the reconstructed value separately to parent access/distance, temporal calibration, morphology, or remaining terminal calibration.

Table 15 gives the SPARC reconstruction. The fixed cubic profile is trained on the 131 development galaxies and evaluated on the untouched 44-galaxy holdout; a ratio below one improves on zero required log-gain.

Table 15: Measurement-inverse required parent-loss diagnostic over 3,389 points in 175 SPARC galaxies. These values are target reconstructions, not physical parent-loss estimates or endpoint scores.

Baseline	Median gain	Median κ_{req}	Median ℓ_{req}	Holdout ratio
Newtonian baryonic	1.6576	0.5054	0.6360	0.5633
TPG/v6	1.0106	0.0105	0.0209	0.9943
fixed MOND	0.9755	-0.0248	-0.0509	0.9655

The baryonic inversion recodes the familiar mass discrepancy. After TPG/v6 or MOND, however, the inferred field is sign-mixed and one universal radius-only cubic profile gives little or no held-out gain. This rejects that simple scalar profile inside the declared shell, but it does not reject a morphology-, channel-, tracer-, or context-conditioned operator. Baseline incompleteness, mass-to-light uncertainty, non-circular motions, inclination/deprojection error, and distance systematics remain direct conventional alternatives. The output is therefore an inverse target atlas for a future source-frozen forward test, not an identification of M_7 , Nature occupation, or a dark-matter replacement law.

11 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer-source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the

BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS} .

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. For galactic inference, these observer constructions do not derive a rotation-curve correction or identify a measured residual as a parent effect. Existing endpoint freezes and scores are unchanged.

12 Finite source-to-readout implementation and its scope

The current finite laboratory is a reproducible implementation, not a new Nature-selection theorem. The modular family search ranks source-side finite conditioning, not galaxy residuals. The separate galaxy-training mode retains an explicitly supplied spectral-log terminal completion; previously exposed or training galaxies are not fresh validation systems. Neither improving a training curve nor changing graph size establishes morphology ownership, a dark-sector explanation or a Tau-specific residual. Published endpoint rules and scores are unchanged.

Ownership and reproducibility. The seed–body companion owns the finite minimum/Hessian example, the body–readout companion its ordered access and contact reduction, and the terminal companion its conditional propagation boundary. The physical-source papers retain the source-selection obligation. Code, declared priors, adverse controls and provenance are available at https://github.com/tau-core-research/tau-core-theory/tree/main/experiments/seed_evolution_lab. Numerical tests are internal evidence only; no published empirical score is changed.

13 Source identification and recovery

The current observer audit separates a conditional forward construction from physical source selection. A frozen body, occupied observer incidence, preparation, complete instrument and stable resolution can determine records without selecting those inputs from Nature. Equal effects do not fix postmeasurement states, and symmetry within a chosen interaction family does not exclude

other symmetric couplings. For galactic inference, these observer constructions do not derive a rotation-curve correction or identify a measured residual as a parent effect. Existing endpoint freezes and scores are unchanged.

Dependency and claim ownership. Foundation VI and the body-readout technical companion own the finite instrument/access proof; Foundation IV and TD-0 retain physical source selection; Foundation V, the temporal synthesis and seed-body companion own the gradient/inertial distinction. This paper inherits only the stated specialization. Canonical assumptions, countermodels and calibration audits are in https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_rqm_instrument_to_record_bridge_001.md. No physical seed, Nature occupation or Tau-exclusive observation is established by this update. Existing empirical endpoint definitions and scores are unchanged.

14 Conditional source preparation and time-resolved access

The common-operator scalar-Dirac completion now supports an explicit conditional chain from interaction vertices to pair transfer and joint readout. Nonzero vertices do not select occupation. Relational coherence can be generated from an occupied number-diagonal pair even while both local mean fields vanish. In a declared resonant rotating-wave and Markov loss model, finite-window emission retains this coherence information, whereas an all-time count discards it. This is standard-compatible quantum recovery, not a Tau-specific observational residual. For galactic inference, these observer constructions do not derive a rotation-curve correction or identify a measured residual as a parent effect. Existing endpoint freezes and scores are unchanged.

Dependency ownership and limits. The P4 common-source assembly owns the scalar/Yukawa benchmark and finite preparation calculations; Foundation VI and the body-readout companion inherit the joint-access mathematics. Foundation IV and TD-0 retain physical source selection, and Foundation V retains clock descent. The parent remains atemporal: the evolution above is a conditional downstream 4D model, not parent memory or a derived morphogenetic time. Physical initial occupation, a gauge-complete realization, detector amplification, absolute calibration and stable observer-relative Q_{OS} remain unselected. No empirical endpoint or score is changed. The canonical derivation and reproducible numerical sources are linked at https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_core_p4_nonabelian_particle_common_source_assembly_001.md.

15 Conclusion

This paper reframes morphology-sensitive rotation-curve work as a forward readout problem. The central claim is not that a new gravity theory has been validated, nor that every galaxy can be rescued by a custom curve. The claim is that a residual-blind morphology/readout state can be used to choose a predeclared readout family, and that this choice can be tested against wrong families, shuffled labels, and conventional baselines.

The current numerical evidence is preliminary but useful. The strongest broad internal signal is matched label-formula compatibility: source-native readout formulas beat wrong formula families in 0.886 of 44 holdout galaxies with a shuffled-label $p \simeq 0.002$ for the mean matched-minus-wrong statistic; the beat-fraction null is reported separately. The same formulas do not yet beat TPG/v6 or MOND-like comparators as a population. Four inspected galaxies are retained

here only as caveated demonstrations of the endpoint workflow. The unchanged NGC4088-class transfer to UGC08490 adds an important counterweight: the correct source onset weakly beats wrong morphology controls, but the full curve is decisively worse than fixed MOND and standard fit-aided halo models. The second replication on UGC03580 then removes the remaining universality claim: the source-correct onset is worse than Newtonian, fixed MOND and earlier wrong-onset controls, while the uncapped continuation overshoots beyond R_{HI} . Hence morphology can carry radial information without the present one-component warp law being a complete or universal rotation law. The endpoint-blind radial reconstruction subsequently replaces the onset-only proxy by a source-complete orientation path on S^2 and reproduces the published inner/outer plane angle. This strengthens the morphology representation but supplies no terminal law and therefore does not alter the negative endpoint verdict. A bounded defect-ratio coordinate can be derived from the two planes, but the explicit $v_{p,a,s}^2 = v_c^2(1 + saK_p)$ counterfamily proves that the same body admits inequivalent terminal powers, amplitudes, signs, and carriers. The standard fixed-inclination control spans 0.99409–1.09685 in squared speed and must be audited before attributing any residual structure to Tau. The common-action pullback reduces these apparent independent freedoms to one signed-descriptor-to-Parent-invariant registration plus the occupied calibrated gravity/observer packet, but that registration is not selected by the current source. The source geometry narrows its hidden fibre to one exact mirror sign: the oriented source record and the standard fixed-inclination projection show that the complete stack is not mirror-even, so the signed triple product must be retained for this lane. This does not establish Parent chirality or a Tau-specific coupling. Normalized-simplex affinity would then select K_1 uniquely within that scalar class, but the source path itself revisits the same K_1 level at terminal-distinct locations. Hence K_1 is not a complete shared descriptor. No alternative signed scalar or physically multicomponent Parent coupling is yet selected. Consequently, any physically occupied bounded two-plane replacement remains a new source-side hypothesis and may not be repaired on these opened endpoints. The more detailed observer/projection-enriched kernel derivation and its four-object audit are separated into a companion paper, because they define a new validation programme rather than the headline population endpoint of the present paper.

Two promotion attempts are now explicitly negative. On the original 45-galaxy label packet, the projection contrast adds no stable cross-fitted proper-score improvement beyond the frozen observability/baryonic and MOND/RAR-common baselines. On the external EDGE-CALIFA packet, the seven-galaxy source-frozen matched-tracer test gives $\bar{D} = -0.0587$, exact one-sided $p = 0.6016$, and only four positive objects. These results preserve the original descriptive association while blocking its promotion to conditional projection specificity or external confirmation.

The theory update sharpens the boundary of that programme. Present morphology, observer/path appearance, morphology-history phase, and time/readout descriptors are source-proxy coordinates of one protected source class. They may improve a kernel only when frozen independently or merged into a joint quotient contribution; the same source evidence may not be counted twice as separate physical corrections. The accompanying reproducibility package records this boundary in a paper-level `KernelRecord_tau^S` status audit. That audit classifies the population lanes, wrong-family controls, shuffled-label tests, and inspected single-galaxy examples as diagnostic/control, source-proxy, or endpoint-active frozen kernels. It does not promote any Paper 8 object to a validated physical readout.

The next step is therefore not another round of residual-driven curve repair. It is a readout morphology catalogue, source-grounding audits for any formula shell promoted beyond proxy status, an independent survey family replacing the now-exhausted clean HALOGAS lane, and a predeclared endpoint campaign. If the correct source-frozen readout family systematically beats wrong families and shuffled labels, and then becomes competitive against Newtonian, MOND/RAR, and TPG-like

baselines, the framework will have a meaningful empirical target. If source-complete endpoint-grade cases fail, those failures should be retained as true negative candidates. The absence of such a case in the present analysis is a limitation of the current source catalogue, not a positive result.

The new representation theorem sharpens that final decision without weakening the negative record. The existing failures constrain the frozen A/C, low-order, S4G, EDGE and PHANGS representations; they do not establish a source-complete falsification of the general body. Any successor must be a genuinely new endpoint-blind Φ_{k+1} , not a post-open re-expression of the same residuals.

A Readout-family derivation ledger

This appendix records the manuscript-level derivations of the readout kernels used in the present population preflight. The purpose is not to prove a final gravity law. It is to make explicit which part is a definition, which part is a source ansatz, which part follows from a standard radial readout/closure calculation, and which part remains an empirical or source-completeness obligation.

This appendix is theory-motivation scaffolding. The numerical test in the main text does not require the reader to accept the full Tau Core bridge ontology. Operationally, the test only requires residual-blind labels, frozen source proxies, predeclared readout families, and matched-versus-wrong/shuffled controls. The bridge equations below explain why these families are grouped as readout specializations, but the population statistics can be audited from the saved artifacts even if the parent theory remains unaccepted.

Dependency boundary: the endpoint statistics do not depend on accepting Appendix A. Appendix A only motivates the grouping of the readout families and records the formula ledger used by the reproducibility package.

A.1 Tau Core bridge equation used by the readout families

The readout families in this manuscript are motivated as reductions of one common conditional bridge object. Let P_D^τ denote the Tau-side ordered-disk readout slice and let P_N^τ denote the Newtonian/metric closure-test slice. Their residual-blind local mismatch is

$$\delta P_\tau = P_N^\tau - P_D^\tau.$$

Only the tangent/off-slice part of this mismatch can generate a readout residual. Denoting that part by $\delta P_\tau^\perp$, the first-order relative generator is

$$\mathcal{M}_\tau = [\delta P_\tau^\perp, P_D^\tau].$$

The bridge then tests whether this generator moves the ordinary diagonal disk sector outside the Newtonian, ordinary-disk, and gauge subspace

$$B_{\text{axis}} = N + D_{\text{diag}} + G.$$

The non-absorption map is

$$\rho_D(\mathcal{M}_\tau) := \pi_B \circ \mathcal{M}_\tau|_{D_{\text{diag}}}, \quad \pi_B : V_{\text{grav}}^\tau \rightarrow V_{\text{grav}}^\tau / B_{\text{axis}}.$$

The common 4D weak-field readout equation used by the present paper is therefore

$$\boxed{\delta g_{\text{morph}}(x) = R_{g,x}^{4D} [\rho_D(\mathcal{M}_\tau(x))(Q_D^\tau(x))].} \quad (8)$$

Here Q_D^τ is the morphology-selected ordered-disk trace-free direction. Equation (8) is the bridge base equation used in this paper. It is not a final law of gravity; it is a conditional readout rule: if the Tau-side slice mismatch is source-defined, survives the quotient, and has a nonzero 4D weak-field readout, then it contributes an effective morphological residual.

For the local disk witness used throughout the bridge, let S_D^τ be the in-disk shear-like partner of Q_D^τ . If the closure-test slice selects

$$Q_N^\tau = \cos \alpha_\tau Q_D^\tau + \sin \alpha_\tau S_D^\tau,$$

then, to first order in the local mismatch angle,

$$\mathcal{M}_\tau = \alpha_\tau J_D, \quad J_D(Q_D^\tau) = S_D^\tau, \quad J_D(S_D^\tau) = -Q_D^\tau.$$

Consequently

$$\rho_D(\mathcal{M}_\tau)(Q_D^\tau) = \alpha_\tau [S_D^\tau]. \quad (9)$$

Equation (9) is the algebraic bridge reduction. It says that the residual is not produced by inserting a new force; it is the quotient-visible part of a readout/closure mismatch. The residual vanishes if $\alpha_\tau = 0$, if S_D^τ is absorbed into B_{axis} , or if the 4D readout map kills this quotient class.

A.2 From bridge residual to a radial family kernel

To compare with one-dimensional rotation curves, the bridge readout must be reduced to a radial observable. In the present preparation-level tests the family label K chooses a residual-blind source functional $\mathcal{S}_K^\tau$ representing the observable part of $\alpha_\tau [S_D^\tau]$ for that morphology/readout class:

$$\alpha_\tau [S_D^\tau] \longrightarrow \mathcal{S}_K^\tau.$$

The 4D readout map then induces a radial operator $\mathcal{R}_K$,

$$\delta g_{K,R}(R) = \mathcal{R}_K[\mathcal{S}_K^\tau](R),$$

and circular-orbit comparison uses

$$\delta v_K^2(R) = R \delta g_{K,R}(R). \quad (10)$$

Thus every endpoint shell below has the common form

$$\delta v_K^2(R) = A_K K_K(R),$$

where $K_K(R)$ is the source-selected radial kernel and A_K is the allowed source amplitude or frozen amplitude rule.

This is the derivation status used in the paper:

- The slice-mismatch and quotient equations (8)–(9) are the conditional Tau Core bridge scaffold.
- The choice of $\mathcal{S}_K^\tau$ is a residual-blind morphology/readout source ansatz, not an endpoint fit.
- A source-frozen ansatz is not yet a full Tau-side source derivation. To promote it beyond proxy status, a separate source-grounding gate must show activation of a Tau-side mode, survival in the null/gauge quotient, stability under the relevant closure test, and survival under the four-dimensional readout map. In schematic form this later gate is

$$O_i^{4D} = R_i^{4D}(\Pi_{\text{stab}}([A_\tau(u)])) .$$

The present paper defines and tests the operational family proxies; it does not claim to have discharged this source-grounding gate.

- The radial kernels $K_K(R)$ follow from applying the chosen radial readout operator $\mathcal{R}_K$ to that source ansatz.
- The current population preflight does not prove the parent Tau Core theory or the exact amplitudes A_K . It tests whether the residual-blind family specialization carries information by comparing matched families, wrong families, shuffled labels, and baseline curves.

Table 16: Logical status of the bridge-to-kernel construction.

Layer	Equation or object	Status in this manuscript
Tau-side slice mismatch	$\delta P_\tau = P_N^\tau - P_D^\tau, \mathcal{M}_\tau = [\delta P_\tau^\perp, P_D^\tau]$	Definition within the conditional bridge scaffold.
Quotient non-absorption	$\rho_D = \pi_B \circ \mathcal{M}_\tau _{D_{\text{diag}}}, \rho_D(\mathcal{M}_\tau)(Q_D^\tau) = \alpha_\tau[S_D^\tau]$	Conditional algebraic bridge result.
4D readout shell	$\delta g_{\text{morph}} = R_g^{4D}[\rho_D(\mathcal{M}_\tau)(Q_D^\tau)]$	Bridge base equation tested operationally here.
Family source specialization	$\alpha_\tau[S_D^\tau] \rightarrow \mathcal{S}_K^\tau$	Residual-blind source ansatz, family by family.
Radial endpoint kernel	$\delta v_K^2 = R \mathcal{R}_K[\mathcal{S}_K^\tau] = A_K K_K(R)$	Derived shell after choosing the source class and radial readout operator.
Amplitude and source completeness	A_K , source radii, thickness, support, projection state	Frozen, train-split, or proxy-level depending on the endpoint row.

A.3 Common normalization and dimensional rule

For every family K , the executable preflight uses a velocity-squared readout of the form

$$v_{gK}^2(R) = v_{\text{base}}^2(R) + \delta v_{gK}^2(R), \quad \delta v_{gK}^2(R) = A_{gK} K_{gK}(R).$$

Here $K_{gK}(R)$ is the source-selected radial kernel and A_{gK} is the corresponding amplitude. The kernel need not be dimensionless. The dimensional requirement is simply

$$[A_{gK}] [K_{gK}] = (\text{velocity})^2.$$

In a strict endpoint, A_{gK} must be source-derived or frozen by a predeclared rule. In the current population preflight, the family shapes are the derived readout shells, while several source scales and amplitudes remain available-data proxies or train-split normalizations. That is why the population result is an internal label–formula compatibility signal, not a conditional or externally replicated morphology-specific result and not a final empirical validation.

A.4 Scale-tail or diffuse LSB readout

Definition/source ansatz. The scale-tail family is used when the source morphology indicates an extended outer load, diffuse low-surface-brightness support, or an H I/tail-like readout component. The residual-blind source inputs are an inner onset radius R_{in} and an outer cutoff/support radius R_{cut} . The minimal $n = 2$ tail window is

$$W_{\text{tail}}(r) = \frac{(r/R_{\text{in}})^2}{1 + (r/R_{\text{in}})^2} \frac{1}{1 + (r/R_{\text{cut}})^2}.$$

The first factor suppresses the inner disk; the second keeps the source finite.

Bridge specialization. In Eq. (8), the scale-tail family corresponds to a source class $\mathcal{S}_{\text{tail}}^\tau = A_{\text{tail}} W_{\text{tail}}(r)$ whose quotient-visible readout is dominated by an extended outer support. The radial operator $\mathcal{R}_{\text{tail}}$ is the one-dimensional accumulated-load readout. Thus the family is obtained by applying $\delta v^2 = R \mathcal{R}_{\text{tail}}[\mathcal{S}_{\text{tail}}^\tau]$, not by fitting an arbitrary tail shape to the rotation residual.

Radial readout. For a one-dimensional preflight, the allowed radial load is the accumulated tail source

$$I_2(R) = \int_0^R W_{\text{tail}}(r) dr.$$

The corresponding scale-tail velocity-squared response is taken to be

$$\boxed{\delta v_{\text{tail}}^2(R) = A_{\text{tail}} \frac{I_2(R)}{R}} \quad K_{\text{tail}}(R) = I_2(R)/R.$$

This is the formula implemented in the reproducibility code as the $n = 2$ scale-tail shell.

Limits. For $R \ll R_{\text{in}}$, $W_{\text{tail}} \sim (R/R_{\text{in}})^2$, so the tail readout is suppressed in the inner disk. For $R_{\text{in}} \ll R \ll R_{\text{cut}}$, $I_2(R) \sim R$, so $I_2(R)/R$ approaches a slowly varying or flat contribution. For $R \gg R_{\text{cut}}$, $I_2(R)$ saturates and the response falls as $1/R$ in velocity squared. Thus the shell has the intended inner-suppressed, outer-supported, finite-tail behavior without reading endpoint residuals.

Claim boundary. This derivation fixes the readout shape once R_{in} and R_{cut} are source-frozen. It does not by itself derive the amplitude A_{tail} , nor does it prove that every diffuse or LSB galaxy belongs to this family.

A.5 Regular exponential-disk readout

Definition/source ansatz. The exponential-disk family is used when the source morphology is a regular axisymmetric disk with a residual-blind scale radius R_s . The source surface readout is

$$\Sigma_{\text{exp}}(R) = \Sigma_0 e^{-R/R_s}.$$

Bridge specialization. In the bridge notation, this family chooses $\mathcal{S}_{\text{exp}}^\tau = A_{\text{exp}} \Sigma_0 e^{-R/R_s}$ as a regular axisymmetric source class. The corresponding radial operator $\mathcal{R}_{\text{exp}}$ is the standard axisymmetric disk Green operator. It is included as a conservative bridge specialization: the Tau-side family selection decides when the regular-disk carrier is the appropriate readout, while the carrier itself is the familiar exponential-disk radial response.

Radial readout. The axisymmetric thin-disk Green-function calculation for an exponential disk gives the standard Freeman radial carrier [11]

$$v_{\text{exp}}^2(R) \propto y^2 [I_0(y)K_0(y) - I_1(y)K_1(y)], \quad y = \frac{R}{2R_s},$$

where I_n and K_n are modified Bessel functions. The preflight kernel keeps this source-fixed shape and absorbs dimensional constants into the amplitude:

$$\boxed{\delta v_{\text{exp}}^2(R) = A_{\text{exp}} R_s y^2 [I_0(y)K_0(y) - I_1(y)K_1(y)]}$$

or

$$K_{\text{exp}}(R) = R_s y^2 [I_0(y)K_0(y) - I_1(y)K_1(y)].$$

Limits. The response is finite at the origin, peaks on disk scales, and declines at large radii. It is therefore a regular-disk carrier, not an arbitrary long-range tail.

Claim boundary. This is the most conservative readout family: it imports the standard axisymmetric exponential-disk radial carrier as the morphology-matched shell. The current paper does not claim that this is a new gravity formula. It tests whether source morphology can decide when this restricted shell is the right readout family.

A.6 Compact finite-source readout

Definition/source ansatz. The compact family is used when the source morphology indicates centrally concentrated or finite support with an outer compact radius R_c . The minimal assumption is that the active readout load is finite:

$$I_{\text{compact}}(R) \longrightarrow I_{\infty} \quad (R > R_c).$$

Bridge specialization. Here $\mathcal{S}_{\text{compact}}^{\tau}$ is the finite-support quotient-visible source class produced by Eq. (8). The radial readout operator $\mathcal{R}_{\text{compact}}$ is used only in its exterior finite-load limit unless stronger source information is available. Therefore the $1/R$ velocity-squared shell is a compact-source consequence, while the inner continuation used by the code is a numerical regularization.

Exterior readout. Outside the support, any finite radial load has the leading exterior behavior

$$\delta g_R(R) \sim -\frac{B_c}{R^2},$$

and therefore

$$\boxed{\delta v_{\text{compact}}^2(R) = R |\delta g_R(R)| \sim \frac{B_c}{R}} \quad (R \geq R_c).$$

The executable preflight uses

$$K_{\text{compact}}(R) = \begin{cases} R^2/R_c^3, & R < R_c, \\ 1/R, & R \geq R_c, \end{cases}$$

so that the kernel is finite in the inner numerical domain and matches the exterior $1/R$ value at $R = R_c$.

Limits. The scientifically meaningful part of this shell is the exterior finite-load falloff. The inner regularization is a numerical preflight convention, not a claim about the detailed central dynamics.

Claim boundary. This family is endpoint-ready only when R_c is source-native or frozen residual-blind. A compact label without a compact-support source field is a candidate or control, not a validation row.

A.7 Thick/flared or projection-sensitive readout

Definition/source ansatz. The thick/flared family is used when the source morphology indicates vertical support, flaring, edge-on projection, or projection-sensitive readout. The source inputs are a disk scale R_s and a dimensionless thickness profile $h(R)/R_s$, or a frozen constant proxy when no radial profile is available.

Bridge specialization. For this family, Eq. (8) is reduced with a three-dimensional/projection-sensitive source class $\mathcal{S}_{\text{flare}}^{\tau}(R, z)$. In a one-dimensional SPARC preflight, the vertical part cannot be fully resolved, so its effect enters the radial operator $\mathcal{R}_{\text{flare}}$ as a damping of short-wavelength Hankel modes. This is why the shell is a projection/readout attenuation, not a generic extra force.

Damped Fourier-Bessel readout. The thin exponential-disk response can be written as a Hankel/Fourier-Bessel radial integral. Vertical thickness damps high spatial-frequency modes. The minimal constant- h damping used in the preflight is

$$D(u, h/R_s) = \frac{1}{1 + u h/R_s}.$$

With $x = R/R_s$, the thick/flared shell is

$$K_{\text{flare}}(R) = R_s \frac{x}{2} \int_0^\infty \frac{u J_1(ux)}{(1 + u^2)^{3/2}} \frac{du}{1 + u h/R_s}$$

and

$$\delta v_{\text{flare}}^2(R) = A_{\text{flare}} K_{\text{flare}}(R).$$

Limits. When $h/R_s \rightarrow 0$, the damping tends to unity and the shell reduces to the thin-disk radial carrier class. Larger h/R_s suppresses short-wavelength radial structure and gives a smoother projected response. This is the desired behavior for vertical/projection-sensitive systems.

Claim boundary. The current SPARC population run mostly uses scalar thickness proxies, so this family is a one-dimensional proxy for a genuinely three-dimensional readout. It becomes endpoint-grade only when vertical/projection observables are source-frozen strongly enough.

A.8 Ring, barred, and lopsided families

The registry also contains ring/resonance, barred $m = 2$, and lopsided $m = 1$ families. Their source forms are

$$\Sigma_{\text{ring}}(R) \propto \exp \left[-\frac{(R - R_{\text{ring}})^2}{2w_{\text{ring}}^2} \right],$$

and

$$\Sigma_m(R, \phi) = \Sigma_m(R) \cos[m(\phi - \phi_m)], \quad m = 1, 2.$$

These are legitimate morphology handles, but they are not promoted here to full one-dimensional endpoint kernels. The ring family can sometimes be used as a localized radial proxy if R_{ring} and w_{ring} are source-frozen. The $m = 1$ and $m = 2$ families are intrinsically velocity-field objects because their phase information is largely lost in a one-dimensional azimuthally averaged rotation curve. Treating them as full SPARC-1D endpoint families would overstate the evidence.

A.9 Summary of derivation status

Table 17: Readout-family derivation status. “Derived shell” means that the kernel shape follows from the stated source ansatz and radial readout calculation. It does not mean that the amplitude or source classification is empirically validated.

Family	Derived kernel	Required source freeze	Status in this paper
$K_{\text{scale_tail_spiral}}$	$K_{\text{tail}} = I_2(R)/R$, $I_2 = \int_0^R W_{\text{tail}}(r)dr$	R_{in} , R_{cut} , amplitude policy	Derived 1D shell; source scales often proxy.
$K_{\text{exponential_disk}}$	$K_{\text{exp}} = R_s y^2 [I_0 K_0 - I_1 K_1]$	R_s , amplitude policy	Derived standard disk carrier; conservative 1D shell.
$K_{\text{compact_finite}}$	$K_{\text{compact}} \sim 1/R$ outside R_c	R_c , finite-load evidence, amplitude policy	Exterior shell derived; inner regularizer is numerical.
$K_{\text{thick_flared}}$	damped Fourier-Bessel kernel with $D = (1 + uh/R_s)^{-1}$	R_s , h/R_s or profile, projection state	Derived 1D proxy for 3D/projection readout.
K_{ring}	localized radial source convolution near R_{ring}	R_{ring} , w_{ring} , phase/context	Proxy only in this paper.
$K_{m=1}, K_{m=2}$	harmonic velocity-field response	$\Sigma_m(R)$, phase ϕ_m , velocity field	Velocity-field preferred; not a full 1D endpoint kernel.

B Holdout per-galaxy freeze ledger

Table 18 is an abbreviated ledger for the 44 holdout galaxies behind the 0.886 source-native formula result. The full machine ledger is `data/derived/source_native_readout_formula_labels.csv`; the score ledger is `data/derived/source_native_readout_formula_scores_by_galaxy.csv`. The source-proxy vector is reported as

$$R_s/R_{\text{in}}/R_{\text{cut}}/R_c/(h/R_s).$$

These are not claimed to be final source-native observables; they are the frozen available-data proxies used by the executable preflight. Caveat codes: FPTs = few rotation points, LINC = low inclination, LDE = large distance error, and VPROXY = vertical geometry proxy only. The final column reports matched-minus-wrong-family mean RMSE, so negative Δ means the matched family beats the wrong-family mean.

Table 18: Abbreviated holdout per-galaxy label/source/score ledger.

Galaxy	Family	Conf.	Role	Caveat	Source vector	Outcome
CamB	tail	1.00	middle	none	0.54/0.32/2.09/0.90/0.23	win; -71.52
ESO116-G012	exp	0.85	middle	LDE	2.46/1.45/10.95/4.13/0.18	win; -10.25
F561-1	tail	0.60	candidate	FPTS,LINC	3.36/1.98/10.89/5.64/0.19	win; -13.86
F563-1	tail	0.65	candidate	LINC,LDE	5.52/3.24/24.89/9.26/0.29	win; -2.90
F568-1	flare	0.70	candidate	LINC,VPROXY	3.09/1.81/14.68/5.19/0.18	loss; 5.40
IC2574	tail	1.00	control	none	3.30/1.94/12.16/5.54/0.25	win; -16.19
IC4202	compact	1.00	candidate	none	7.69/4.52/26.52/18.33/0.10	win; -0.43
NGC0024	flare	0.90	middle	VPROXY	1.01/0.59/11.84/1.70/0.13	win; -26.51
NGC2683	flare	0.90	middle	VPROXY	10.32/6.06/35.26/18.95/0.10	win; -2.99
NGC3726	flare	0.90	candidate	VPROXY	7.79/4.57/34.51/13.06/0.14	win; -4.36
NGC3949	flare	0.70	middle	FPTS,VPROXY	2.60/1.53/7.29/4.37/0.11	win; -14.09
NGC3972	flare	0.90	middle	VPROXY	2.86/1.68/9.07/4.80/0.12	win; -11.38
NGC4013	compact	1.00	middle	none	10.86/6.38/32.08/22.26/0.11	loss; 5.33
NGC4088	flare	0.90	candidate	VPROXY	6.75/3.97/22.44/11.33/0.12	win; -4.72
NGC4183	exp	1.00	control	none	6.24/3.66/24.28/10.47/0.22	win; -7.54
NGC4214	tail	0.80	middle	LINC	1.74/1.02/6.14/2.92/0.16	win; -23.65
NGC4389	flare	0.70	candidate	FPTS,VPROXY	1.87/1.10/5.38/3.15/0.09	win; -21.07
NGC5907	flare	0.90	middle	VPROXY	16.50/9.69/54.29/27.68/0.15	win; -1.22
NGC5985	compact	0.85	candidate	LDE	9.32/5.47/35.23/17.80/0.09	win; -7.42
NGC6789	tail	0.80	middle	FPTS	0.24/0.14/0.82/0.41/0.22	win; -107.71
NGC6946	compact	0.65	middle	LINC,LDE	6.13/3.60/21.14/11.86/0.11	loss; 13.68
NGC7331	flare	0.90	candidate	VPROXY	10.49/6.16/37.61/18.84/0.11	win; -1.84
UGC00731	tail	0.85	middle	LDE	3.52/2.07/14.72/5.91/0.39	win; -16.98
UGC00891	tail	0.65	control	FPTS,LDE	2.65/1.56/9.79/4.45/0.37	win; -23.48
UGC02023	tail	0.45	middle	FPTS,LINC,LDE	1.35/0.79/4.17/2.27/0.17	win; -37.05
UGC02259	exp	0.85	middle	LDE	2.73/1.60/9.43/4.58/0.22	win; -8.92
UGC03580	compact	0.85	candidate	LDE	1.64/0.96/28.70/4.07/0.13	loss; 28.48
UGC04305	tail	1.00	candidate	none	1.72/1.01/6.42/2.88/0.23	win; -28.17
UGC04499	tail	0.85	control	LDE	2.71/1.59/9.82/4.55/0.26	win; -18.61
UGC05414	tail	0.65	control	FPTS,LDE	1.43/0.84/4.78/2.40/0.23	win; -33.83
UGC05716	tail	0.85	control	LDE	4.00/2.35/16.03/6.71/0.35	win; -17.56
UGC05829	tail	0.65	control	LINC,LDE	2.25/1.32/8.32/3.77/0.26	win; -25.96

Continued on next page

Galaxy	Family	Conf.	Role	Caveat	Source vector	Outcome
UGC05986	tail	0.85	middle	LDE	2.99/1.76/10.17/5.02/0.15	win; -9.47
UGC06818	tail	1.00	middle	none	2.35/1.38/7.62/3.94/0.16	win; -21.42
UGC06917	tail	1.00	control	none	3.64/2.14/11.44/6.11/0.16	win; -14.68
UGC06983	tail	1.00	middle	none	5.20/3.05/18.45/8.72/0.24	win; -8.10
UGC07151	exp	1.00	control	none	1.79/1.05/6.04/3.00/0.17	win; -29.87
UGC07261	exp	0.45	control	FPTS,LINC,LDE	2.27/1.33/7.36/3.81/0.17	win; -22.60
UGC07577	tail	1.00	middle	none	0.56/0.33/1.78/0.94/0.13	win; -69.53
UGC07603	exp	0.85	middle	LDE	1.32/0.78/4.80/2.22/0.23	win; -32.16
UGC08699	compact	0.85	middle	LDE	4.27/2.51/26.33/10.75/0.10	win; -2.94
UGC11914	compact	0.65	candidate	LINC,LDE	2.28/1.34/9.85/6.01/0.08	win; -10.70
UGC12506	exp	1.00	candidate	none	15.14/8.89/55.56/25.41/0.18	loss; 4.99
UGCA444	tail	1.00	control	none	0.81/0.48/3.74/1.36/0.47	win; -54.28

C Source-reference ledger

Table 19 records how the main external references enter the present paper. The entries are provenance for residual-blind source fields and baseline context. They do not by themselves validate any readout formula.

Table 19: Reference ledger for source fields used or proposed by the morphology-readout protocol.

Reference group	Role in this paper	Claim boundary
[19]	SPARC rotation curves and baryonic mass-model arena.	Common numerical data source; not a readout-morphology catalogue.
[26, 25, 22]	MOND-like and empirical RAR comparison context.	Baseline comparators; not used to choose morphology-readout families.
[11]	Standard exponential-disk radial carrier used in the regular-disk readout derivation.	Formula-context reference; not evidence for the morphology-matched endpoint protocol by itself.
[41, 44, 39, 43]	Examples of H I, velocity-field, late-type photometric/kinematic, and lopsidedness data products suitable for readout morphology fields.	Support feasibility of a catalogue; object-specific endpoint gates still require per-galaxy source freeze.
[27, 38, 6, 20]	Catalogue, decomposition, multiwavelength, and resolved-gas infrastructure needed for a residual-blind readout morphology catalogue.	Source-infrastructure references; they support data availability and provenance, not endpoint validation by themselves.
[21, 47]	Matched-resolution EDGE CO and CALIFA optical maps, velocity-convention calibration, and the external seven-galaxy confirmation packet.	The frozen morphology-alignment proxy fails its promotion gate; the data source does not identify a Tau mechanism or exhaust standard astrophysics.
[1, 12, 36]	NGC3198 H I extent, extraplanar gas, and radial-flow/tail-onset source context.	Used to illustrate source acquisition and failure-mode routing; not promoted as an accepted endpoint in this paper.
[48, 5]	NGC4013 H I and vertical-structure context.	Supports mixed/vertical-overlay interpretation; NGC4013 remains retrospective and replay-needed.
[35, 37, 46]	NGC5907 warp, edge-on, and projection/ISM context.	Supports the projection/mixed readout lane; single-galaxy result remains preliminary.
[2, 29]	NGC7331 H I warp/history and vertical-scale context.	Supports caveated mixed endpoint and pure-branch negative-control interpretation.
[44]	NGC4088 H I geometry, strong disk distortion, position-velocity asymmetry, asymmetric warp behaviour, and near-companion context.	Supports the warp/history/asymmetric-projection readout lane; current NGC4088 result remains source-review and normalization-law caveated.
[7, 45, 34, 40]	Exploratory disturbed/dwarf H I morphology source classes.	Included as examples for future catalogue lanes; not scored as endpoint evidence here.

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